慢性疼痛管理與老年寵物復健

國立臺灣大學附設動物醫院

復健及整合醫學科

主治獸醫師

劉金鳴

CVA, CVCH, CVFT, CVTP,

CCRP, CKTP, CVPP

DACVSMR (Canine) Candidate

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慢性疼痛管理

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8 signs that your dog is in pain

Pain is an evolutionary advantage. So how do you get the important messages that your dog is trying to hide from you?

Study up on these general, pain behaviors!



1 **Loss of appetite**

If your dog isn't feeling well, he might not be up to the task of digesting food.

Obsessive grooming 2

If your dog keeps licking part of his body, he probably has an ache there.





3 **Weird posture or gait**

You've probably seen your pet strike some adorably awkward poses, especially during nap time.

Pained noises 4

Whimpering, whining, yelping, and groaning are all sure to get your attention.





5 **Lethargy**

Take note of any dramatic changes. They could indicate that your dog is feeling low.

Antisocial behavior 6

Just like people, many dogs become antisocial when they're feeling poorly.





7 **Panting**

If your dog is panting for no apparent reason, he could be stressed or sick.

Shivering, shaking or trembling 8

he might be shivering in pain.

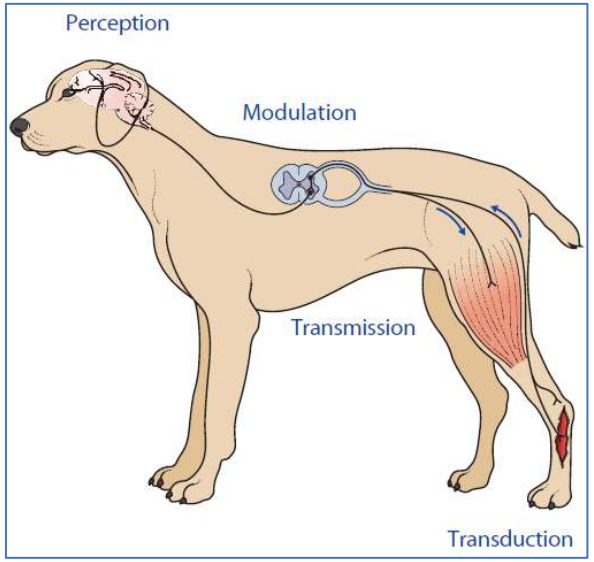


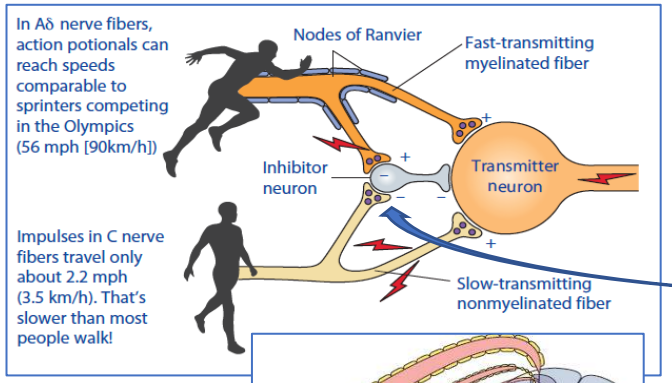
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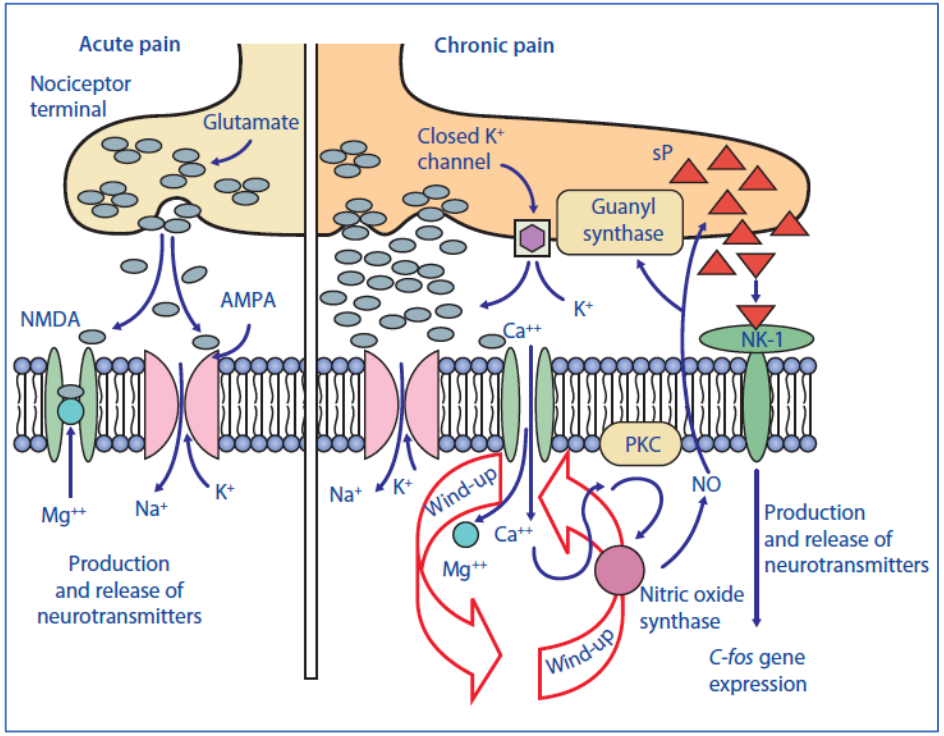
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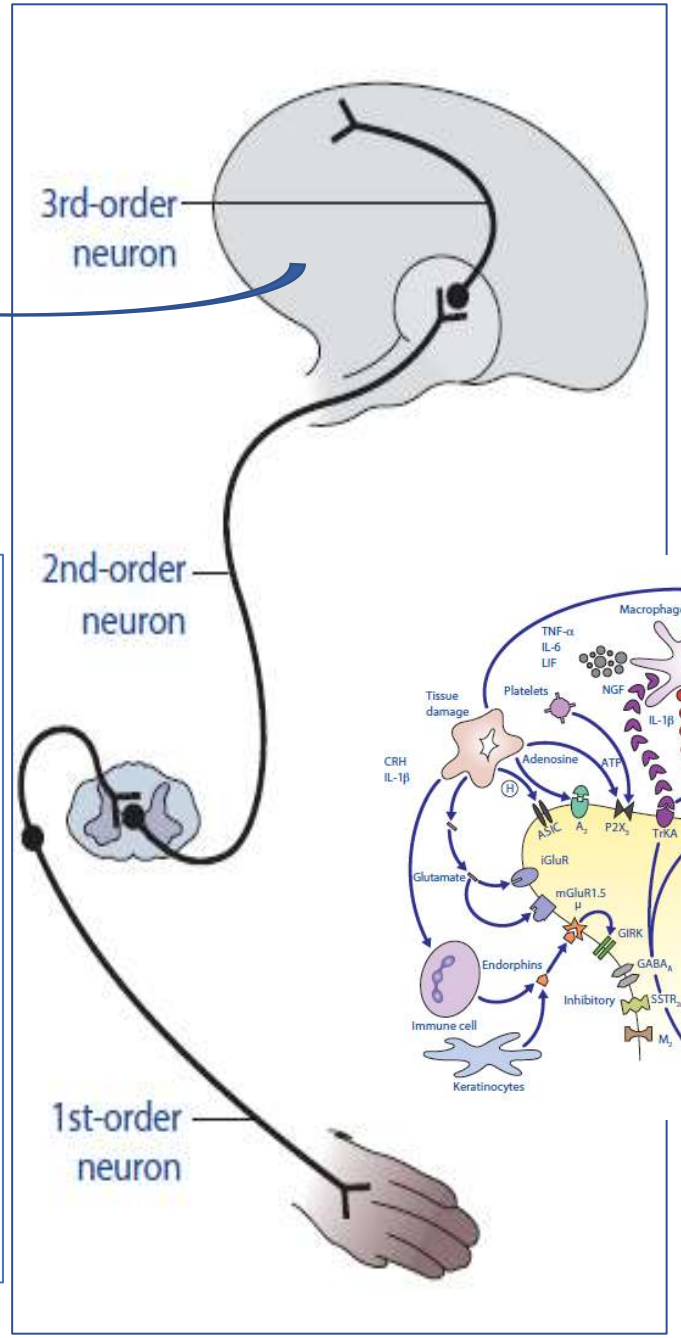
疼痛途徑

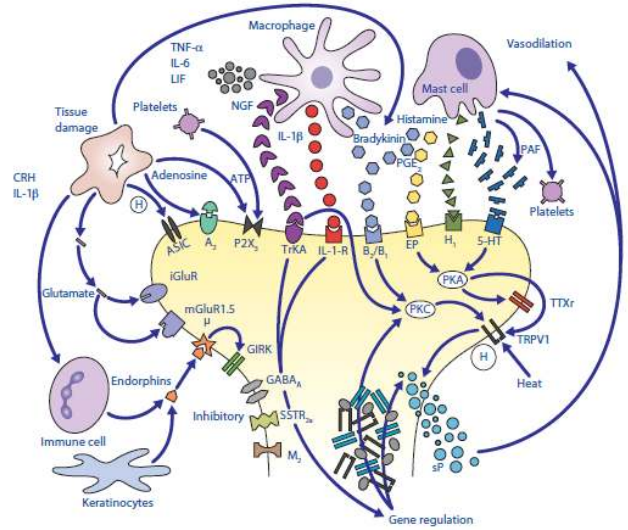








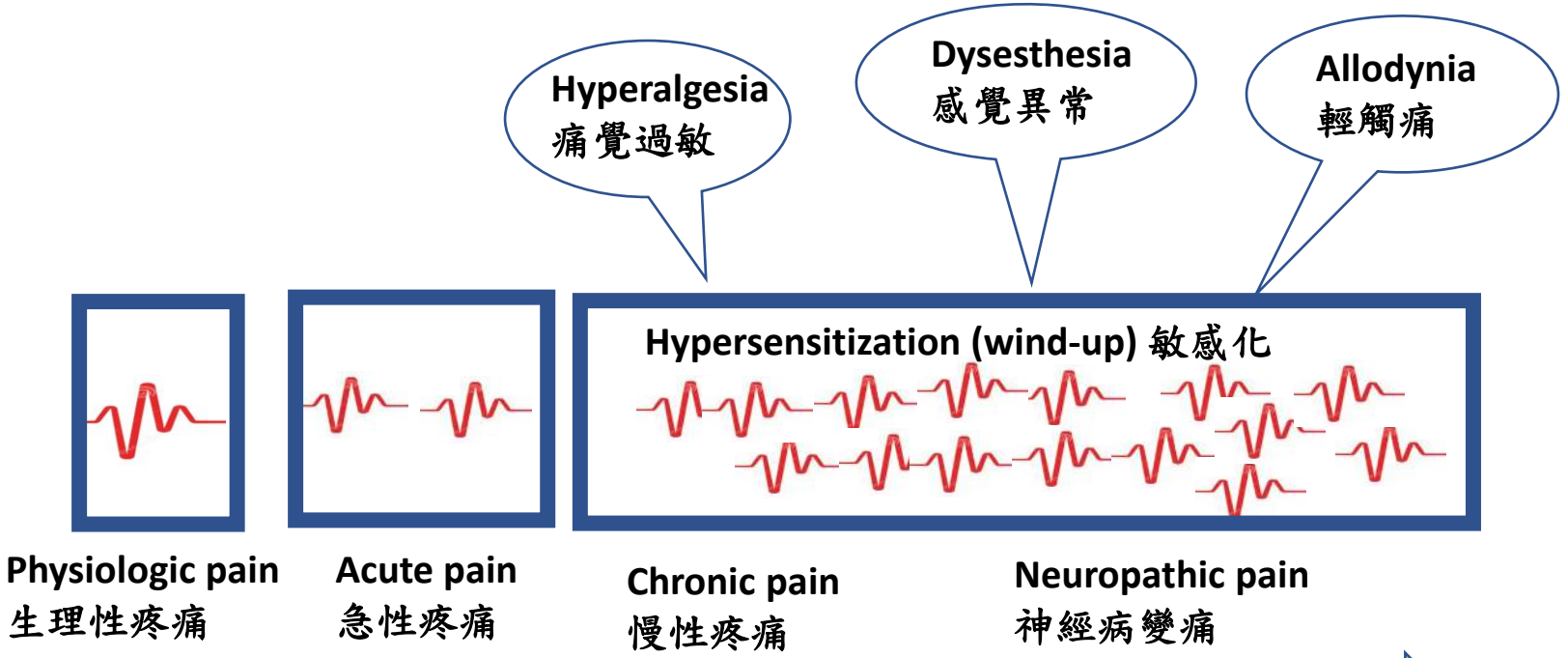




Fox, 2014. Pain Management in Small Animal Medicine

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疼痛種類



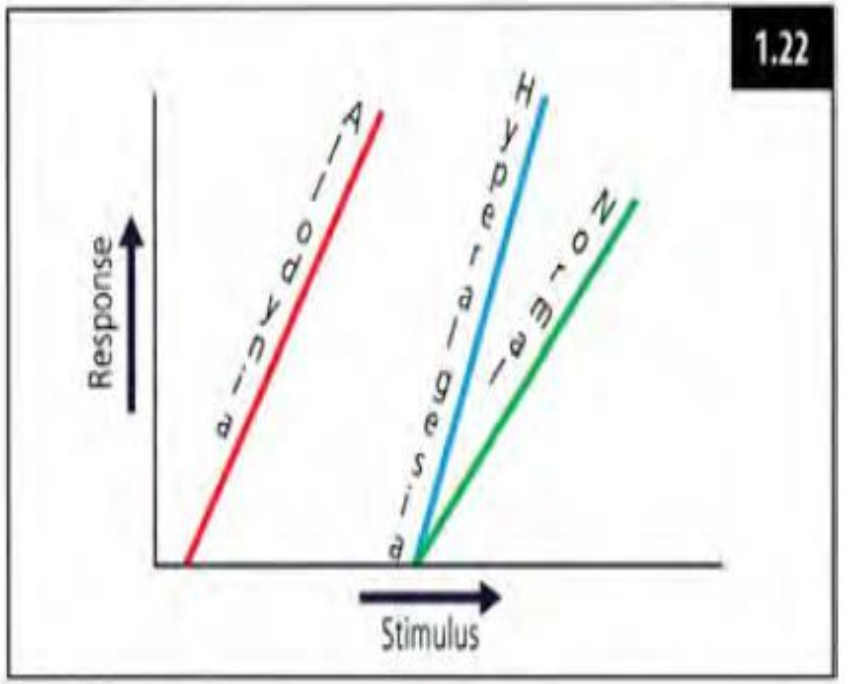


FIGURE 1.22 Stimulus-response curve of noxious insult.

Fox. 2024, Chronic Pain in Small Animal Medicine, 2nd Ed

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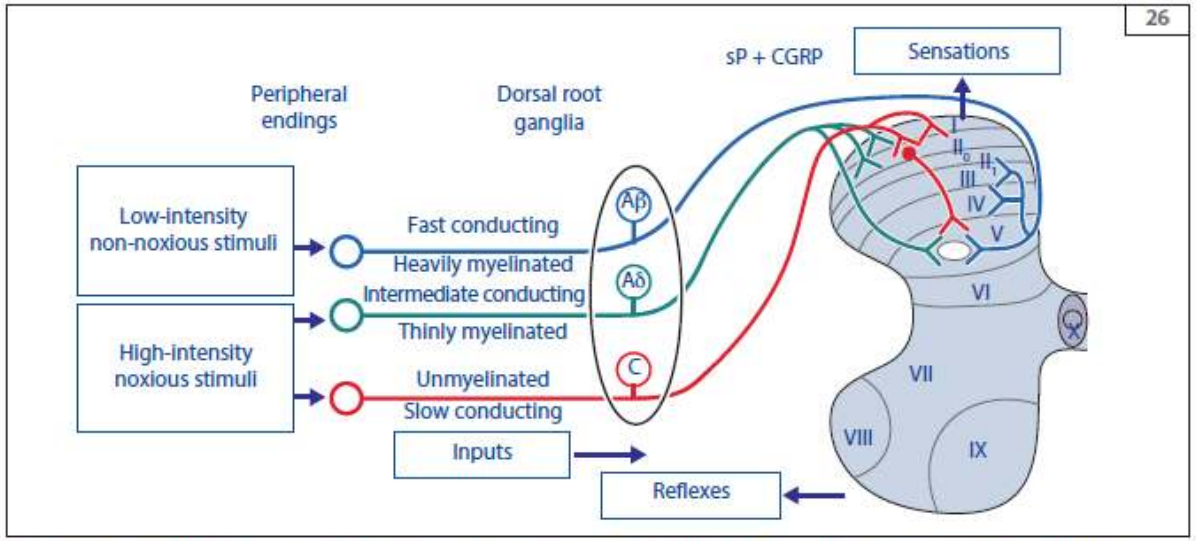


Fig. 26 Pathway for noxious stimuli through the substantia gelatinosa in the dorsal horn. CGRP: calcitonin gene-related peptide; sP: substance P.

Fox, 2014, Pain Management in Small Animal Medicine

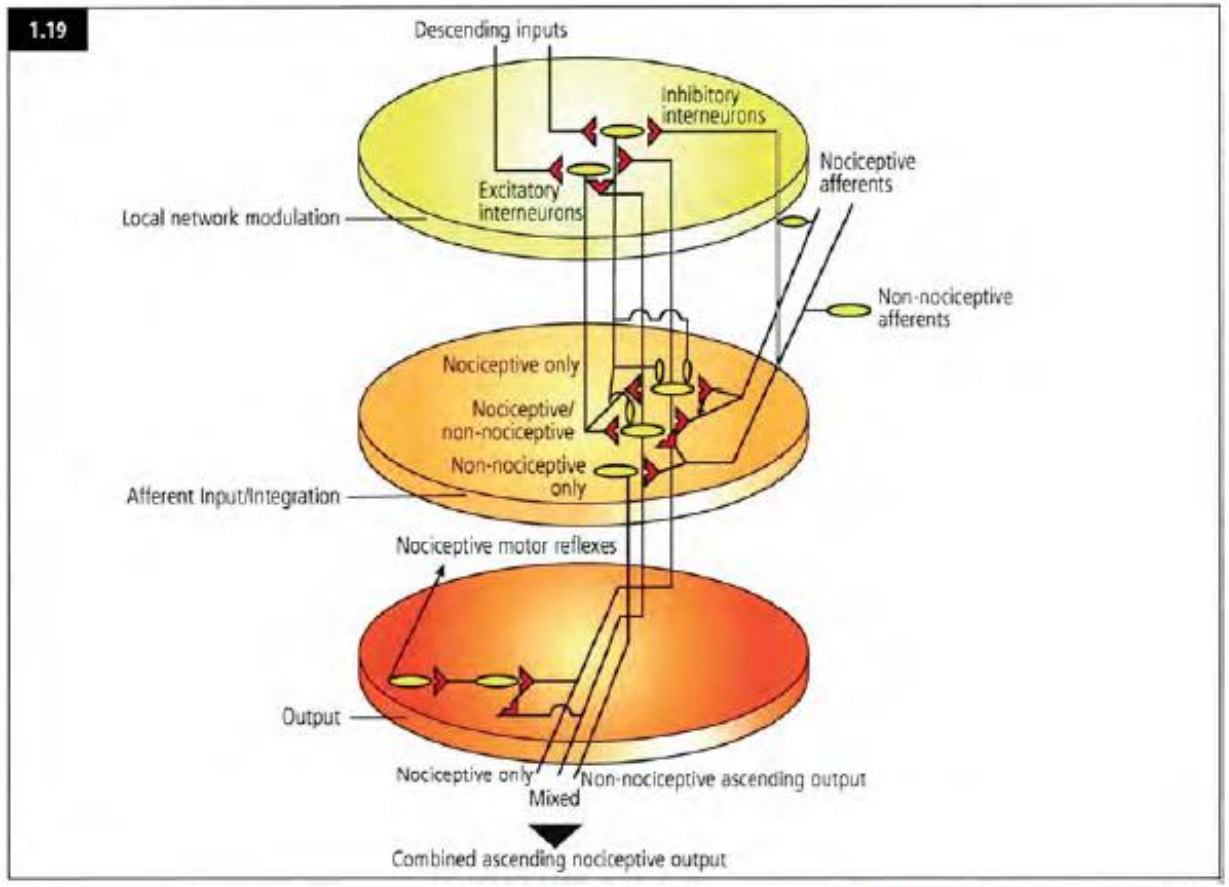


FIGURE 1.19 Dorsal horn neurons consist of projecting neurons, propriospinal neurons, and local interneurons.

Fox. 2024, Chronic Pain in Small Animal Medicine, 2nd Ed.

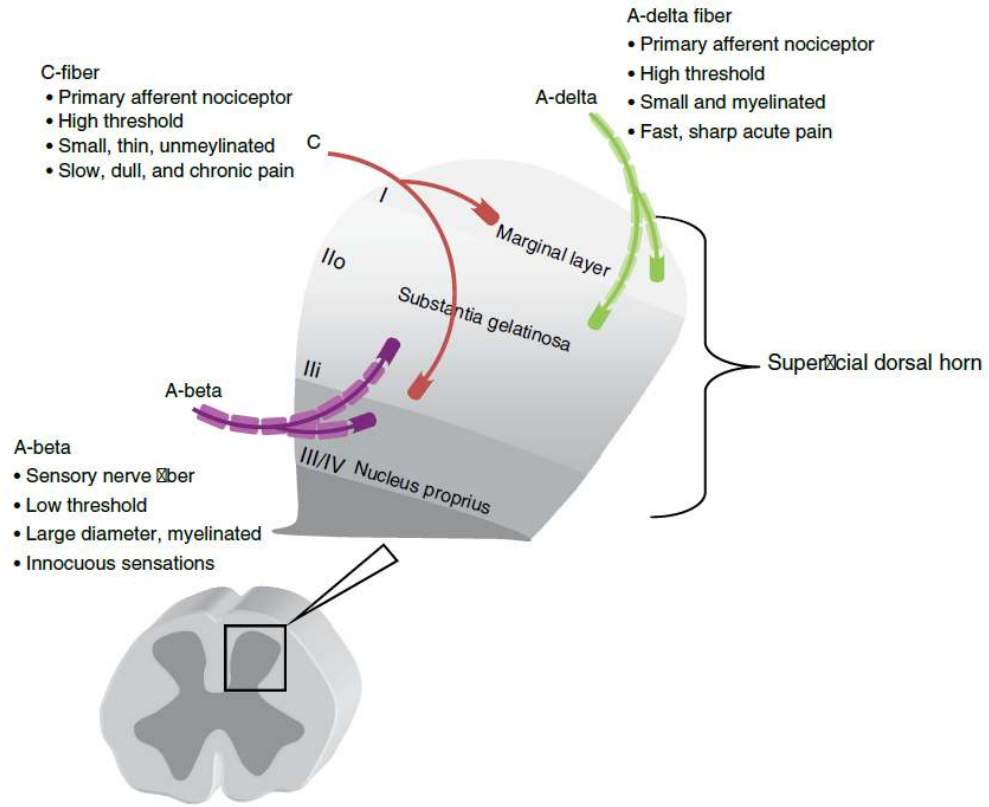


Figure 4.3 A-delta, A-beta and C fiber synapse within the superficial laminae of the dorsal horn of the spinal cord. Source: Drawn by Kristen Cooley.

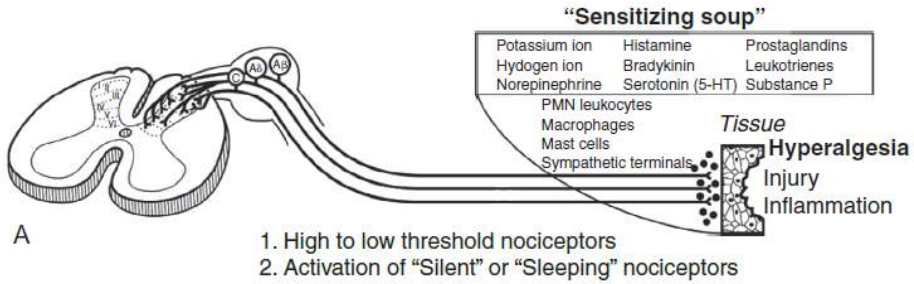


FIGURE 2-10 **A**, Peripheral sensitization. Tissue damage and inflammation produce nociceptor-sensitizing substances, including prostaglandins, histamine, serotonin, bradykinin, proteases, cytokines (tumor necrosis factor–α), and nerve growth factor. This "sensitizing soup" lowers the nociceptor threshold to noxious and non-noxious stimuli and activates "silent" or "sleeping" nociceptors, resulting in hyperalgesia and allodynia.

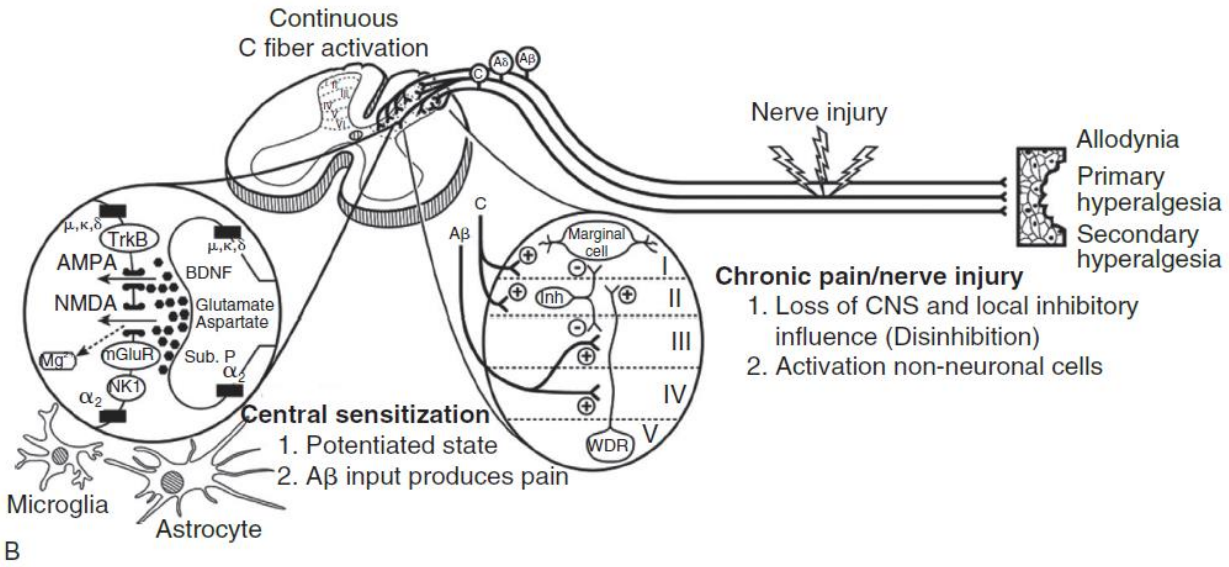


FIGURE 2-10—cont'd **B**, Central sensitization. Strong (high-intensity) persistent stimuli activate C fibers, causing the release of a variety of transmitters, including glutamate, substance P (sP), and other factors such as brain-derived neurotrophic factor (BDNF) at central nerve terminals; this results in the activation of AMPA, NMDA, Neurokinin 1 and tyrosine receptor kinase B (Trk B) receptors, respectively, producing acute and long-lasting dull, aching, burning pain sensations. The activation of these receptors increases the activity of signaling molecules that alter gene expression and change the responsiveness (sensitize) of the spinal cord to subsequent input. Chronic stimulation may result in neurochemical changes (neuroplasticity) in the spinal cord such that all stimuli produce pain. AMPA, a-Amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid; NMDA, N-Methyl-D-aspartate.

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疼痛評估：

- 經認證過之疼痛量表CSOM (Client Specific Outcome Measurement)
  - Canine brief pain inventory (CBPI)犬簡要疼痛清單
  - Helsinki chronic pain index (HCPI)赫爾辛基慢性疼痛量表
  - Canine orthopedic index (COI)犬骨科活動量表
  - Liverpool osteoarthritis in dogs (LOAD)利物浦犬慢性關節炎量表
  - Visual analog scale (VAS)視覺類比量表
  - Glasgow composite measure pain scale short form (CMPS-SF)簡式格拉斯哥複合式疼痛量表
  - Canine osteoarthritis staging tool (COAST)犬慢性關節炎分級工具

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2015 AAHA/AAFP Pain Management Guidelines for Dogs and Cats*

Mark Epstein, DVM, DABVP, CVPP (co-chairperson), Ilona Rodan, DVM, DABVP (co-chairperson), Gregg Griffenhagen, DVM, MS, Jamie Kadrlik, CVT, Michael Petty, DVM, MAV, CCRT, CVPP, DAAPM, Sheilah Robertson, BVMS, PhD, DACVAA, MRCVS, DECVAA, Wendy Simpson, DVM

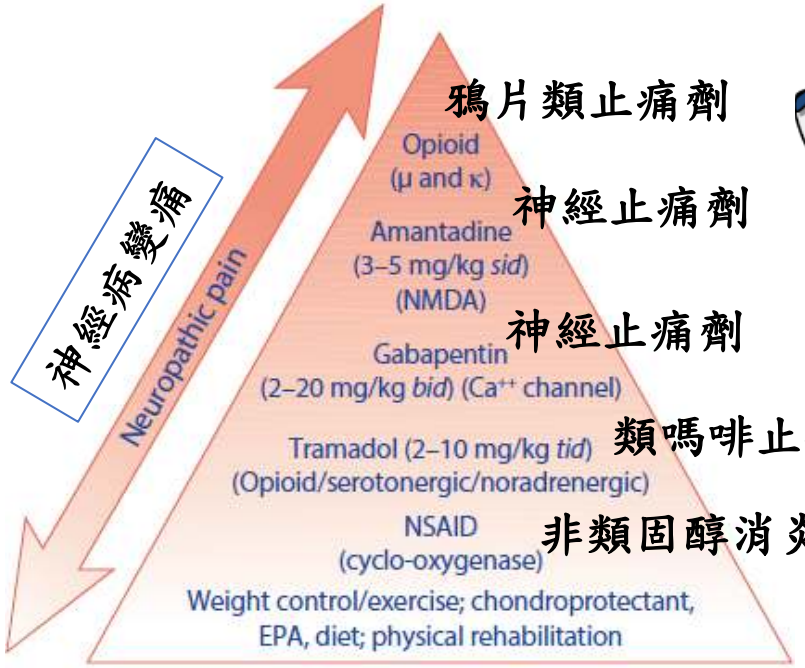
2022 AAHA Pain Management Guidelines for Dogs and Cats*

Margaret E. Gruen, DVM, MVPH, PhD, DACVB†, B. Duncan X.Lascelles, BSc, BVSc, PhD, CertVA, DSAS(ST), DECVS, DACVS, FRCVS†, Elizabeth Colleran, DVM, MS, DABVP (Feline), Alison Gottlieb, BS, CVT, VTS (ECC), Jennifer Johnson, VMD, CVPP, Peter Lotsikas, DVM, DACVS-SA, DACVSMR, Denis Marcellin-Little, DEDV, DACVS, DACVSMR, Bonnie Wright, DVM, DACVAA

- Pharmacological Agents in the Treatment of Chronic Pain
  - NSAIDs/Galliprant ®
  - Gabapentin- “new tramadol”
  - Amantadine
  - DMOAD
  - Nutraceuticals- omega 3 fatty acid, glucosamine…
  - CBD
  - Intra-articular injection
  - anti-NGF monoclonal antibodies
  - TRPV1 agonists
- Non-pharmacologic Modalities for Chronic Pain Management
  - Weight Optimization
  - Exercise and Rehabilitation Therapy
  - Acupuncture
  - MTrP
  - Environmental Modification

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慢性疼痛止痛劑



鴉片類止痛劑

神經止痛劑

神經止痛劑

類嗎啡止痛劑

非類固醇消炎止痛劑

體重控制/運動；軟骨保護劑，營養補充劑；物理治療復健

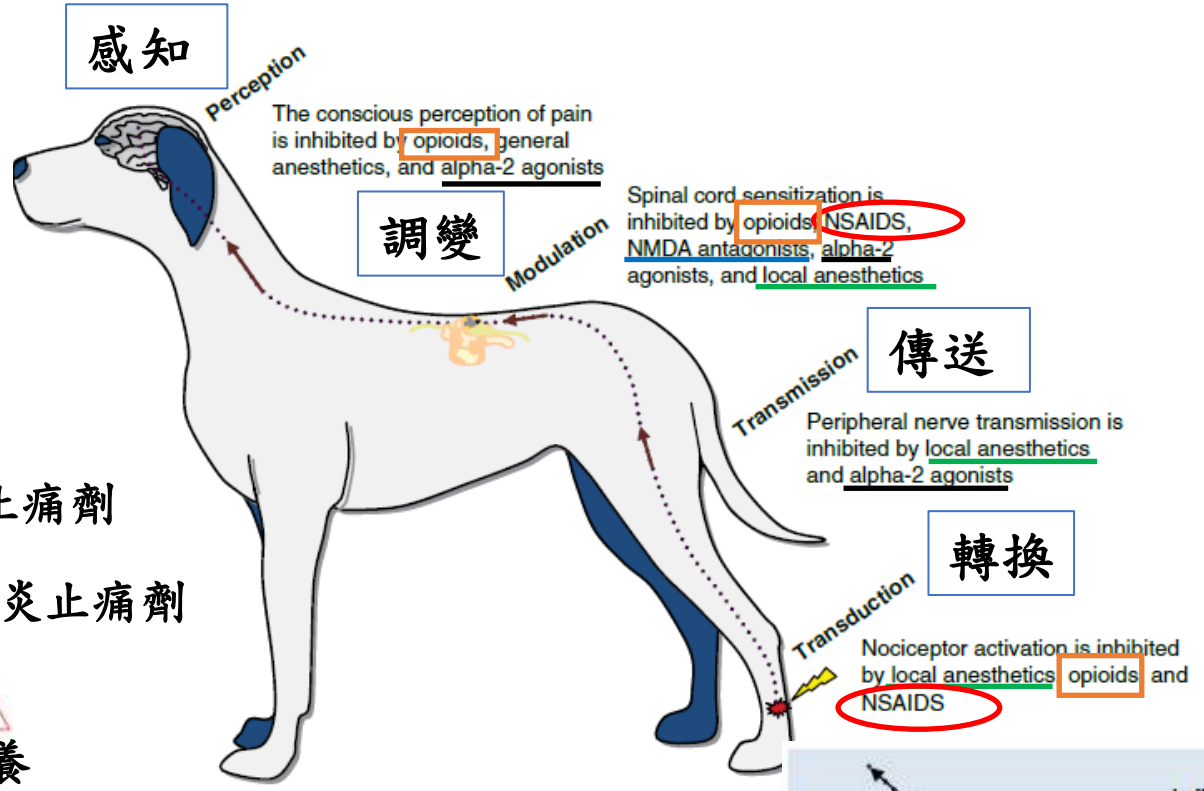


Figure 7.1 The pain pathway. Source: Courtesy of Kristen Cooley.

Goldberg, 2015. Pain Management for Veterinary Technicians and Nurses

Examples

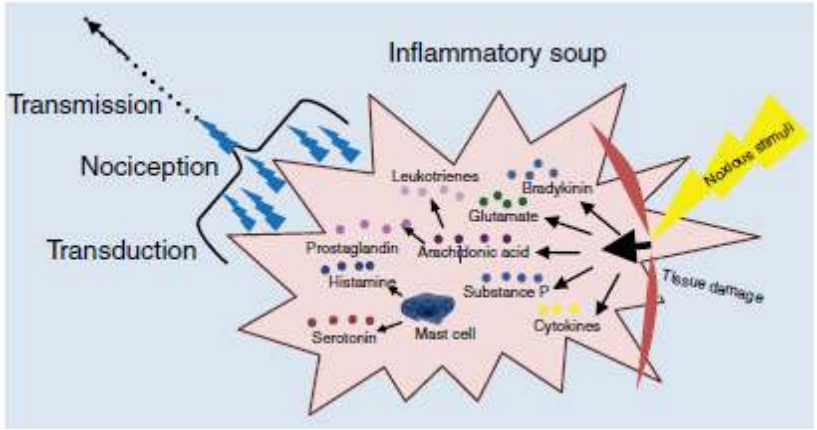
NSAIDs:
Meloxicam, Carprofen, Onsior, Previcox, Galliprant, diclofenac

Opioids:
Codeine, Buprenorphine, Tramadol, Butorphanol,

Alpha-2 agonists:
Dexmedetomidine

NMDA antagonists:
Ketamine, Amantadine,

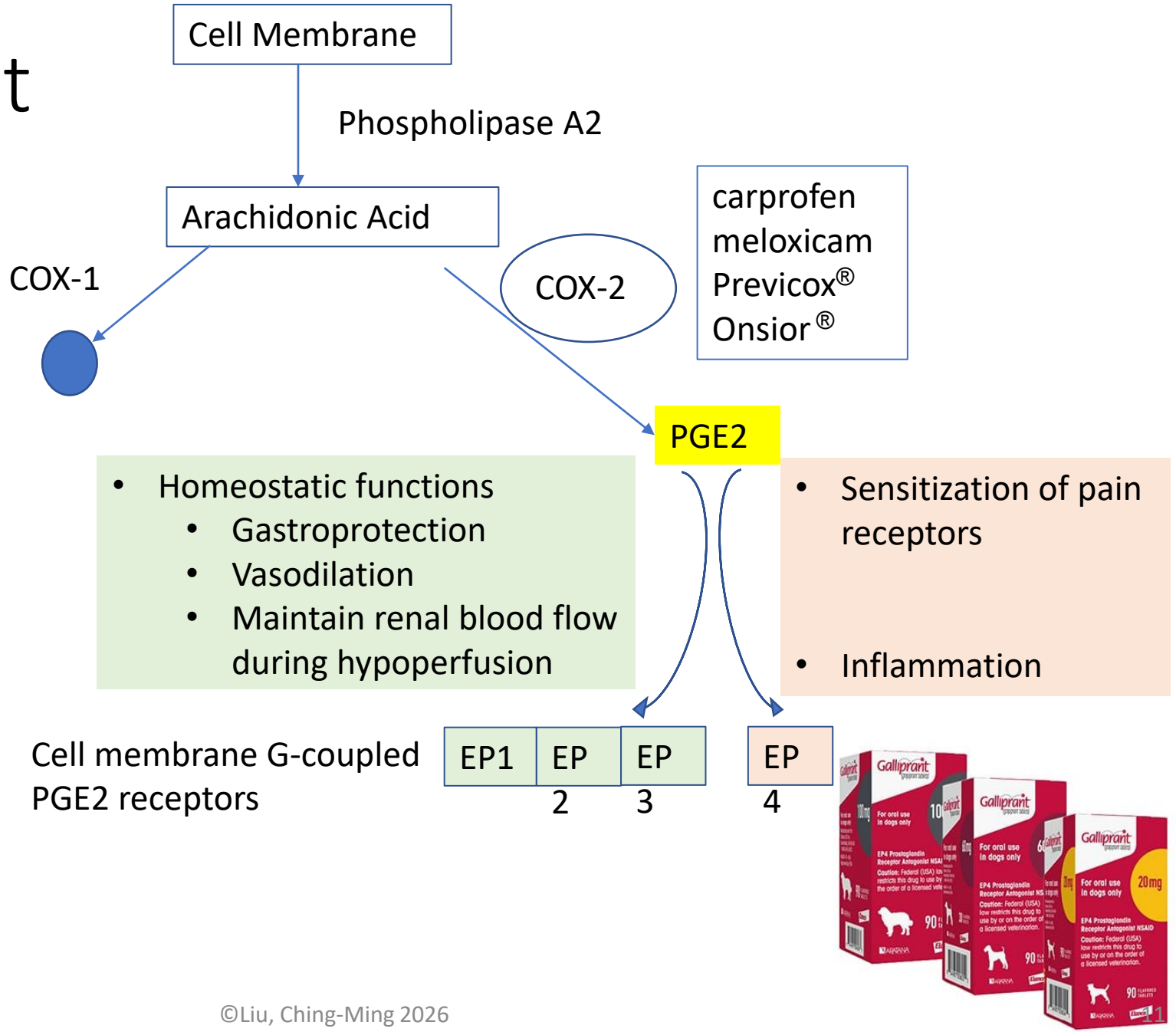
Local anesthetics:
Nocita, Lidocaine



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NSAIDs/Galliprant

- Anti-inflammatory
- Inhibit Peripheral sensitization
- Inhibit Central sensitization



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Tramadol

- Weak mu-receptor agonist (opioid-like effects)
- analgesic effects may be due to serotonin and norepinephrine reuptake inhibition (Serotonin ↑, Norepinephrine ↑)
- Inhibitory neurotransmitters: serotonin, norepinephrine

- Metabolite (M1, o-demethyltramadol; ODT)
  - Human, cat----√
  - Dog ----x (elimination in the dog appears to be more rapid than in the cat)



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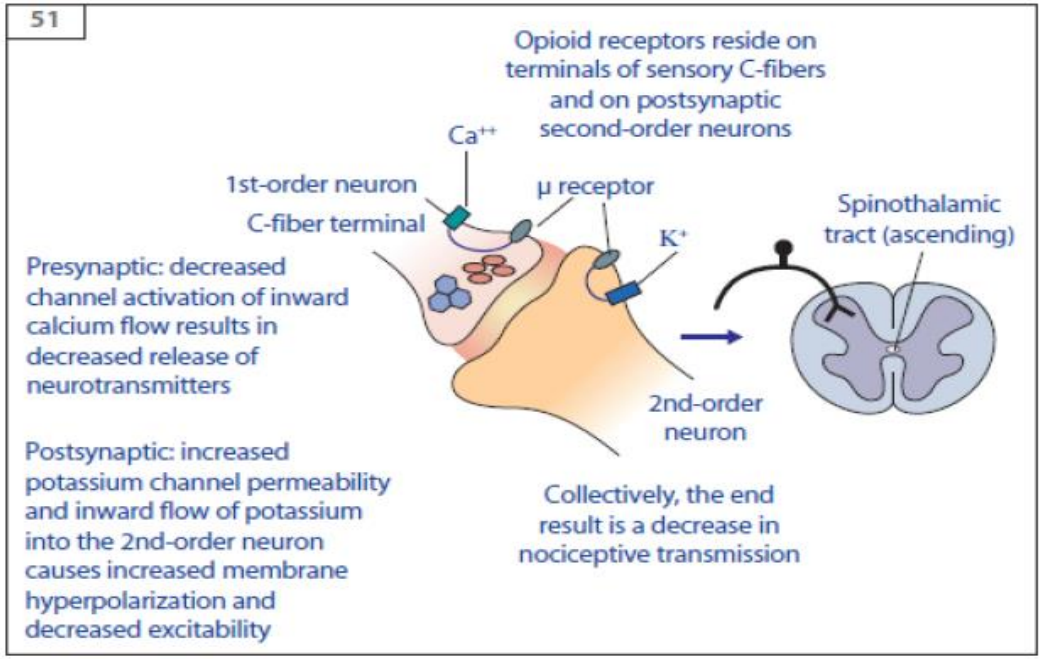


Fig. 51 Opioid-induced antinociception is mainly C-fiber mediated[5], where opioids act at both the pre- and postsynaptic sides of nerve junctions.

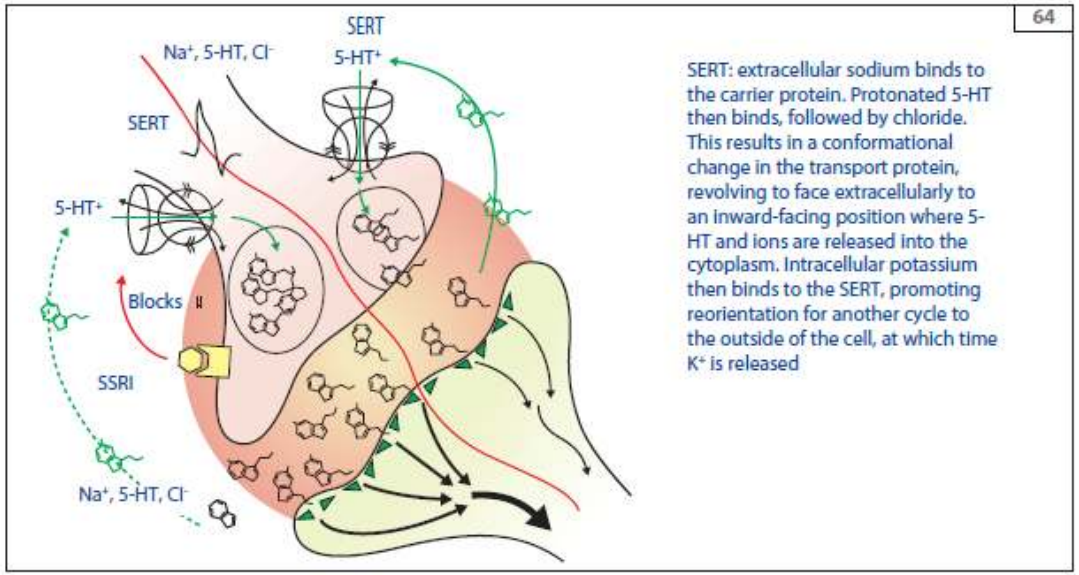
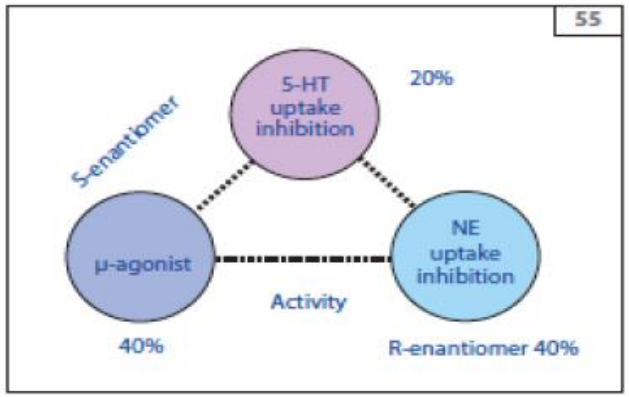


Fig. 64 Selective serotonin reuptake inhibitors (SSRIs) increase the extracellular level of the neurotransmitter serotonin by inhibiting its reuptake into the presynaptic cell, increasing the level of serotonin available to bind to the postsynaptic receptor. Left side of figure illustrates the effect of a SSRI, whereas the right side depicts the normal reuptake of serotonin. 5-HT: 5-hydroxytryptamine; SERT: serotonin transporter.

Fig. 55 Most (60%) of the activity associated with tramadol is as a tricyclic antidepressant/serotonin (5-HT) reuptake inhibitor. Less than half of its activity is as an opioid. NE: norepinephrine.



Pain Management in Small Animal Medicine, Steven M Fox, 2014

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Gabapentin

- Ca++ channels ↓
- Serotonin ↑
- Norepinephrine ↑
- Na+ channel ↓
- NMDA receptor antagonist

- analgesic
- weakness

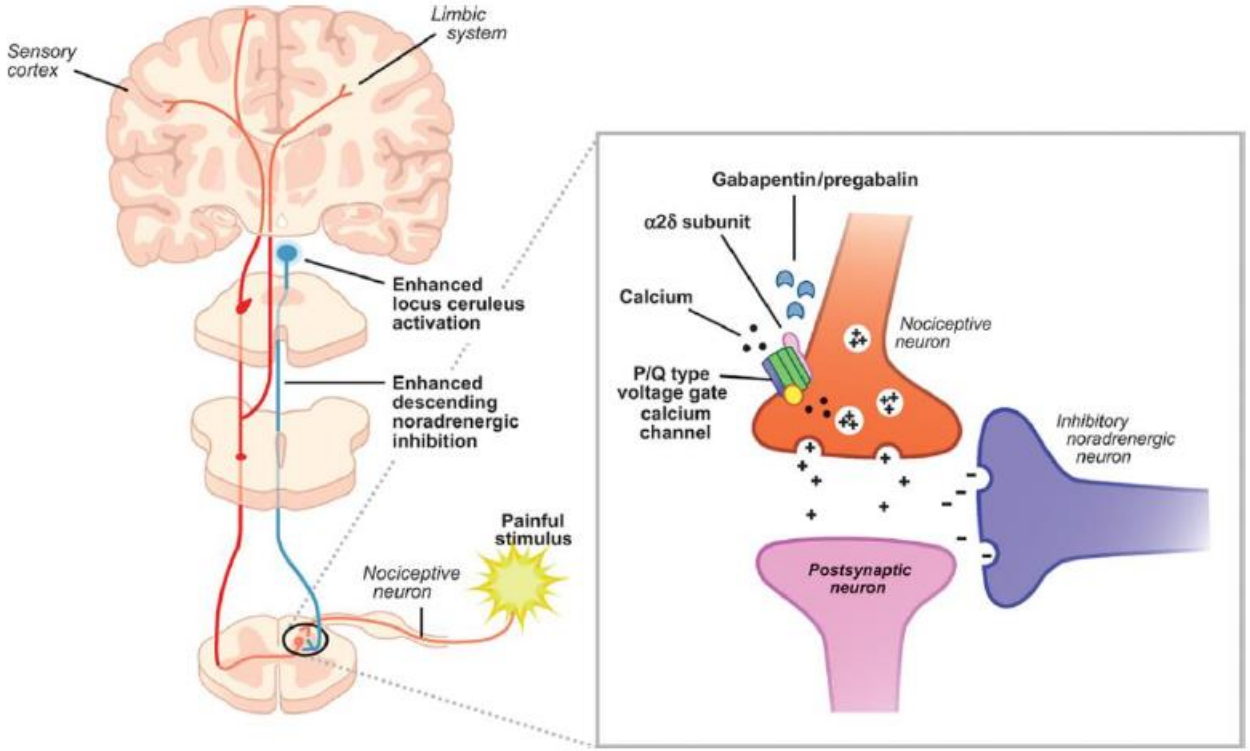


Amantadine

- Anti- virus
- NMDA receptor antagonist
- Helper for gabapentin
- Excreted via kidneys



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From google

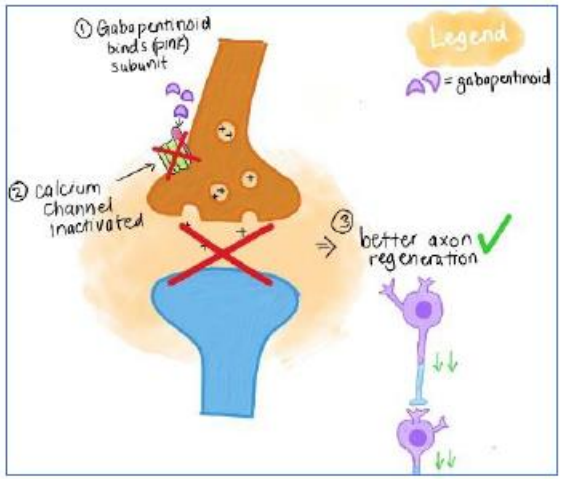
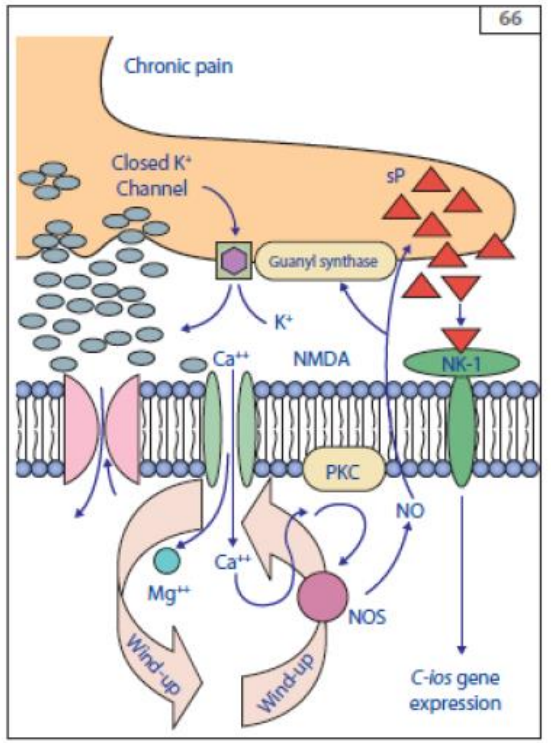


Fig. 66 Central nervous system wind-up involves the N-methyl-D-aspartate (NMDA) receptor which, when activated, allows the conduction of calcium into the neuronal axon through a channel previously blocked by a magnesium 'plug'. NK-1: neurokinin-1; NO: nitric oxide; NOS: nitric oxide synthase; PKC: protein kinase C; sP: substance P.



Pain Management in Small Animal Medicine, Steven M Fox, 2014

慢性關節炎營養補充品

- 疾病修飾藥物(DMOAD)
  - Adequan® 美國FDA唯一認可對骨性關節炎有療效之補充品
  - 每週2次皮下或肌肉注射連續4週，之後每週1次連續4週，以後視需要時使用
  - Cartrophen® (pentosan)
  - 每週1次皮下注射連續4週，之後視需要每2週1次，以後視需要時使用
- Nutraceuticals
  - Omega 3 fatty acid
  - Glucosamine
  - Chondroitin
  - UC-II













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CBD大麻二酚



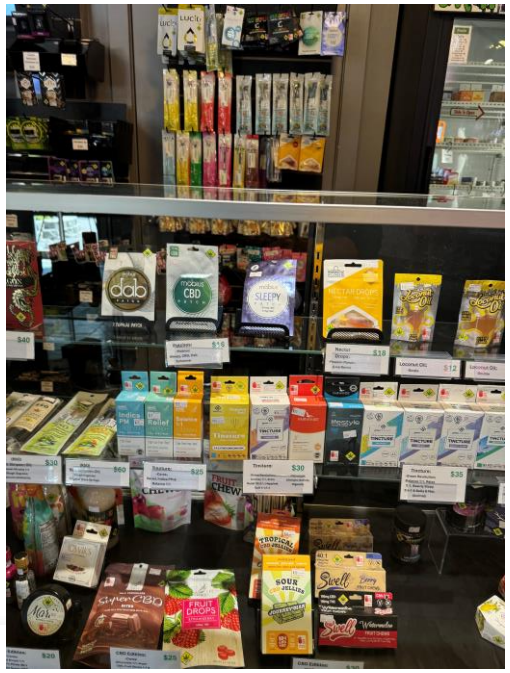
www.caninearthritis.co.uk

- CBD是甚麼?
  - Marijuana（大麻)醫療用大麻
  - Hemp(漢麻; 工業大麻)娛樂用大麻

  - CBD（cannabidiol; 大麻二酚）
  - THC（Tetrahydrocannabinol; 四氫大麻酚）

  - Pure oil（純油）
  - Broad spectrum（廣譜油）
  - Full spectrum（全譜油）















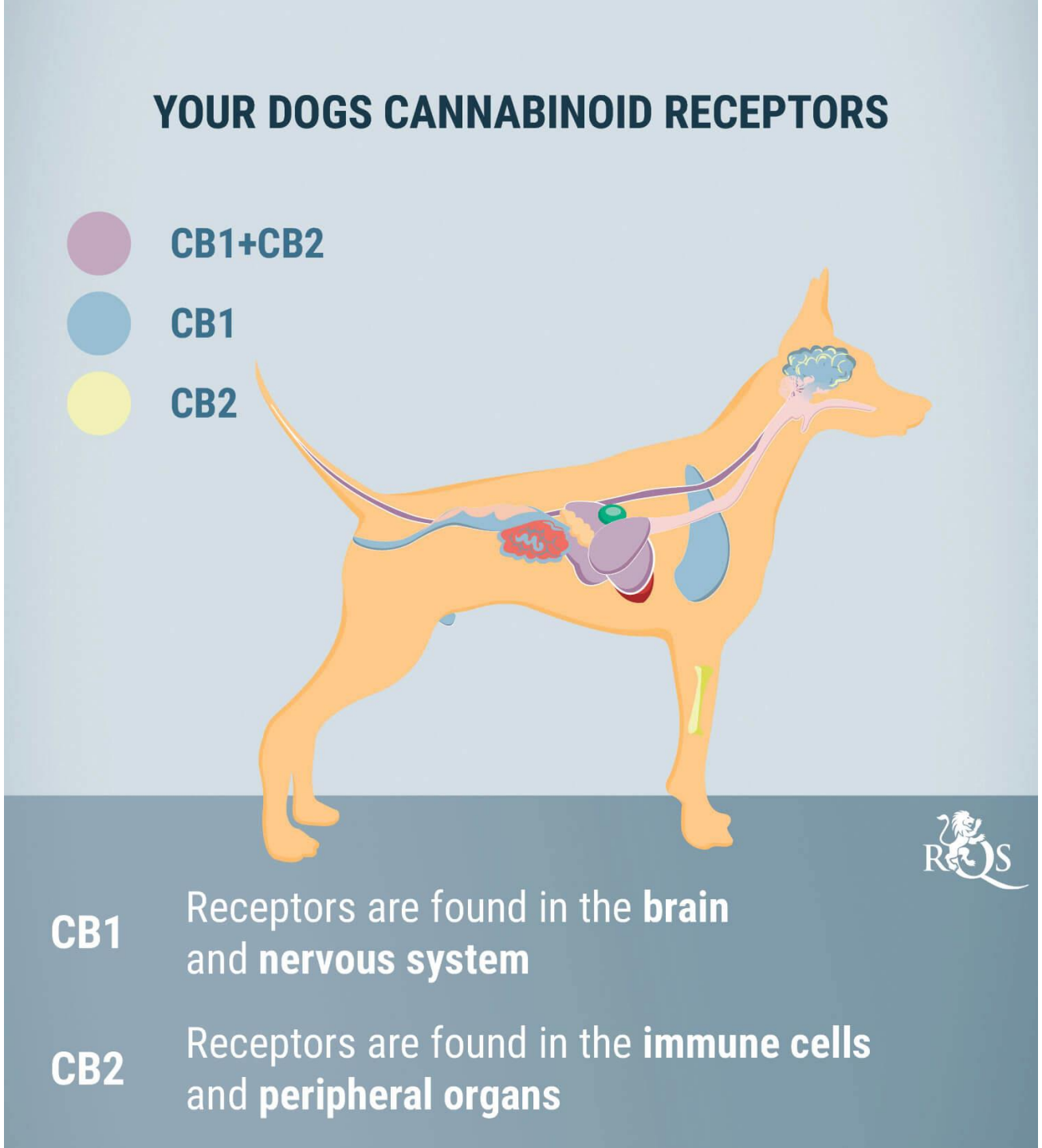


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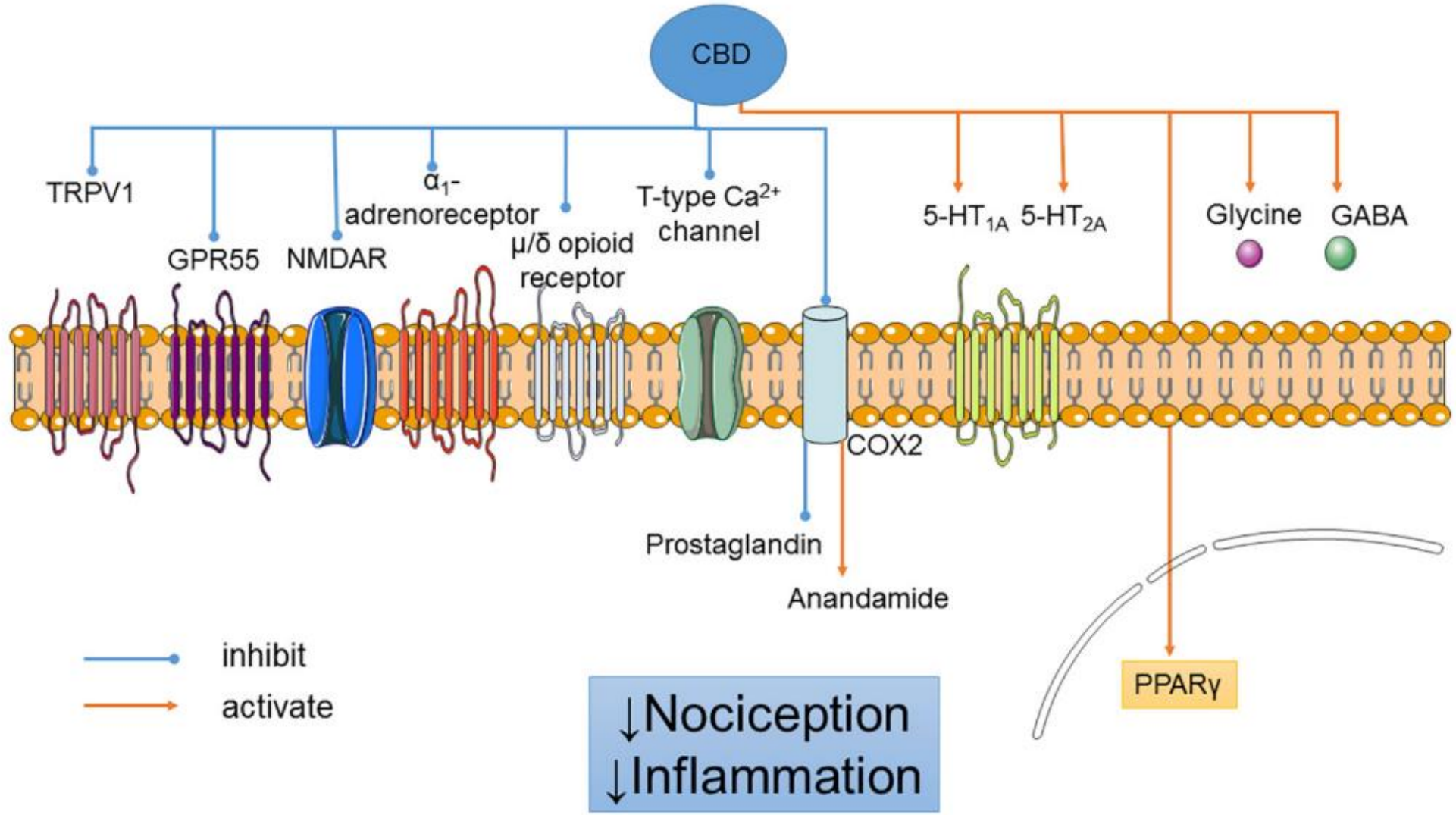
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- EBS（endocannabinoid system; 內源麻素系統）
- CB1, CB2受體，CB1主要分布在大腦及其他周邊神經，而 CB2 則主要分布於免疫調節的器官中



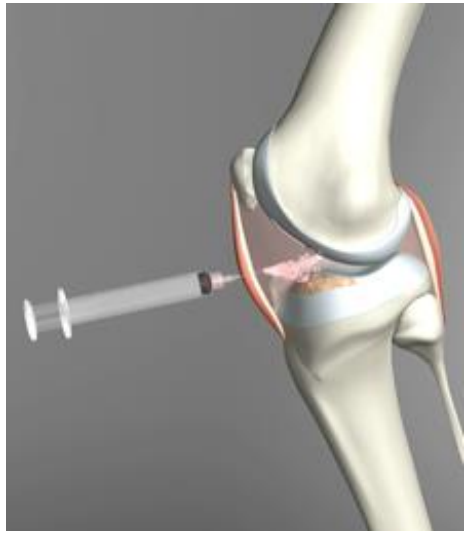
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The molecular mechanisms that underpin the biological benefits of full spectrum cannabis extract in the treatment of neuropathic pain and inflammation

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- 關節注射
  - **Steroid 類固醇**
    - Triamcinolone 1-3 mg/joint (10mg/ml)
    - Methylprednisolone 10-20 mg/joint (20 mg/ml or 40mg/ml)
  - Every 3 months for up to 2 years
  - Max. of 4 inj. Per year







  - **HA(hyaluronic acid) 玻尿酸**
    - **Viscosupplementation黏彈性補充劑**
    - Commonly used in horses and human
    - Synovial fluid of HA molecular weight
      - **Human** 6.0 x10^6 Dalton
      - **Horses** 3.0 x10^6 Dalton
      - **Dogs** 1-2 x10^6 Dalton
    - Weekly for 3 weeks







  - **PRP(platelet rich plasma)高濃度血小板血漿**
  - **Stem cell 幹細胞**
  - **Secretome - Exosome 外泌體**
  - **Hydrogel**
    - **Spryng**
    - **Arthamid- 2.5% Polyacrylamide(聚丙烯醯胺)**
  - **Tin-117m**





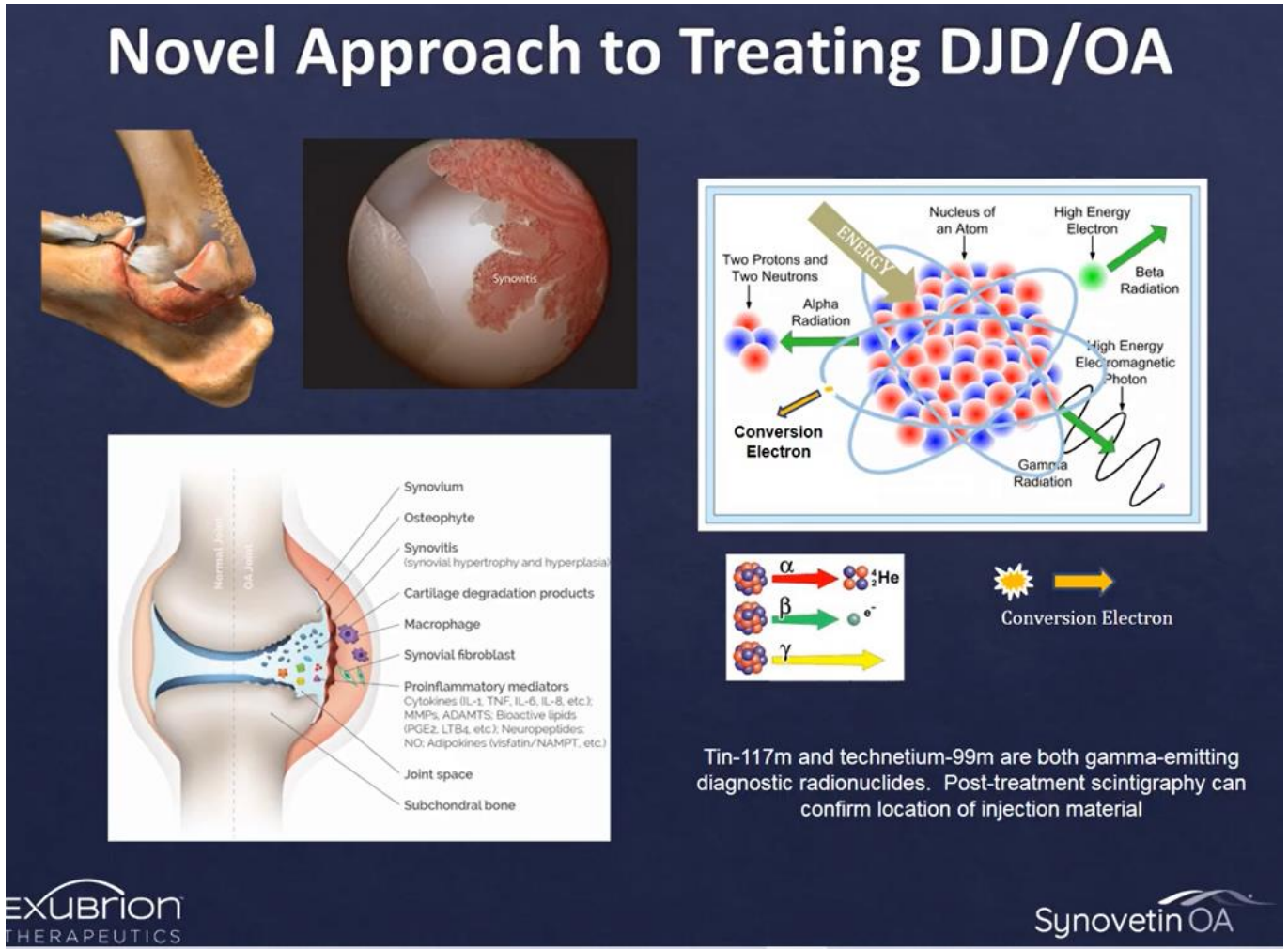
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Radiosynoviorthesis Tin- 117m
放射性同位素錫

153Sm (釤) as bone-seeking radiopharmaceutical

90Y(釔) as brachytherapy to treat osteosarcoma

117mSn(錫) as radiosynoviorthesis (放射線滑膜切除術)to treat osteoarthritis



Richard E Wendt 3rd et al., Radiation Safety Considerations in the Treatment of Canine Skeletal Conditions Using 153Sm, 90Y, and 117mSn. *Health Phys*. 2020 Jun;118(6):702-710. doi: 10.1097/HP.0000000000001222

Barnard, Zuber, Moore. Samarium Sm^{153} lexidronan for palliative treatment of dogs with primary bone tumours. *J Am Vet Med Assoc*. 2007;230:1877–1881.

Radiosynoviorthesis Tin- 117m 放射性同位素錫



https://www.youtube.com/watch?v=V6Uc6JFbEkk&t=84s

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Radiosynoviorthesis 放射線滑膜切除術
Tin- 117m 放射性同位素錫

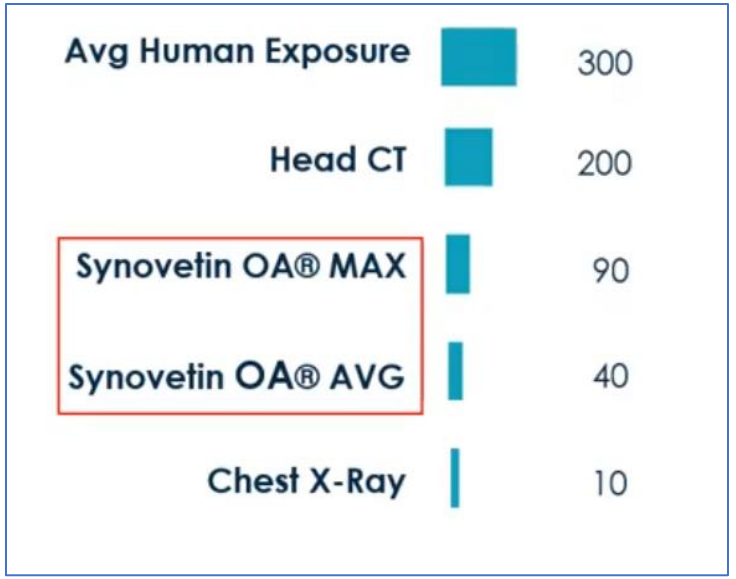
- 輻射劑量：mrem（毫倫目）
- 平均人類暴露接受量為300 mrem
- 人類病人接受放射線治療允許釋放到環境輻射量上限為 500 mrem/年
- 獸醫接受游離輻射暴露量上限為 100 mrem/年
- 頭部CT 200 mrem，胸部X光10 mrem
- 接受Synovetin OA注射後釋放輻射量介於40~90 mrem/整年

- 相關文獻

Health Phys. 2020 Jun;118(6):702-710. doi: 10.1097/HP.0000000000001222.

Radiation Safety Considerations in the Treatment of Canine Skeletal Conditions Using 153Sm, 90Y, and 117mSn

Richard E Wendt 3rd [1], Kimberly A Selting [2], Jimmy C Lattimer [3], Janine Wong [4], Jaime Simón [5], Nigel R Stevenson [6], Stanley D Stearns [7]



From XUBRION THERAPEUTICS

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Radiosynoviorthesis Tin- 117m 放射性同位素錫



- 使用低劑量放射線治療直接對synovium有抗炎症作用
- 操作簡易且低侵入性
- 可避免長期服用止痛劑或使用類固醇
- 改善疼痛症狀顯著且慢性關節疼痛每年一次關節內注射
- 目前美國FDA核准犬肘關節使用
- 操作者需 Radiation Safety License 並需遵守使用規範
- 注射後不需要住院，2-4週不要與主人同床
- 顆粒大小2-20μm適合被macrophage 辨識及吞噬且不漏出關節腔並且無異物感
- 最終經由淋巴到肝臟代謝

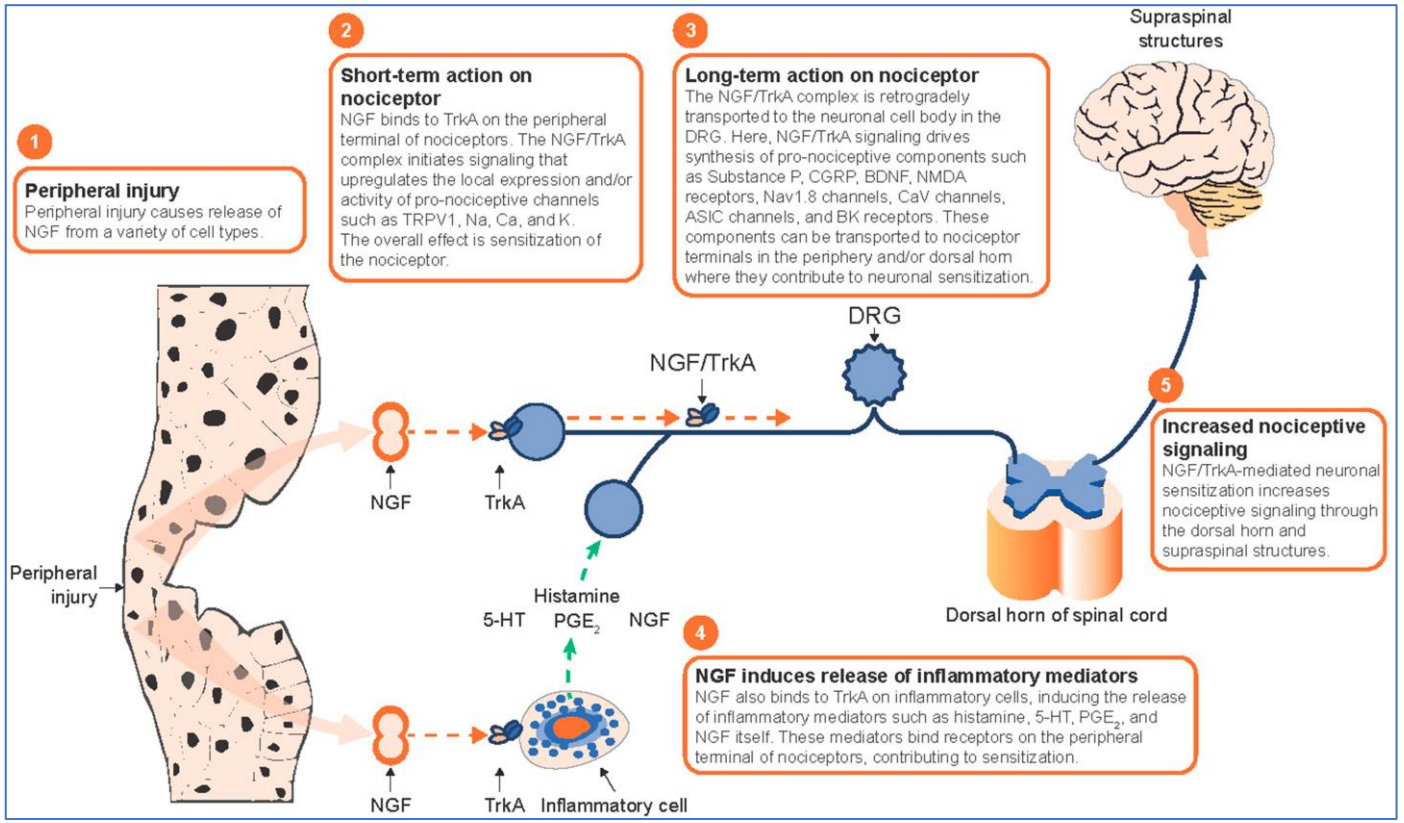


Reference from https://www.synovetin.com/, Dr. Brunke's experience and Dr. Fox's talk 2021

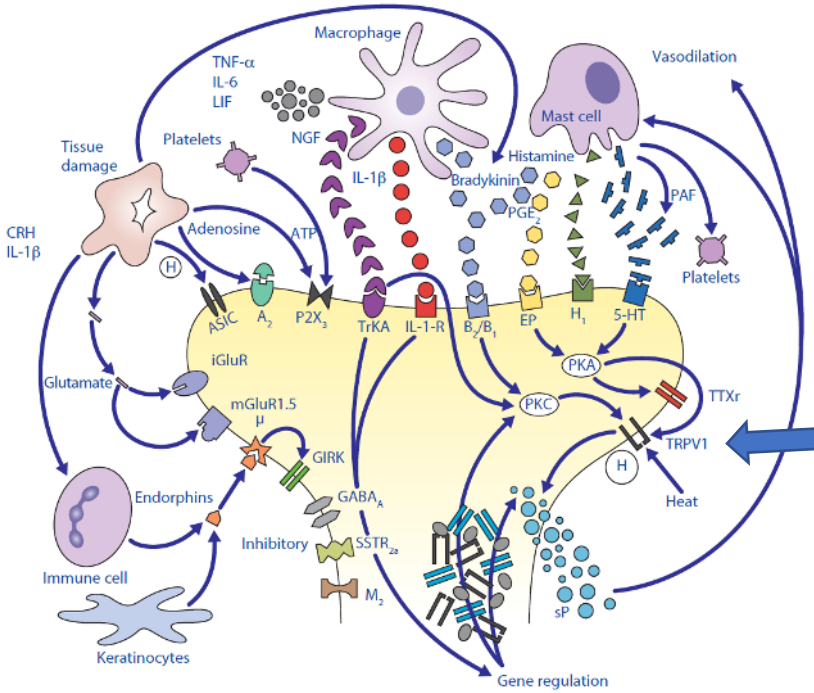
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- Anti-NGF monoclonal antibodies





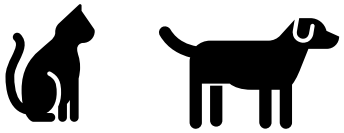
- TRPV1 agonists (capsaicin)

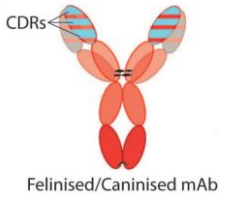




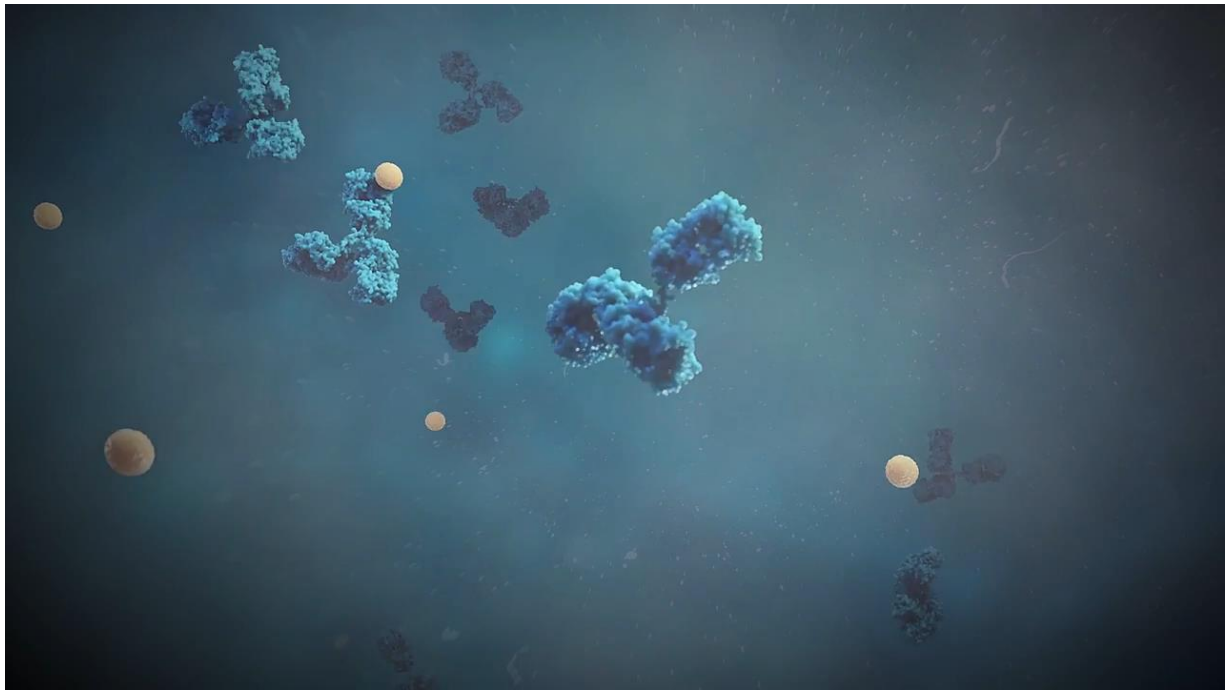
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Anti-NGF mAb 抗神經生長因子單株抗體









https://www.zoetisus.com/oa-pain/first-time-resolution.aspx

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JAVMA AVMA

Bedinvetmab (Librela/Beransa) in dogs raises safety concerns, including rapidly progressive osteoarthritis, and warrants vigilant adverse event reporting

Dirsko J. F. von Pfeil, Dr med vet[1-3*]; Robert J. Adams, BVM&S[4]; Louise Clark, BVMS[5]; Curtis W. Dewey, DVM[6,7]; Matthew W. Brunke, DVM[6,8]; Michael Garner, Dr med vet[9]; Otto I. Lanz, DVM[2]; Nate Miller, DVM[2]; Lee Breshears, DVM[2]; Loïc M. Déjardin, DVM[10]; Andrew Armitage, BVM&S[11]; Kei Hayashi, PhD[8]; Aldo Vezzoni, Med.Vet.[12]

[1]Small Animal Surgery Locum, Dallas, TX
[2]Spry Companions Veterinary Orthopedic Specialists, Denver, CO
[3]Department of Small Animal Clinical Sciences, College of Veterinary Medicine, Michigan State University, East Lansing, MI
[4]Veterinary Specialists of Sydney, Miranda, NSW, Australia
[5]Davies Veterinary Specialists, Shillington, UK
[6]Elemental Pet Vets, Freeville, NY
[7]Global Pet Project, Freeville, NY
[8]Hayashi Laboratory, College of Veterinary Medicine, Cornell University, Ithaca, NY
[9]CatsOnly, Hamburg, Germany
[10]Department of Small Animal Clinical Sciences, College of Veterinary Medicine, Michigan State University, East Lansing, MI
[11]Greenside Referrals, Greenside Veterinary Practice, Melrose, Roxburghshire, UK
[12]Clinica Veterinaria Vezzoni, Cremona, Italy

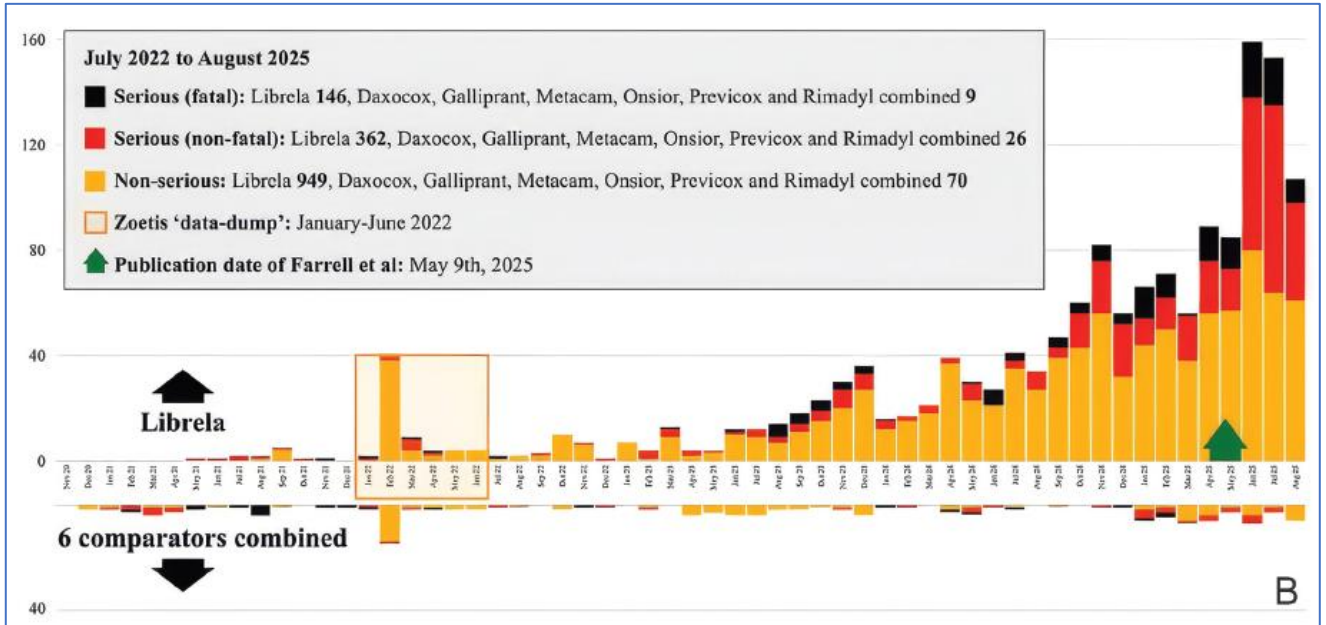
*Corresponding author: Dr. von Pfeil (dvpfeil@yahoo.com)

2026 June



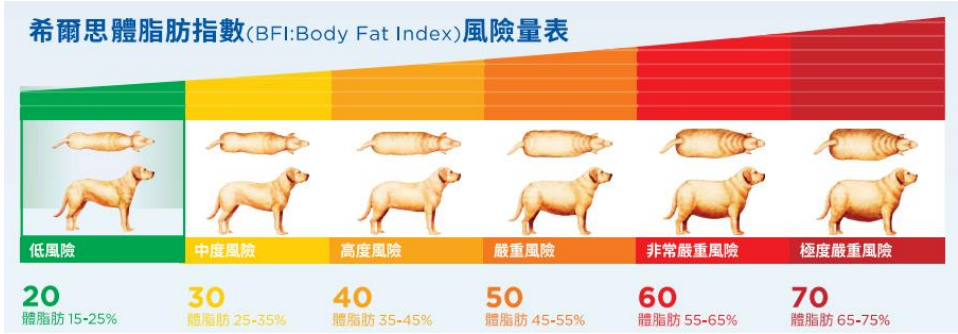
Figure 1—Radiographic appearance of canine rapidly progressive osteoarthritis (RPOA) in an 8.5-year-old Staffordshire Bull Terrier with no history of trauma. A to D—Pretreatment CT scans show mild elbow arthrosis and excluded humeral intercondylar fissure. E to H—Post-bedinvetmab (Librela/Beransa) radiographs demonstrate severe left elbow arthrosis and a right lateral humeral condylar fissure with an atypical medial epicondylar palisading periosteal reaction. Given the patient's age and breed, which is not predisposed to humeral condylar fissure, a report to the manufacturer specifying a suspicion of RPOA was filed as all other pathologies were ruled out. The manufacturer filed a report with an incorrect diagnosis of "osteosarcoma." (Reproduced from Farrell M, Waibel FWA, Carrera I, et al. Musculoskeletal adverse events in dogs receiving bedinvetmab [Librela]. *Front Vet Sci*. 2025;12:1581490. doi:10.3389/fvets.2025.1581490. Reprinted with permission.)

- NGF對骨及軟骨以及關節環境恆定有一定影響，由於aNGFmAb (anti-nerve growth factor monoclonal antibody) 會造成RPOA(rapidly progressive OA)因此美國FDA以及歐洲EMA拒絕核准人類aNGFmAb上市
- 犬用aNGFmAb是以32隻健康，年輕米格魯成犬，為期26週安全試驗。以5隻受試犬放射線影像資料為主。
- 除了罹患神經疾病犬隻勿使用，長期使用仍須留意RPOA之發生
- 結論:
- 對OA緩解疼痛有其效果，須告知主人ADE，仍需小心使用時機，長期使用需定期評估動物狀況



Non-pharmacologic Modalities for Chronic Pain Management

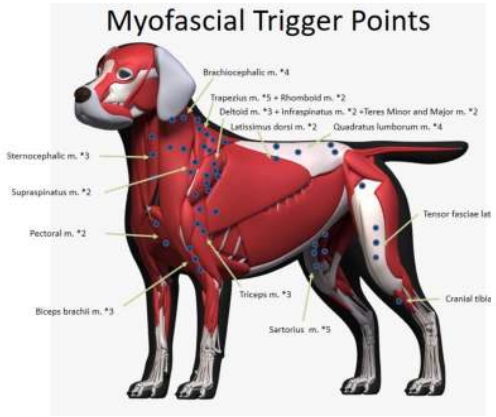
- **Weight Optimization 體重最佳化**
- **Exercise and Rehabilitation Therapy 運動與復健治療**
- **Acupuncture, massages 針灸，按摩**
- **MTrP (myofascial trigger point) 激痛點**
- **Environmental Modification 環境改善**













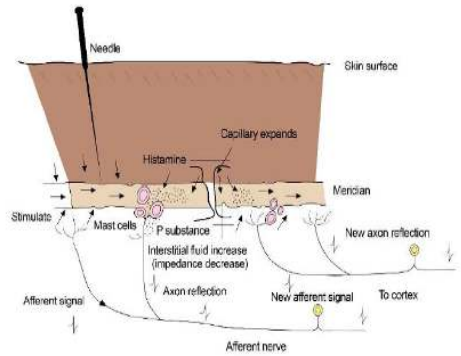


Fig. 1. Schematic illustration depicting the anatomic components of an acupuncture point.

Open Veterinary Journal, (2021), Vol. 11(2): 203-209



推拿按摩

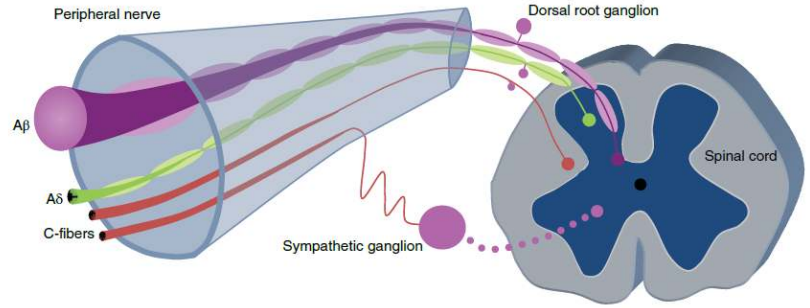
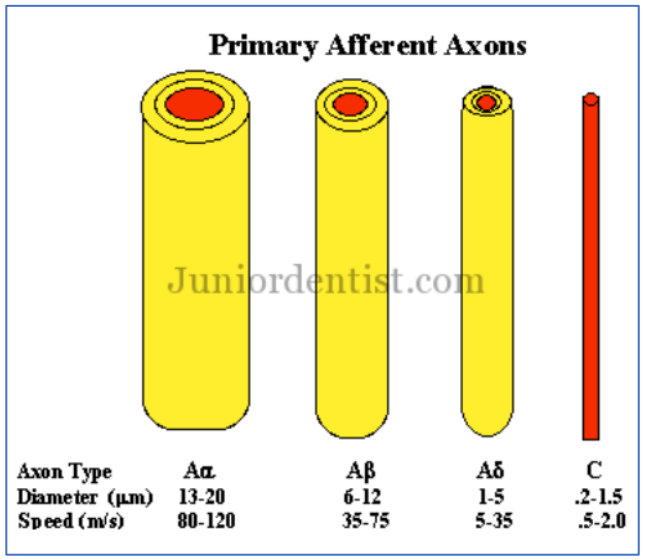
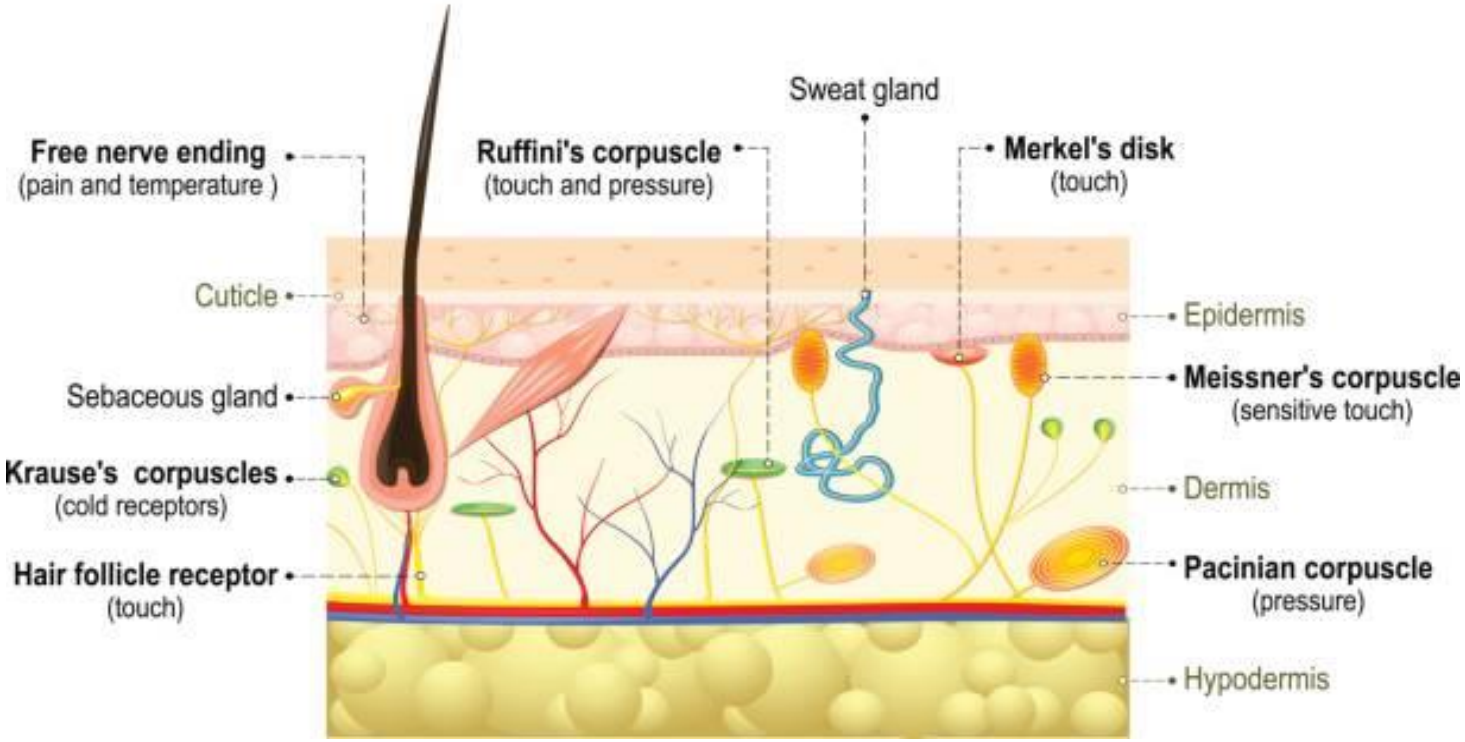


Figure 4.2 Low-threshold A-beta nerve fibers; high-threshold, myelinated A-delta fibers; and small, unmyelinated C fibers lie within the peripheral nerve sheath and synapse in the dorsal horn of the spinal cord. Source: Drawn by Kristen Cooley.

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- **Massages 按摩**
  - **Effleurage 輕撫法**
  - **Petrissage 揉撫法**
  - **Tapotement 敲擊法**
  - **Cross-friction 交錯摩擦法**





https://media.istockphoto.com/

MTrP (myofascial trigger point)激痛點

- **TREATMENT OF MTrP 治療激痛點**
  - ACUPUNCTURE 針灸
  - PHOTOBIOMODULATION 雷射
    - Clinically, in the authors' experience, delivering 8 to 10 J/cm^2 is an effective dose for treating TPs
  - MANUAL THERAPY 徒手治療
    - ischemic compression
    - apply firm digital pressure along the length of the taut band, or "strum" the taut band
  - EXTRACORPOREAL SHOCKWAVE THERAPY 體外震波
    - The recommended dosage in dogs is 0.12 to 0.15 mj/mm^2; the number of shockwaves per treatment is 200 to 500 and frequency of treatment: one treatment weekly for 3 to 4 weeks
  - THERMAL THERAPY 溫熱治療
    - Applying superficial heat over the area of MTrP results in muscle pain relief and contributes to relaxation of muscle fibres

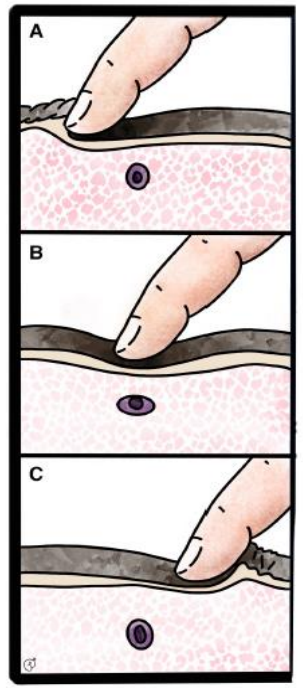


FIG 1. Diagram of the flat finger technique used to diagnose myofascial pain syndrome in muscles that cannot be easily grasped (e.g., infraspinatus muscle). It involves placing gentle pressure on one edge of the belly of the muscle (A) and moving from one side to the other while maintaining pressure (B and C). The dark circles within the muscle band indicate a trigger point (the dot) within a taut band (the circle). See Video S1 for a demonstration. Figure credit: Lauren Petty

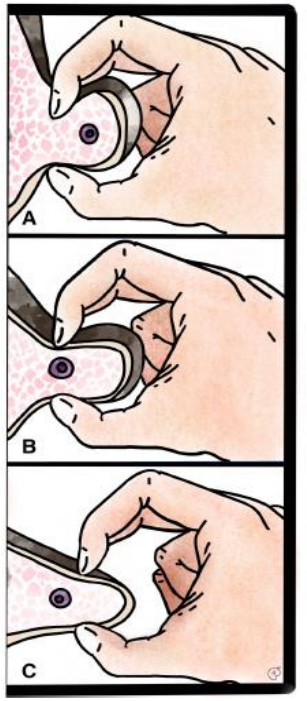
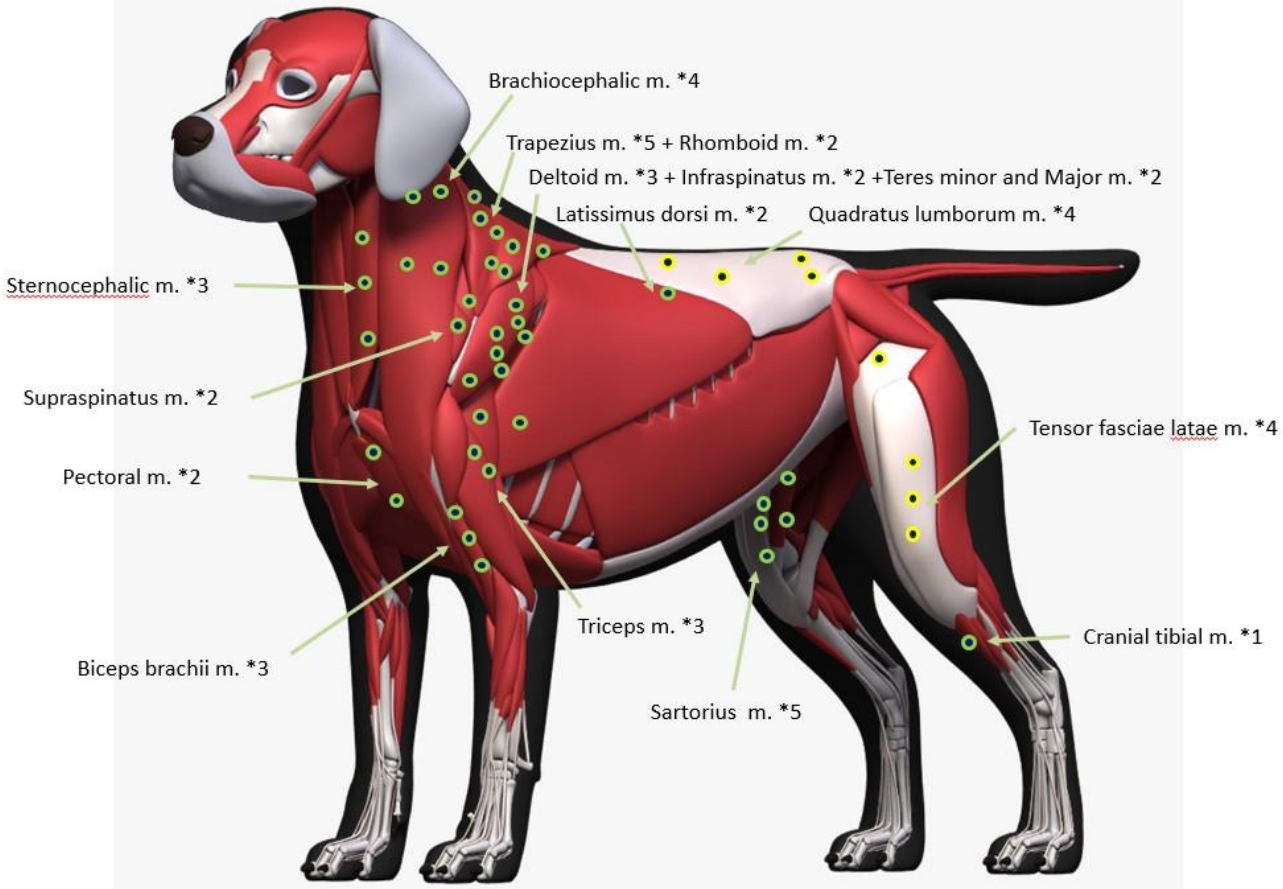


FIG 2. Diagram of the pinch technique used to diagnose myofascial pain syndrome in muscles that can be easily grasped (e.g., triceps or biceps). It involves grasping the entire muscle (A) and letting it slide through the fingers while applying gentle but steady pressure (B and C). The dark circles within the muscle band indicate a trigger point (the dot) within a taut band (the circle). See Video S1 for a demonstration. Figure credit: Lauren Petty

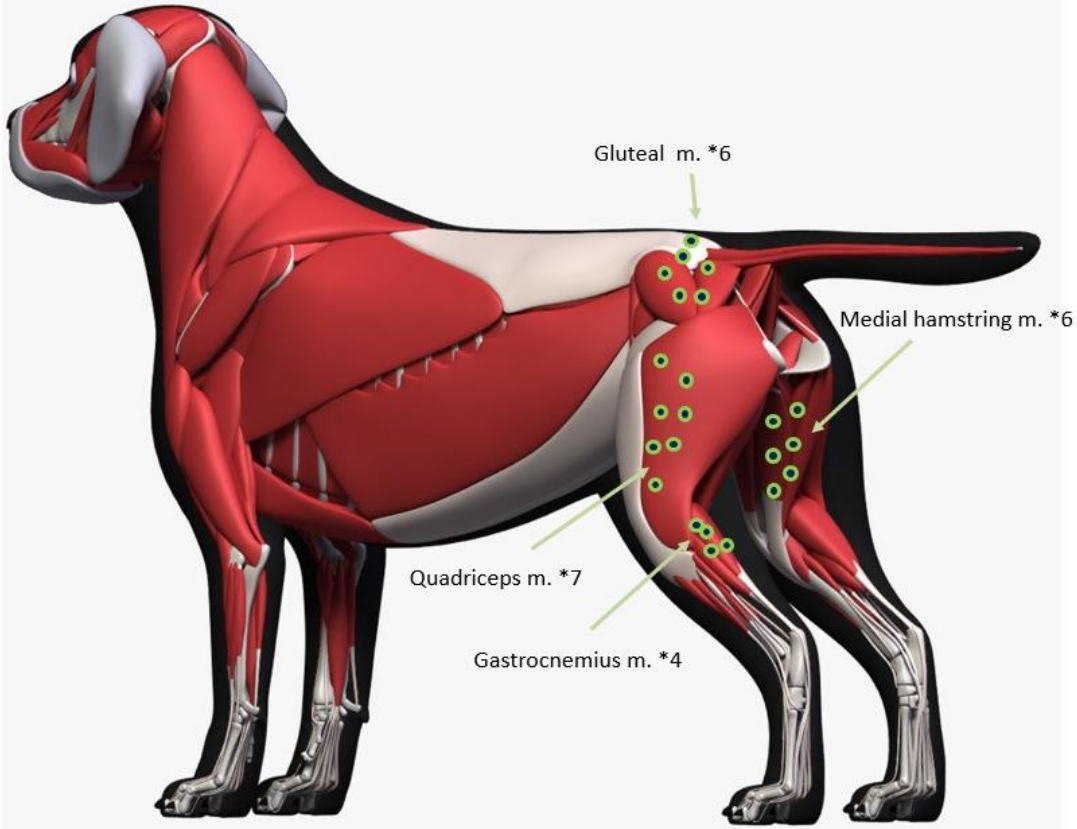
Petty 2024, Myofascial pain syndrome in small animal practice, *Journal of Small Animal Practice* (2024); 1–6

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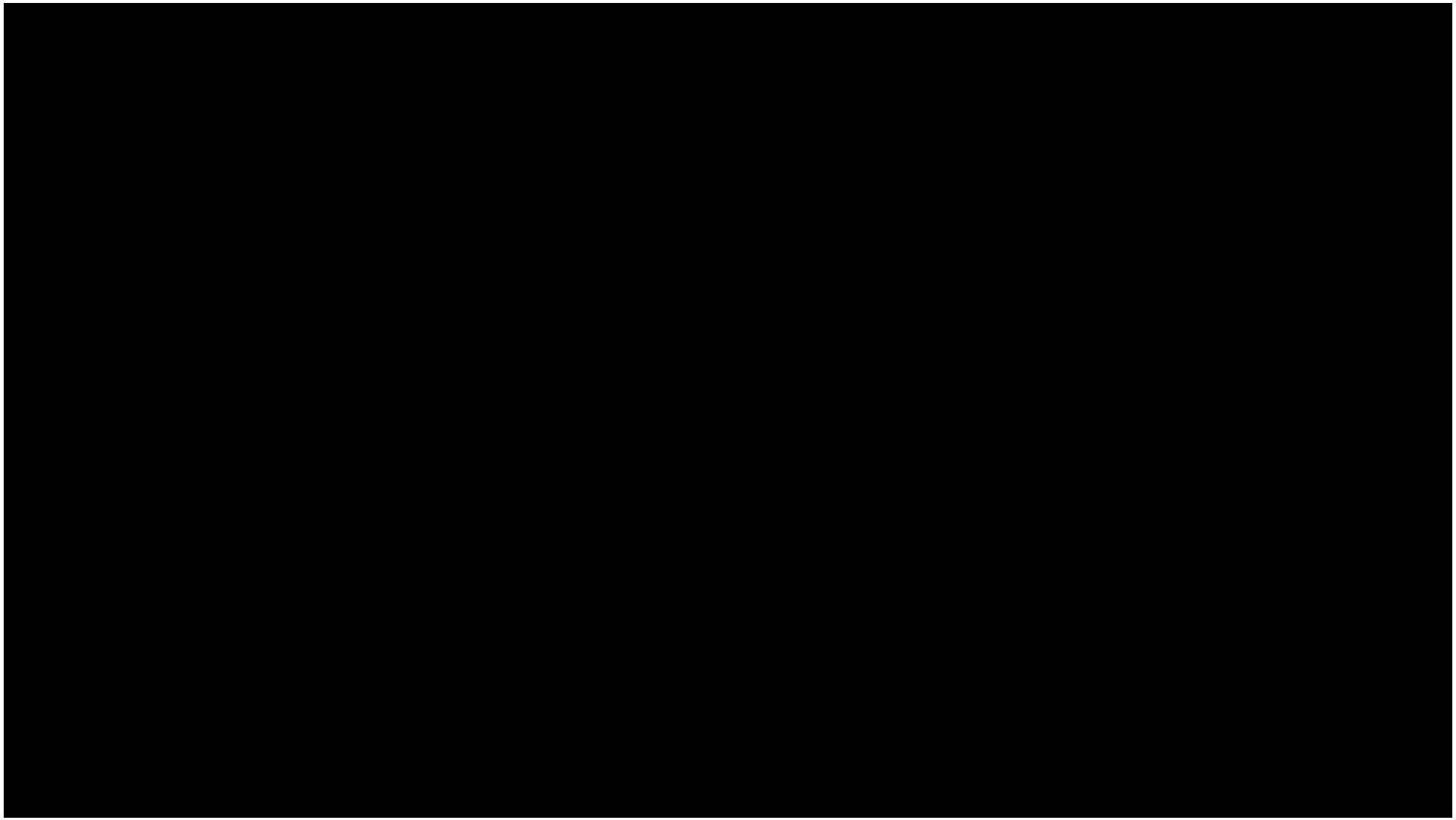
Myofascial Trigger Points



Myofascial Trigger Points



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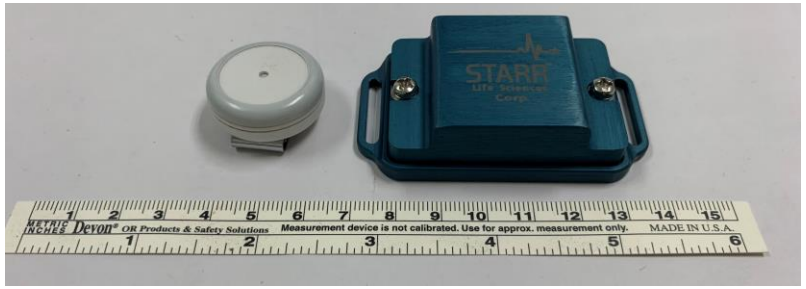
Petty 2024, Myofascial pain syndrome in small animal practice, *Journal of Small Animal Practice* (2024); 1–6

Pain assessment tools

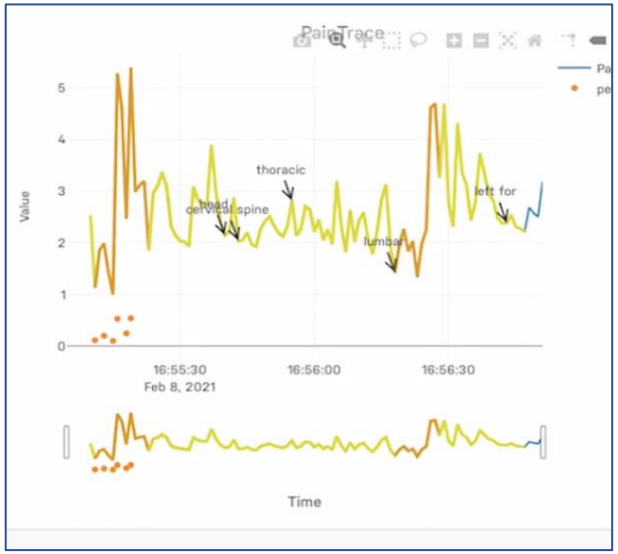
- Pain score
  - CBPI
  - CODI
  - LOAD
- Activity monitor
  - Actical
  - PLUS CYCLE
- Pain Trace device

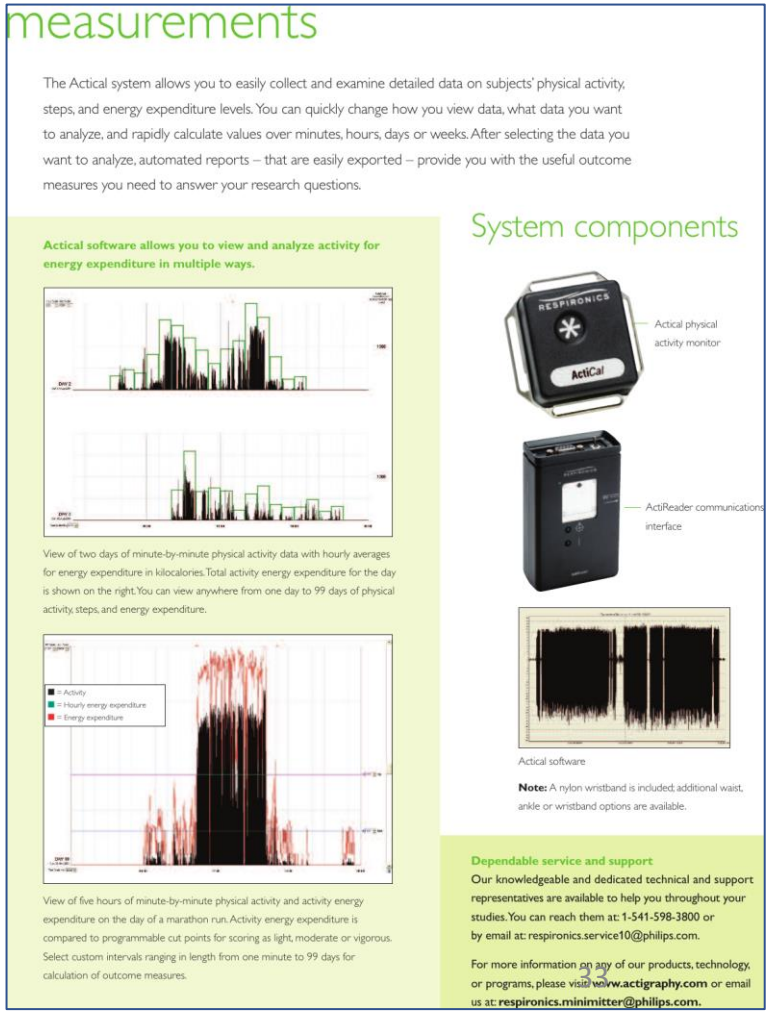












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老年寵物復健

Senior care guideline

AAHA Senior Care Guidelines for Dogs and Cats

This paper provides a working framework for enhancing the well-being of senior pet dogs and cats. Approaches to screening the medical status of senior pets are described in detail, with particular emphasis on establishing baseline data in healthy animals, the testing of clinically ill animals, and assessing senior pets prior to anesthesia and surgery. The management of pain and distress and the application of hospice and palliative care are addressed. Advice on ways to approach euthanasia and dealing with end-of-life issues is also provided. **J Am Anim Hosp Assoc 2005;41:81-91.**

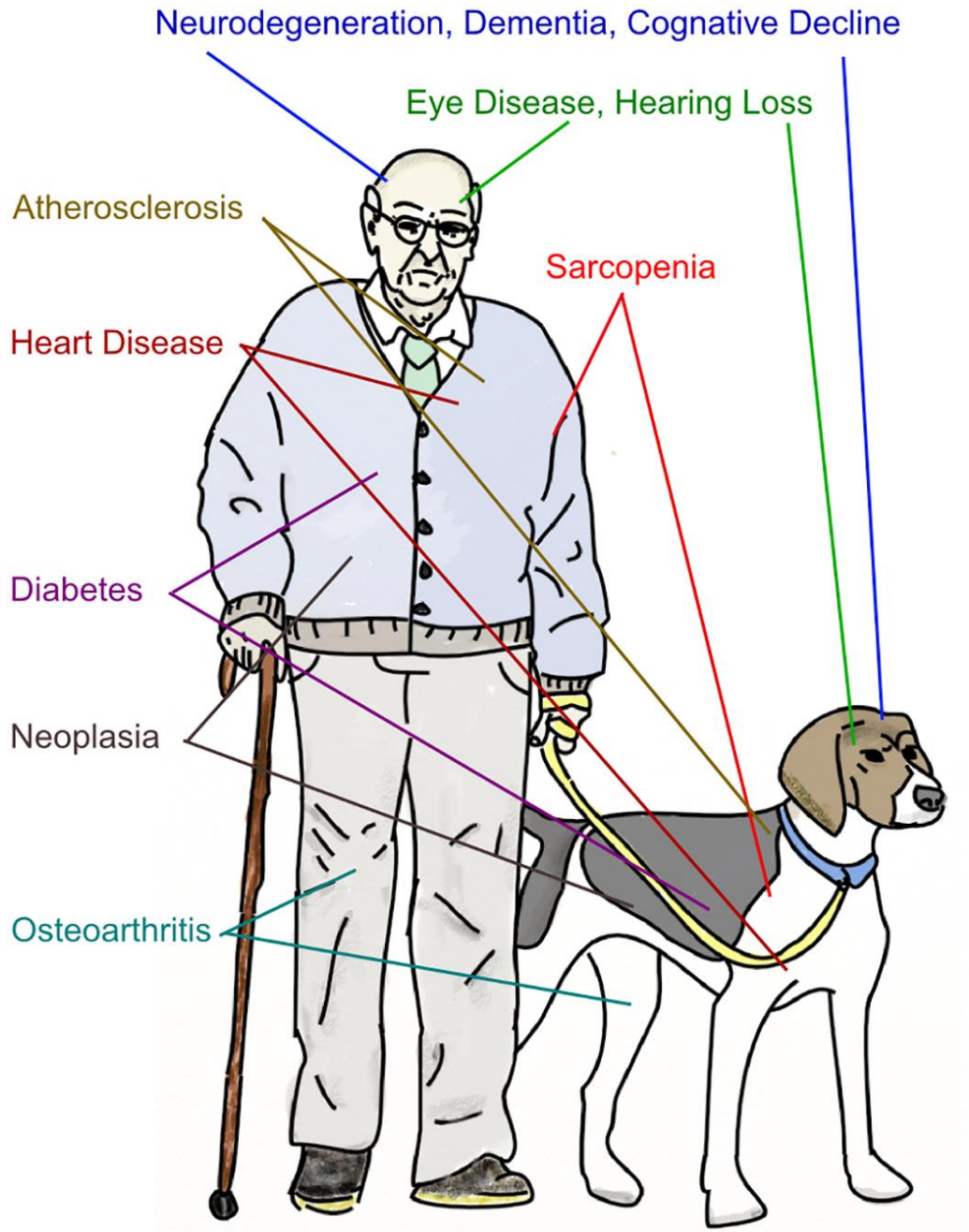
2023 AAHA Senior Care Guidelines for Dogs and Cats

Ravinder Dhaliwal, DVM, MS, DACVIM (Oncology), DABVP (Canine and Feline)†, Elizabeth Boynton, DVM, Sheila Carrera-Justiz, DVM, DACVIM (Neurology), Nicole Cruise, CVT, Mary Gardner, DVM, Janice Huntingford, DVM, DACVSMR, MS, CVPP, CVA, Heidi Lobprise, DVM, DAVDC, Elizabeth Rozanski, DVM, DACVECC, DACVIM (Small Animal Internal Medicine)

ABSTRACT

Veterinary professionals are taught to recognize that "old age is not a disease." However, clients may have the perception that older dogs and cats undergo an unavoidable physical, mental, and behavioral decline attributable simply to old age. The veterinary team's role includes providing medical care and support to senior pets to maintain their quality of life, as well as supporting and educating clients on proper senior animal care and addressing any misconceptions about the aging process. These Guidelines describe a systematic approach to the healthcare of the senior pet that is based on an evidence-guided assessment of both healthy and unhealthy canine and feline patients. By using evidence-guided medicine, which may include conventional and integrative approaches as well as environmental management for the senior pet, the patient's quality of life can be enhanced and potentially extended during this vulnerable life stage. Developing a senior program for the veterinary practice requires appropriate training of the entire healthcare team and includes a robust client education program that focuses on the wellbeing of the senior pet. Palliative and hospice care should be included in the education and information provided for both the veterinary team and the families of senior pets. (*J Am Anim Hosp Assoc* 2023; 59:1–21. DOI 10.5326/JAAHA-MS-7343)

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Pitt JN, Kaeberlein M (2015) Correction: Why Is Aging Conserved and What Can We Do about it?. PLOS Biology 13(5): e1002176.

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骨刺
(Bone spur; **Spondylosis deformans**; vertebral osteophytosis)

- **Spondylosis deformans** (Vertebral spinal osteophytes) 變形性脊椎病
- Diffuse idiopathic skeletal hyperostosis (DISH) 瀰漫性自發性骨肥厚
- Discospondilitis 脊椎椎間盤炎
- Intervertebral disc diseases (IVDD) 椎間盤疾病

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骨刺 (spondylosis deformans; vertebral osteophytosis)

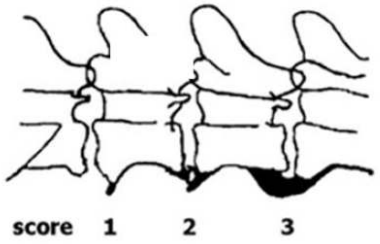
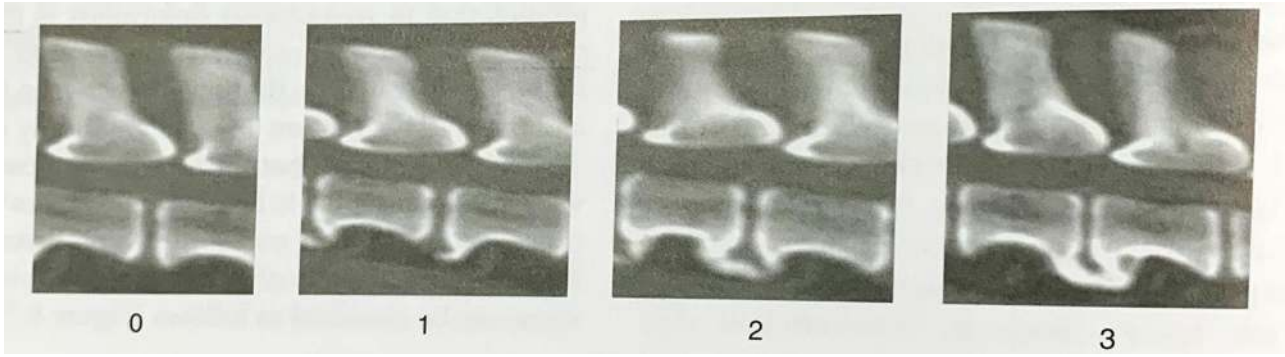
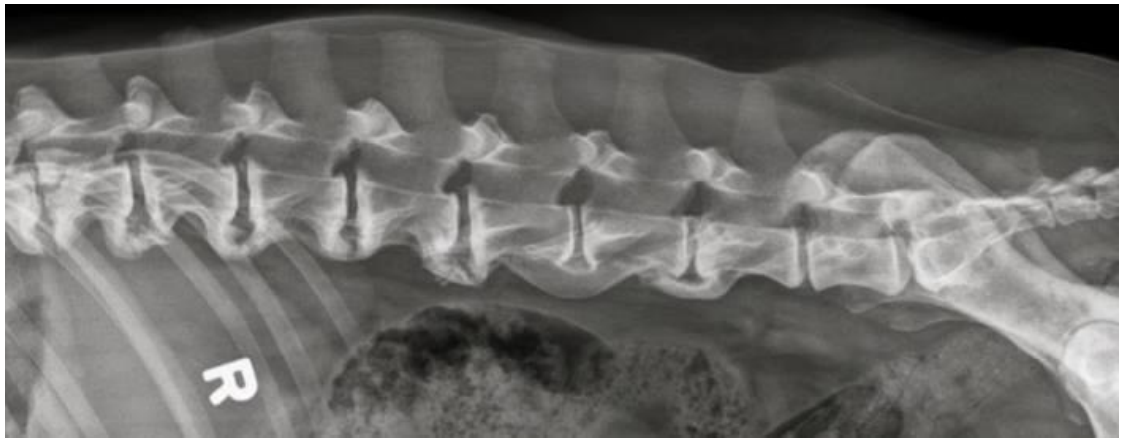
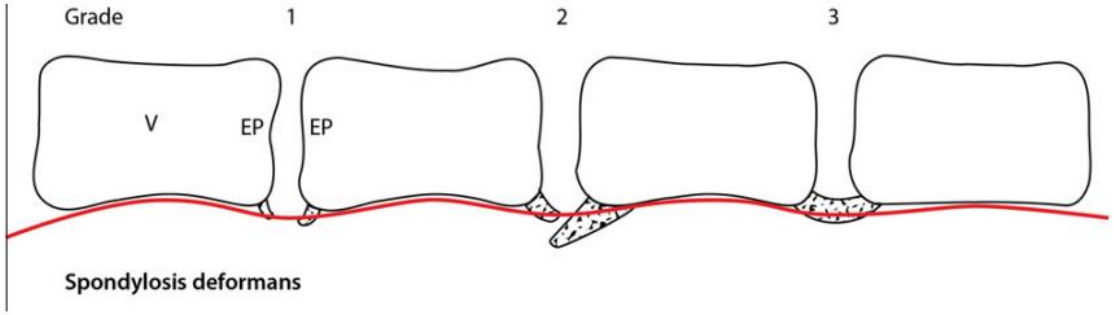


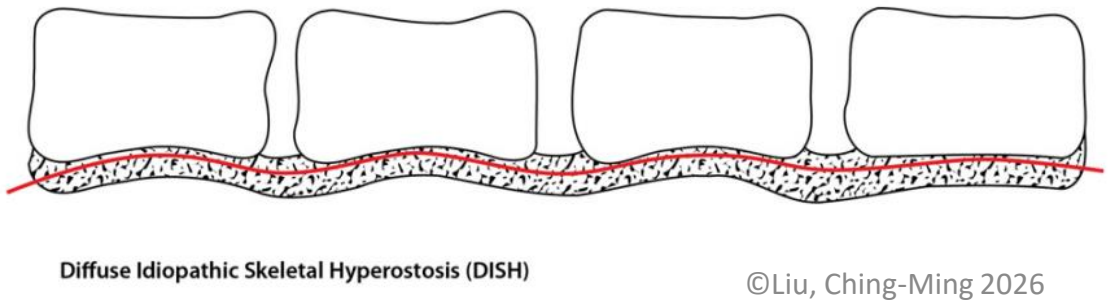
Figure 1. The Norwegian scoring system of enthesophyte development in dogs (reproduced from [10] with permission from *J Small Anim Pract*). A score of 0 (not shown) corresponds to no enthesophyte development.

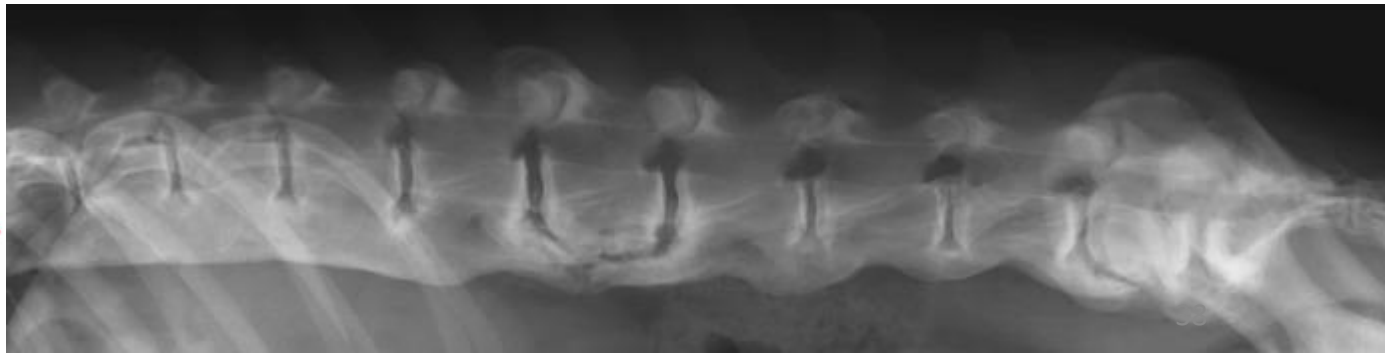




H.C. Kranenburg et al./The Veterinary Journal 190 (2011) e84–e90







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- 9歲以上發生率高，大型犬居高風險
- Spondylosis is more prevalent in some breeds than in others. In an early study (1967) of spondylosis in Boxers from the USA, UK, and Sweden
- The disease was found in about 50% of dogs by six years of age, and in 75% at an age of nine years.
- The prevalence and the grade of spondylosis increase with age.
- By the age of 9 years, approximately 25-70% of dogs are affected. Large-breed dogs are at increased risk.

- 拳師狗發生率高
- The prevalence of spondylosis deformans is especially high in boxers (approximately 40-85% depending on age)

- 經常發生位置： L7-S1, T12-L3, 頸椎較少
  - The L7-S1 disc space is the most commonly affected and the thoracolumbar region (T12-L3) being the next. Spondylosis deformans in the cervical region is less common.

- Remember that “-itis” means inflammation, and spondylosis deformans is a non-inflammatory degenerative disease

Kranenburg et al. 2011. Diffuse idiopathic skeletal hyperostosis (DISH) and spondylosis deformans in purebred dogs: A retrospective radiographic study, The Veterinary Journal, Vol. 190 (2), pe84-e90.
Fingeroth and Thomas, 2015, Spondylosis deformans. Advances in Intervertebral disc disease in dogs and cats.

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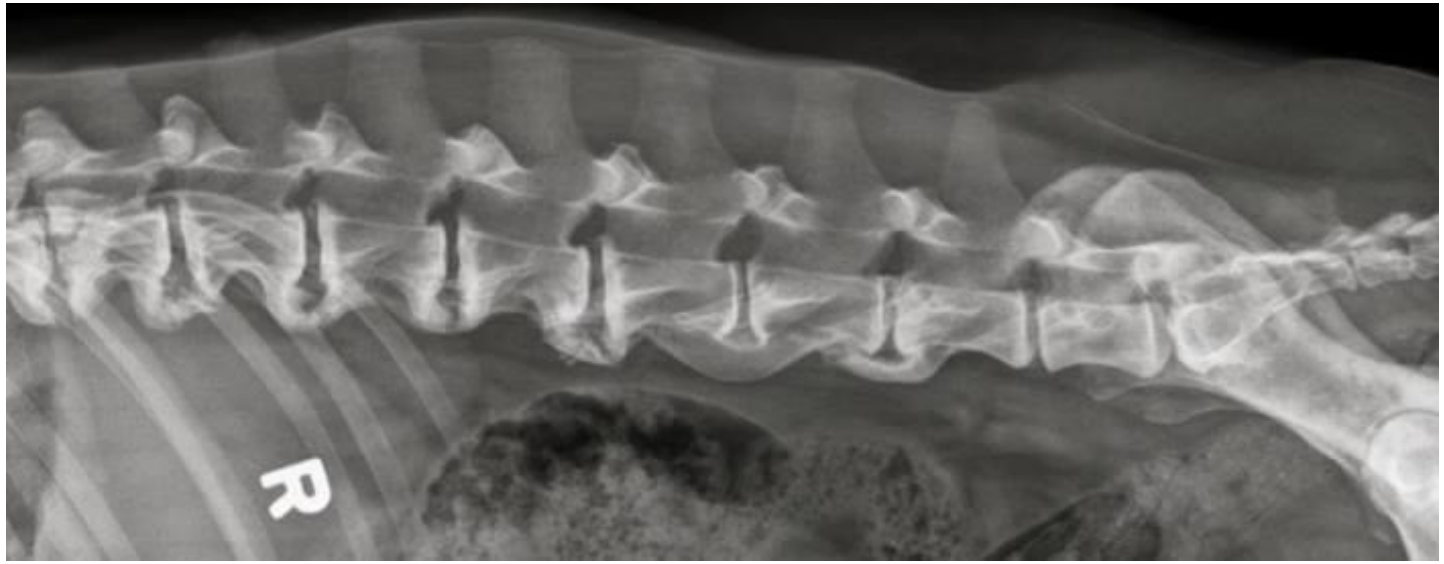




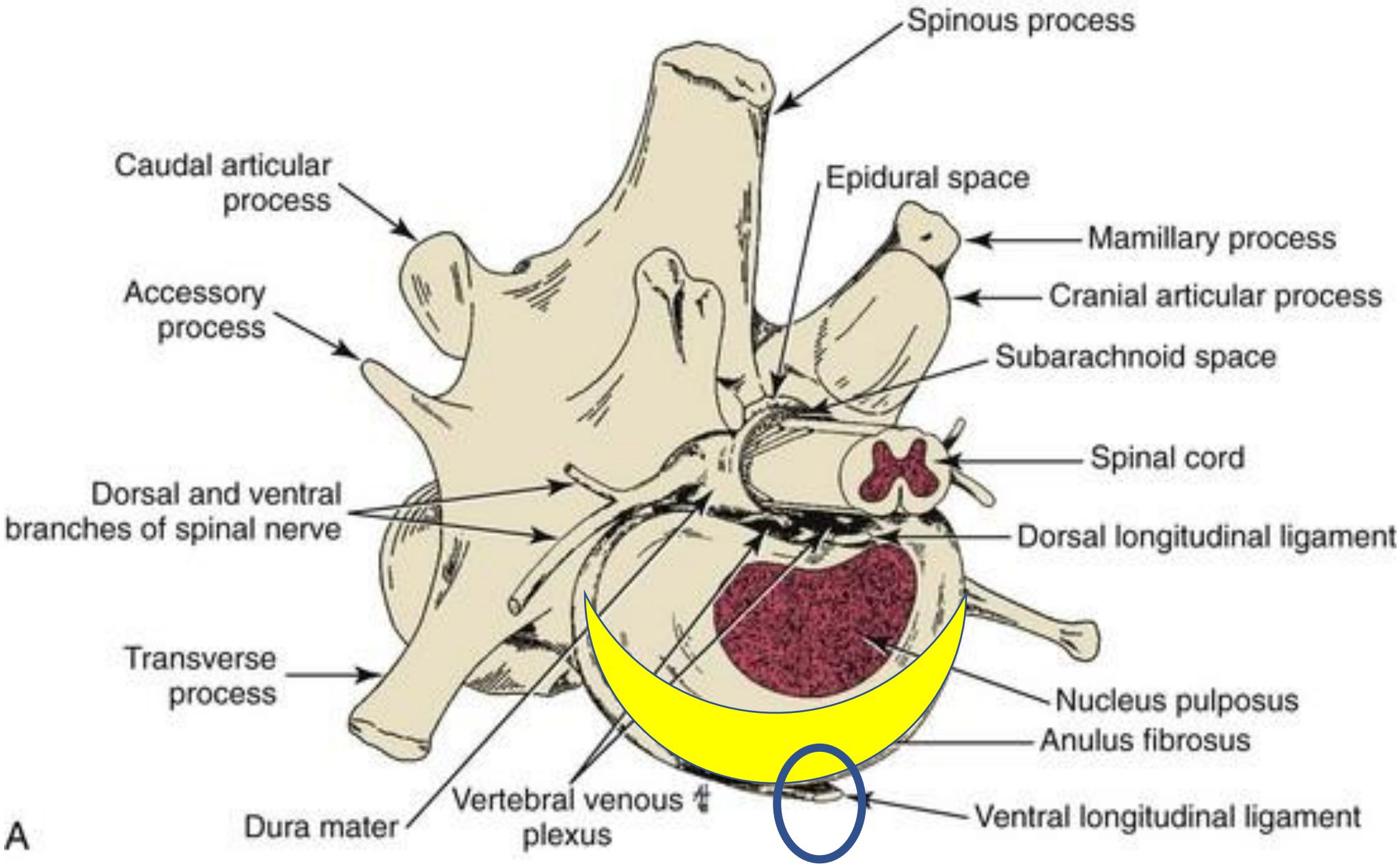


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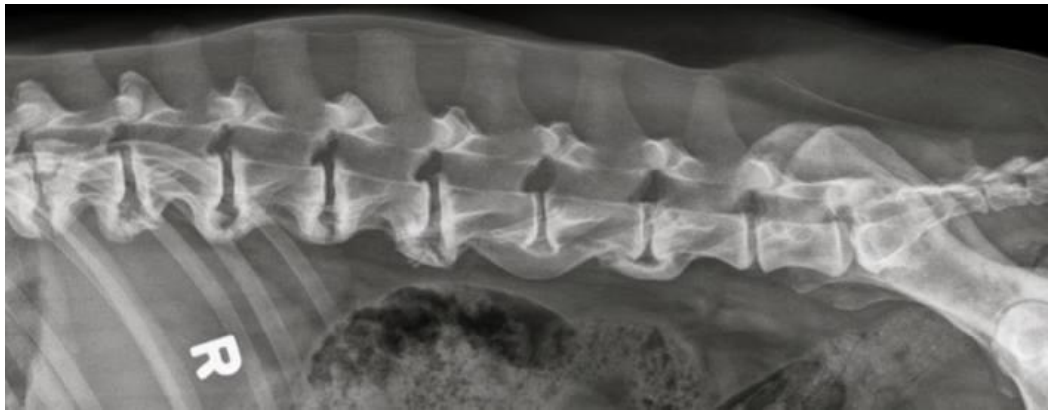
- 脊椎骨骨質增生(骨刺)
  - 一般 X 光就可判定
  - 通常為腹側兩相鄰脊椎骨間逐漸搭橋
  - 不造成脊髓直接壓迫
  - 嚴重者逐漸脊椎僵直
  - 疼痛大多起因於側枝神經根受壓迫
  - 行動不便或行動緩慢



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- Treatments

  止痛藥

  - 以NSAIDs為主

  營養補充品

  - Supplements such as glucosamine, methylsulphonylmethane (MSM), chondroitin sulphate, omega 3 fatty acids and more may have an effect.

  非藥物治療

  - Treatments like veterinary chiropractic, osteopathy, acupuncture, physiotherapy, and massage have also shown encouraging results

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慢性關節炎；骨性關節炎(OA)

- Osteoarthritis (OA) 骨性關節炎

- Degenerative Joint Diseases (DJD) 退化性關節病

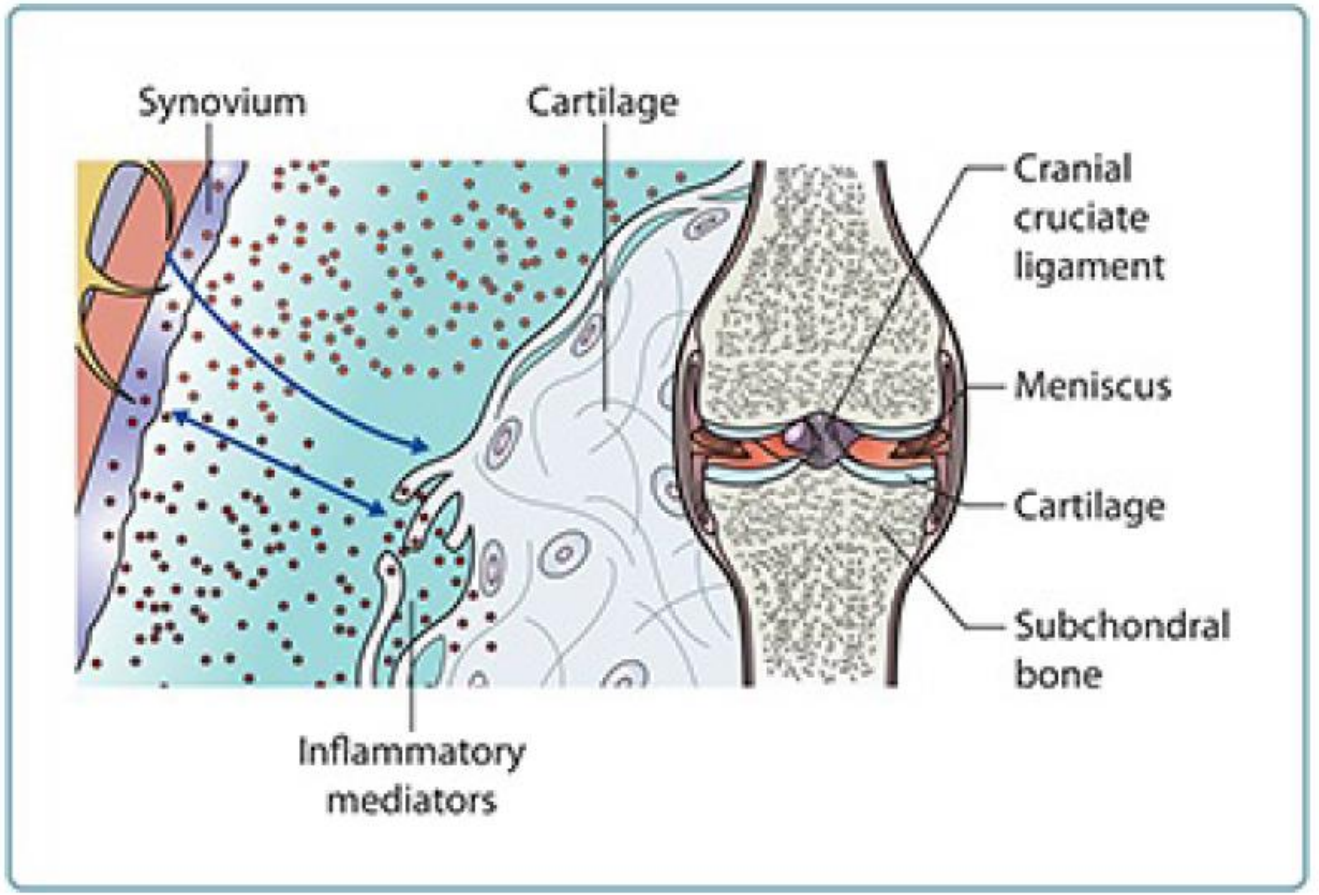
- 可動關節(diarthrosis; movable joint)
- 微動關節(amphiarthroses; slightly-movable joint)
- 不動關節(synarthrosis; immovable joint)

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- Osteoarthritis is estimated to affect approximately 20% of dogs ≥ 1 year of age and 90% of dogs > 5 years of age.

- The joint can be considered as an ‘organ’

- OA 骨性關節炎是一種進行性疾病
  - Cartilage destruction 軟骨損壞
  - subchondral bone sclerosis 軟骨下骨硬化
  - synovial inflammation 滑液膜發炎
  - periarticular osteophytes formation 骨贅生成
  - Shoulder, elbow, hip, stifle, spine
  - Clinical signs: joint pain, joint effusion, lameness, ROM ↓

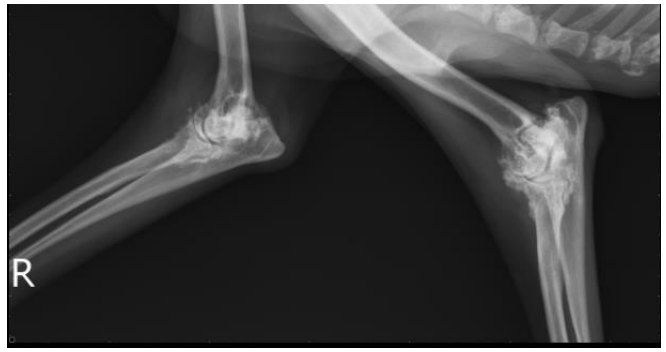


Steven M Fox, 2017, Multimodal Management of Canine Osteoarthritis, 2nd Edition

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- 原因
  - Primary Causes
    - Idiopathic
  - Secondary Causes
    - Developmental disorders
    - Joint instability
    - Trauma
- 症狀
  - 關節疼痛、關節腫脹、摩擦音
  - 行動僵硬、行動不便
- 診斷
  - 骨科學檢查
  - 神經學檢查
  - X光
- 治療
- 預後





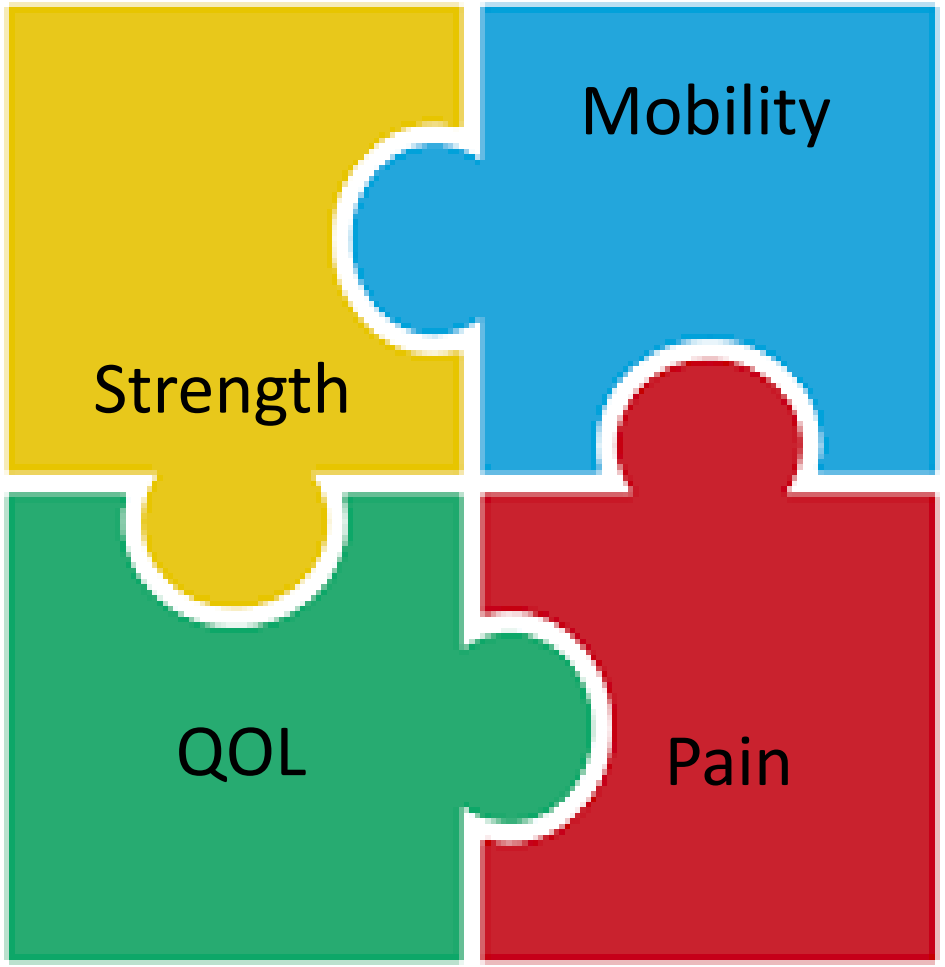




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- OA Treatment 目標
  - ➢Pain management 疼痛管理
  - ➢Strengthening 加強肌力
  - ➢Mobility and ROM 活動力及關節活動
  - ➢QOL (Quality of Life) 生活品質

- Pain management
  - Drug
  - Supplements
  - Disease modifying
  - Intra-articular injection
- Rehabilitation
  - Modalities
  - Manual therapy
  - Hydro therapy
  - Home exercises
  - Obese control
  - Environmental modification
- 評估CMI(臨床計量工具), QOL(生活品質)



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- 評估CMI (Clinical Metrology Instrument)
  - CODI (Cincinnati Orthopedic Disability Index, 2003)
  - CBPI (Canine Brief Pain Inventory, 2009)
  - LOAD (Liverpool Osteoarthritis in Dogs, 2013)
  - COAST (Canine Osteoarthritis Staging Tool, 2018)

- 評估 QOL (Quality of Life)
  - HHHHHMM (5H2M; 2007)
    - Hurt, Hunger, Hydration, Hygiene, Happiness,
    - Mobility, More good days than bad



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- Pain management
  - Pharmaceuticals
    - NSAIDs
    - Non-NSAIDs
    - Joint injection





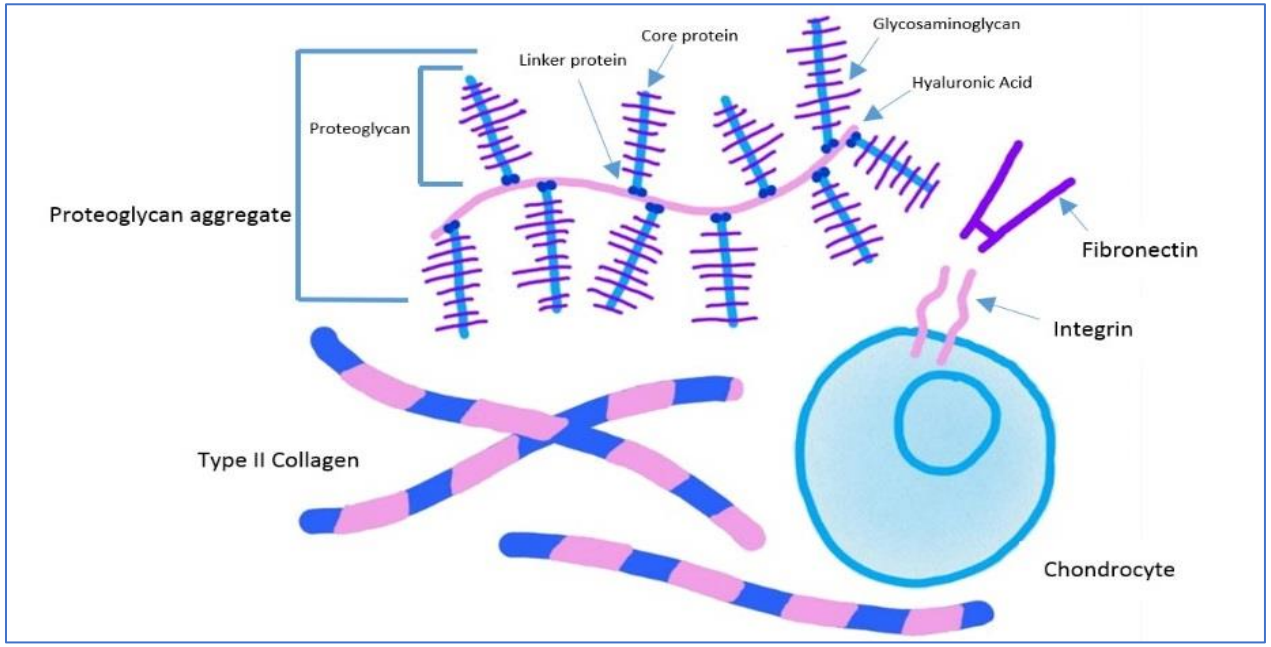
- Nutraceuticals supplement
  - Omega 3 fatty acid
  - Glucosamine
  - Chondroitin
  - UC II



- Disease modifying drugs
  - Polysulfated glycosaminoglycans
    - Adequan®
    - Pentosan

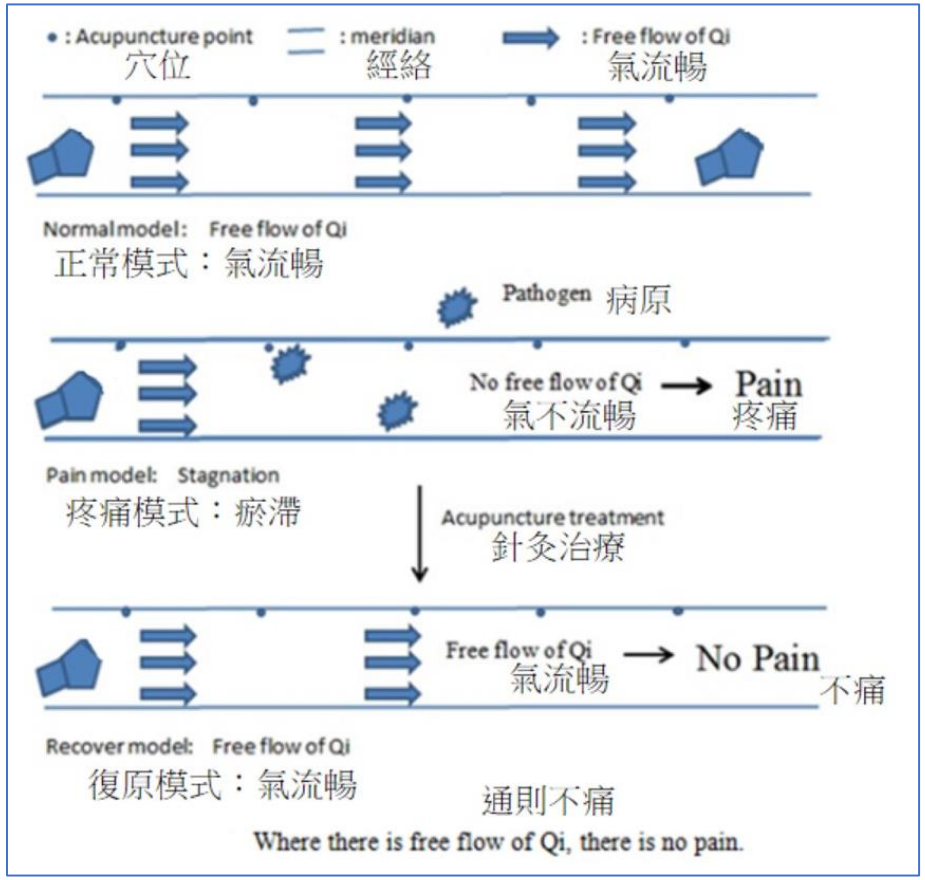






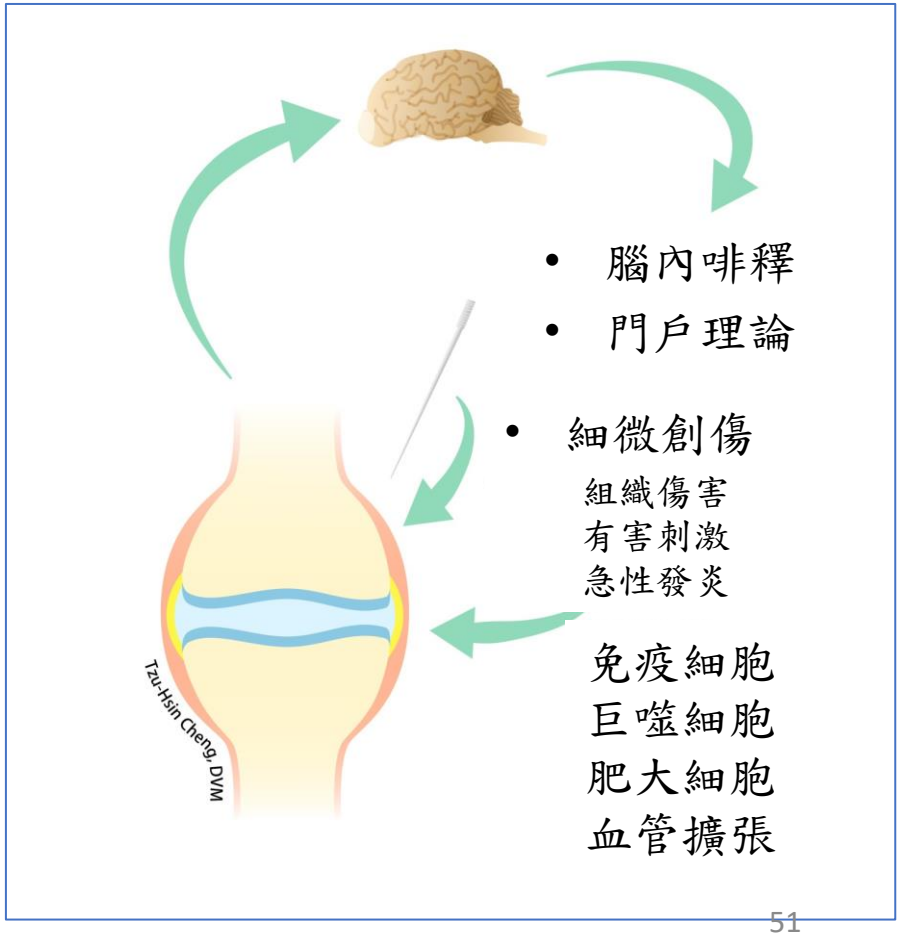
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- 中獸醫
  - 針灸：
    - 止痛
    - 幫助修復
    - 恢復功能





針灸



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- 介紹針灸穴位選擇

Table 2 Acupuncture points used in the present study based on the affected joint(s)

Affected Joint	Acupuncture points in all dogs (placed bilaterally if paired)	Points for all pelvic/ thoracic limb OA (placed bilaterally)	Electro-acupuncture over the primarily affected joint (unilateral EA)	Additional points used for each primarily affected joint (bilaterally placed)
Hip	GV-20[a] 頭百會	BL-40, ST-36, BL-54, BL-60, KID-3, LIV-3 委中，後三里，秩邊，崑崙 太溪，太衝	GB-29—GB-30 居髎--還跳	Jian jiao, GB-41 薦角，足臨泣
Stifle	BL-11—BL-23 大杼--腎俞 GV-14[a]—Bai hui[a] 大椎--腰百會		SP-10—GB-34 血海—陽陵泉	SP-9, GB-33, KID-3 血海，膝陽關，太溪
Tarsus			BL-60—LIV-3 崑崙—太衝	ST-41, ST-44, SP-6 解溪，內庭，三陰交
Shoulder		BL-13, GB-21, LI-4, SI-9, BL-10 肺俞，膊中，合谷，搶風，天柱	TH-14—LU-1 肩髎—中府	LI-15, SI-3, TH-3 肩井，後溪，中渚
Elbow			LI-11—LU-5 曲池—尺澤	LI-10, TH-10, SI-8 前三里，天井，小海
Carpus			PC-6—LI-4 內關—合谷	TH-5, LU-7, SI-3 外關，列缺，後溪

Baker-Meuten et al. BMC Veterinary Research (2020) 16:357

- 理療復健
  - 雷射
  - 治療超音波
  - 脈衝磁圈
  - 震波
  - 水療
  - 電療

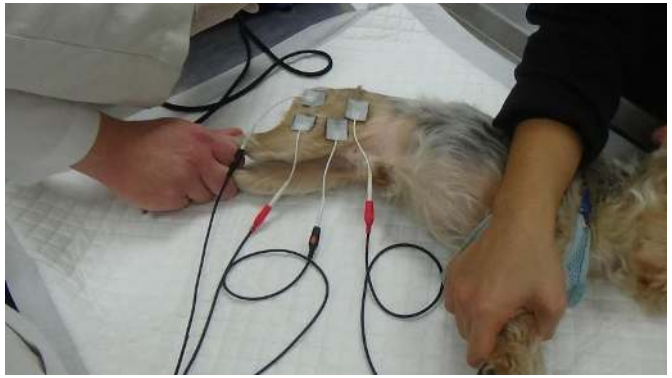








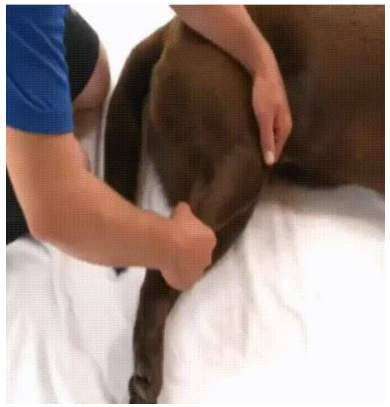


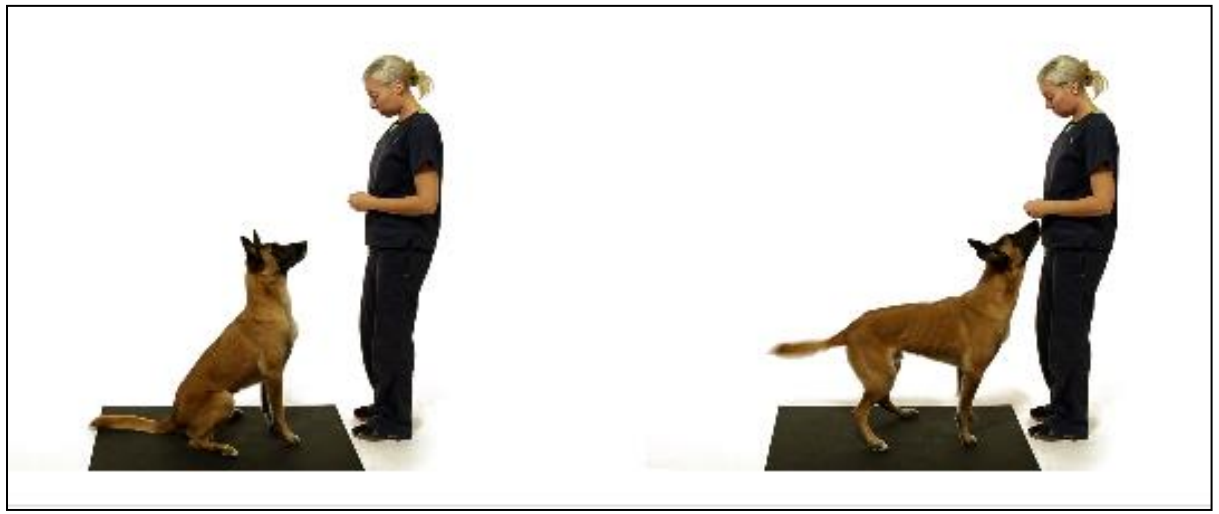




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- 居家運動復健
  - 上下緩坡
  - 坐站練習
  - 關節活動









https://canineexercise.com/

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- 手技復健
  - 推拿按摩
    - 體表刺激、經絡刺激、穴位刺激、激痛點、肌肉刺激
  - 關節活動與伸展
  - 關節鬆動術

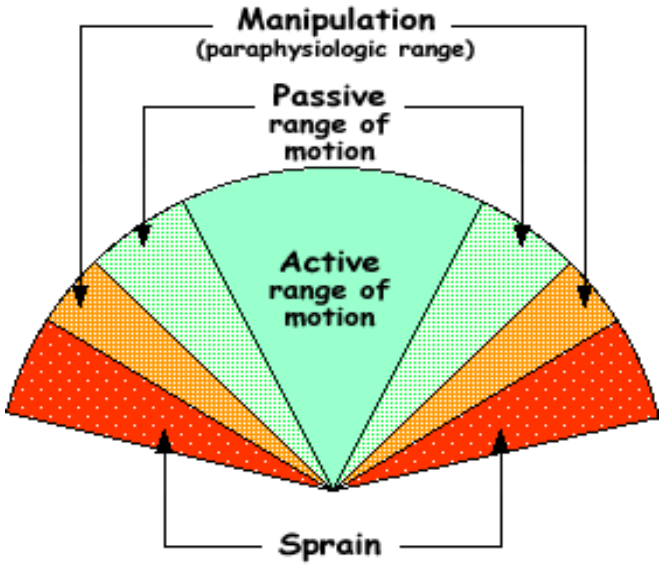




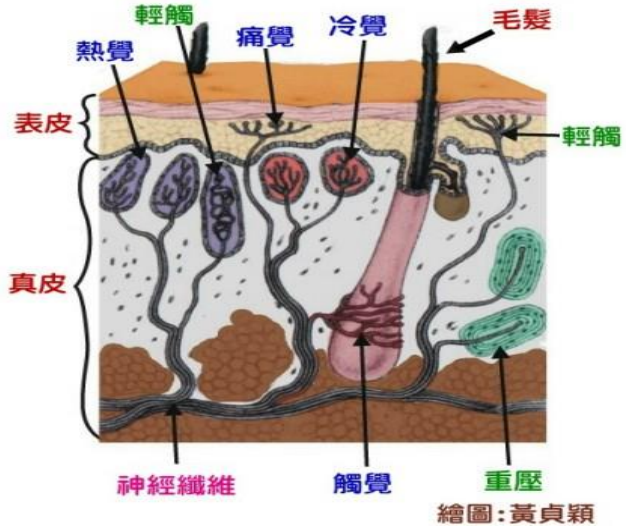




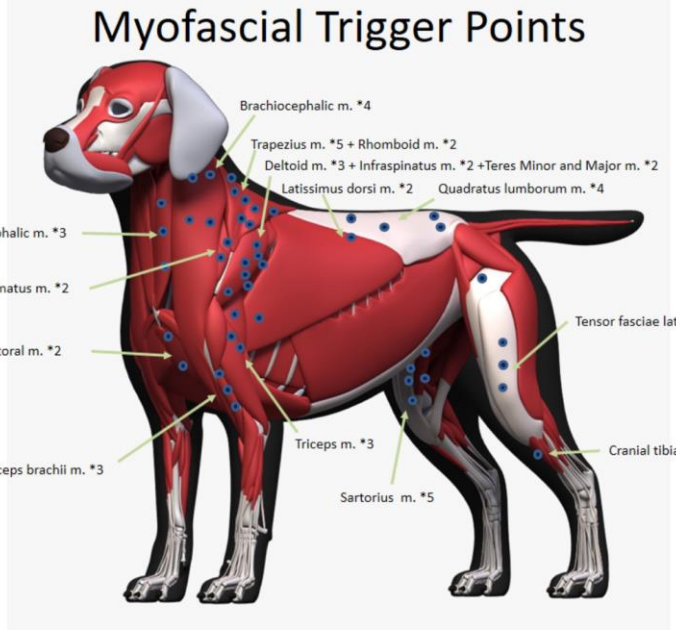
推拿按摩



https://images.app.goo.gl/QzPcfsyURujWJHy48



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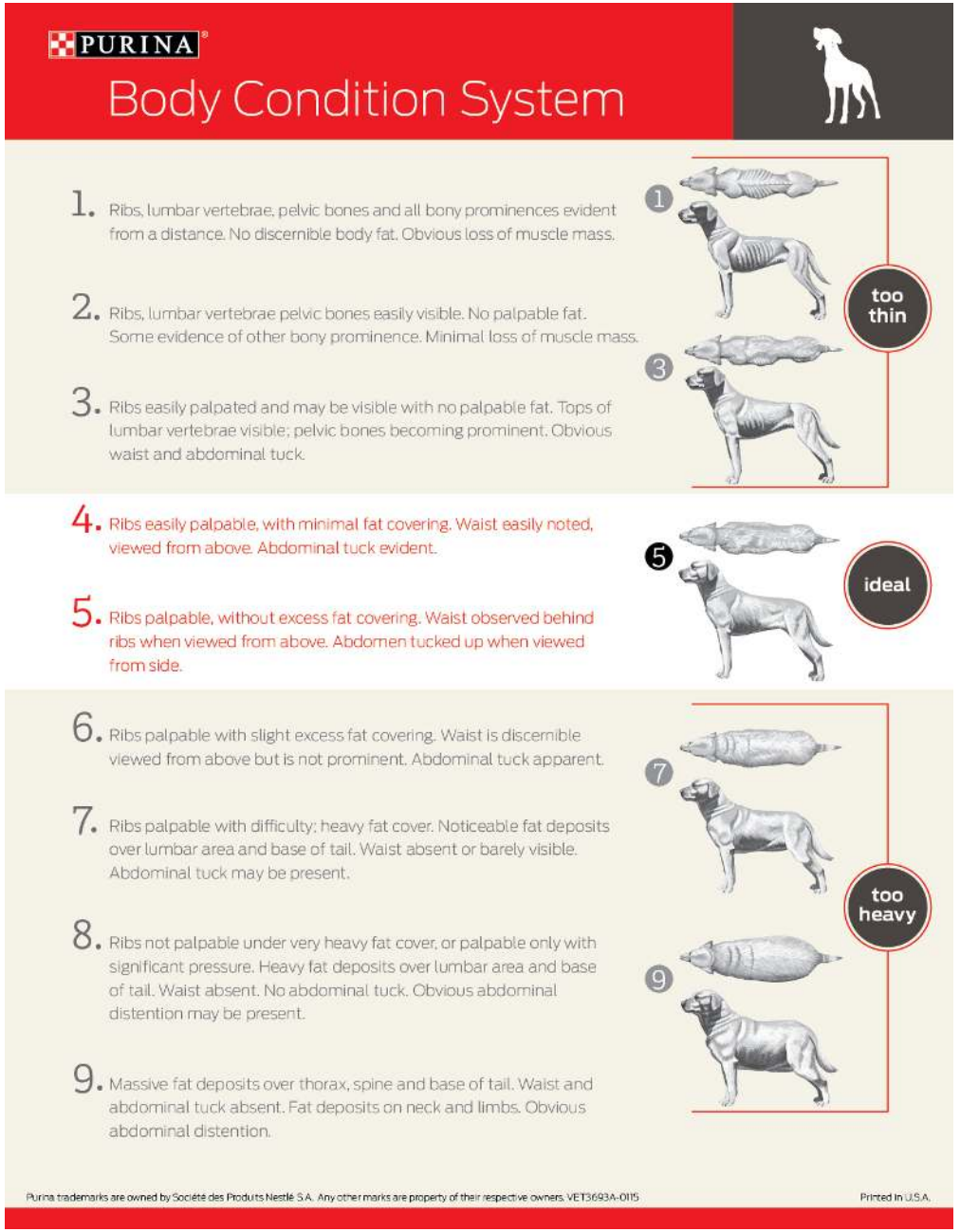
- Weight control
- Environmental modification











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肌少症(sarcopenia)

Sarcopenia (肌少症)

- Aging

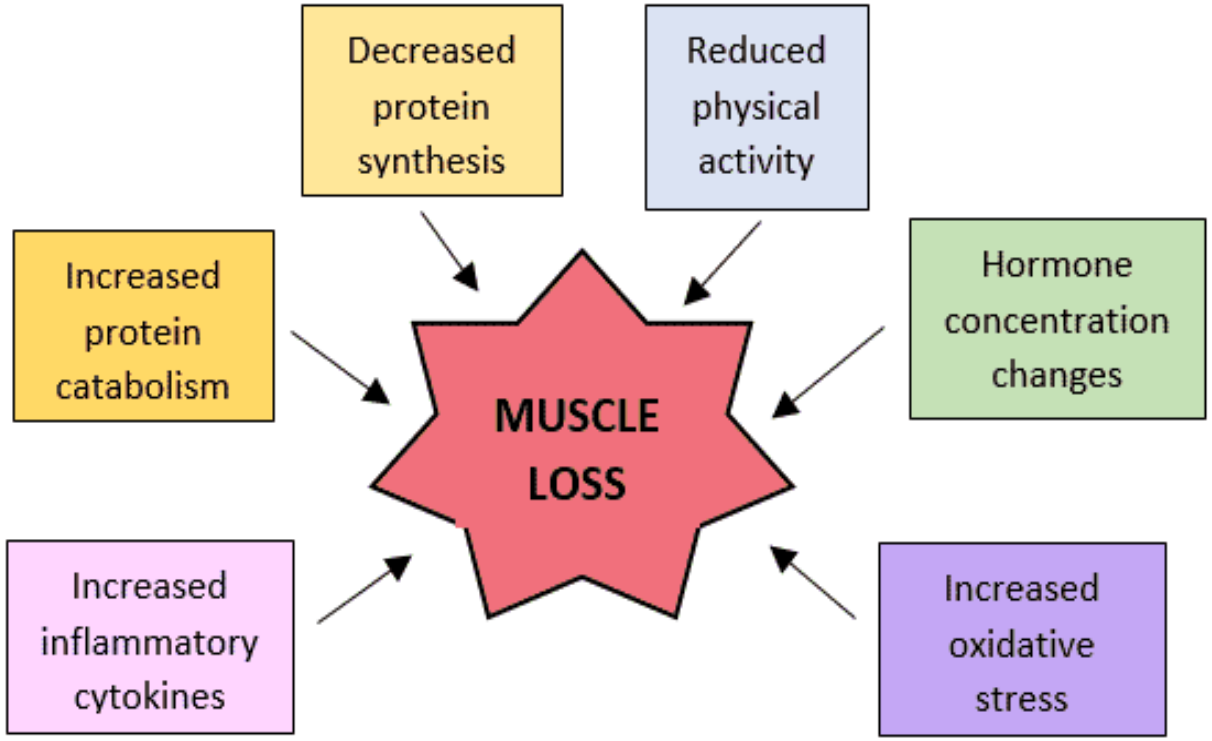
Cachexia (惡病質)

- Cardiac cachexia
- Cancer cachexia
- Renal cachexia
- Acute illness and injuries

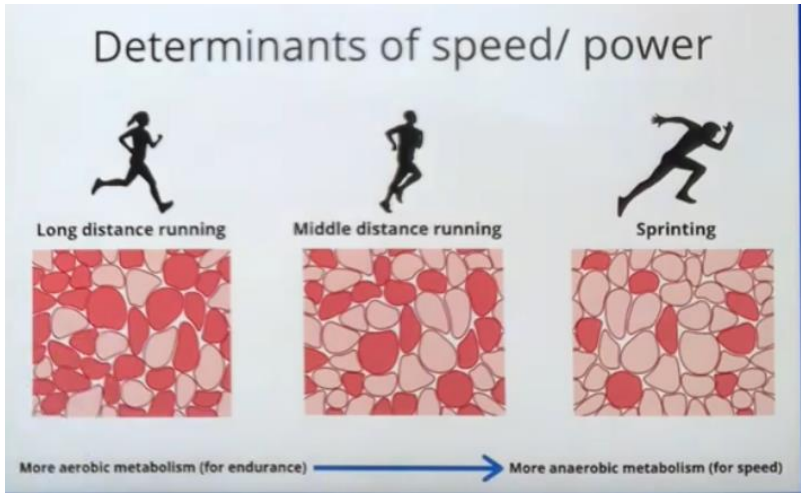
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肌少症(sarcopenia)

- 肌少症是指瘦肉組織(lean-body mass)流失
- 在人，20至80歲期間肌肉逐漸流失達30%，由於瘦肉消失伴隨肥肉累積因此不易由體重變化看出肌少症
- 隨年齡增加逐漸老化顯著瘦肉組織逐漸喪失，並無其他疾病引起
- 原因：蛋白質代謝異常，內分泌異常，慢性發炎，氧化反應異常，活動量減少
- 第1，2型肌肉纖維比例



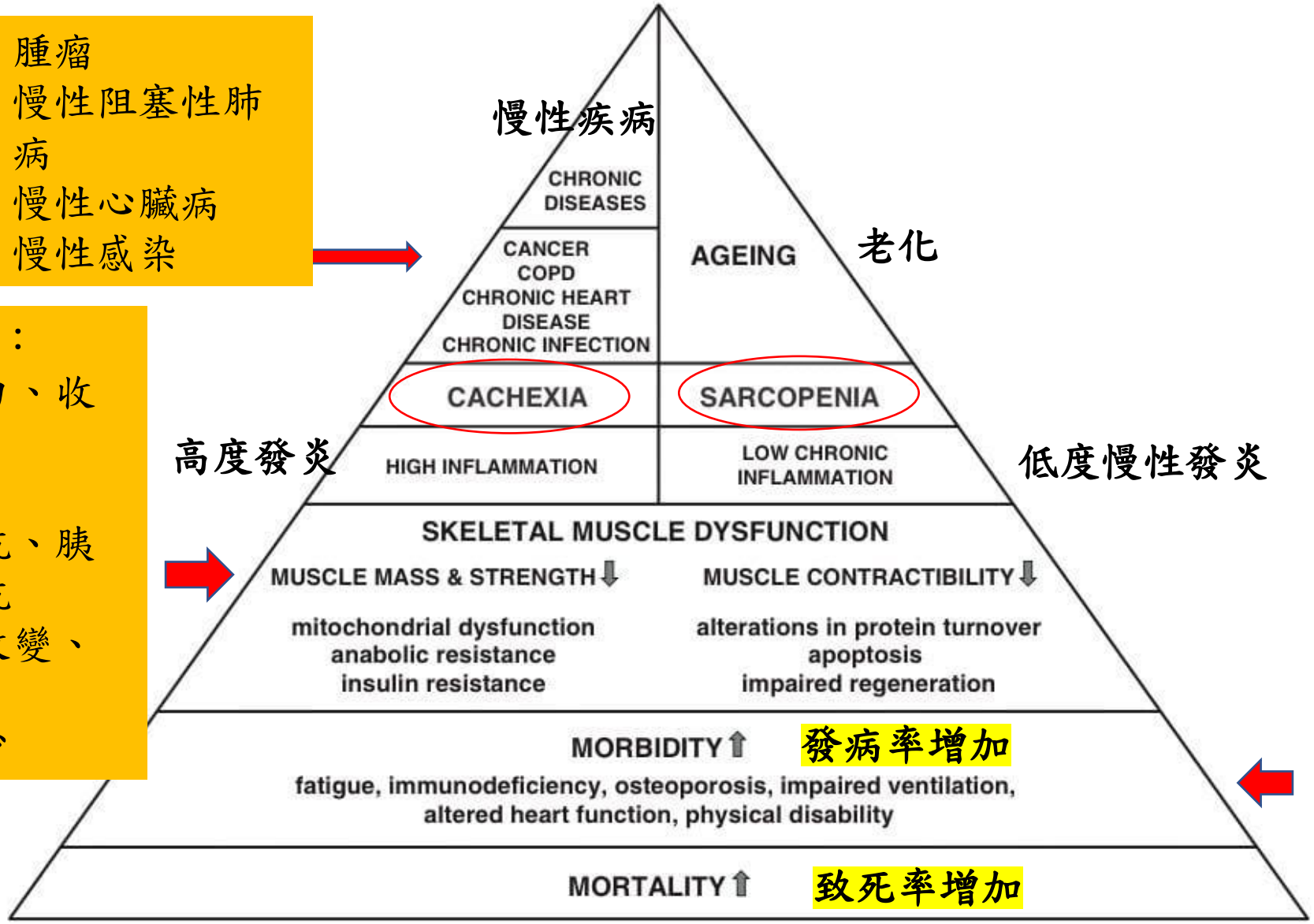
https://www.veterinary-practice.com/article/nutrition-requirements-of-senior-pets



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- 腫瘤
- 慢性阻塞性肺病
- 慢性心臟病
- 慢性感染

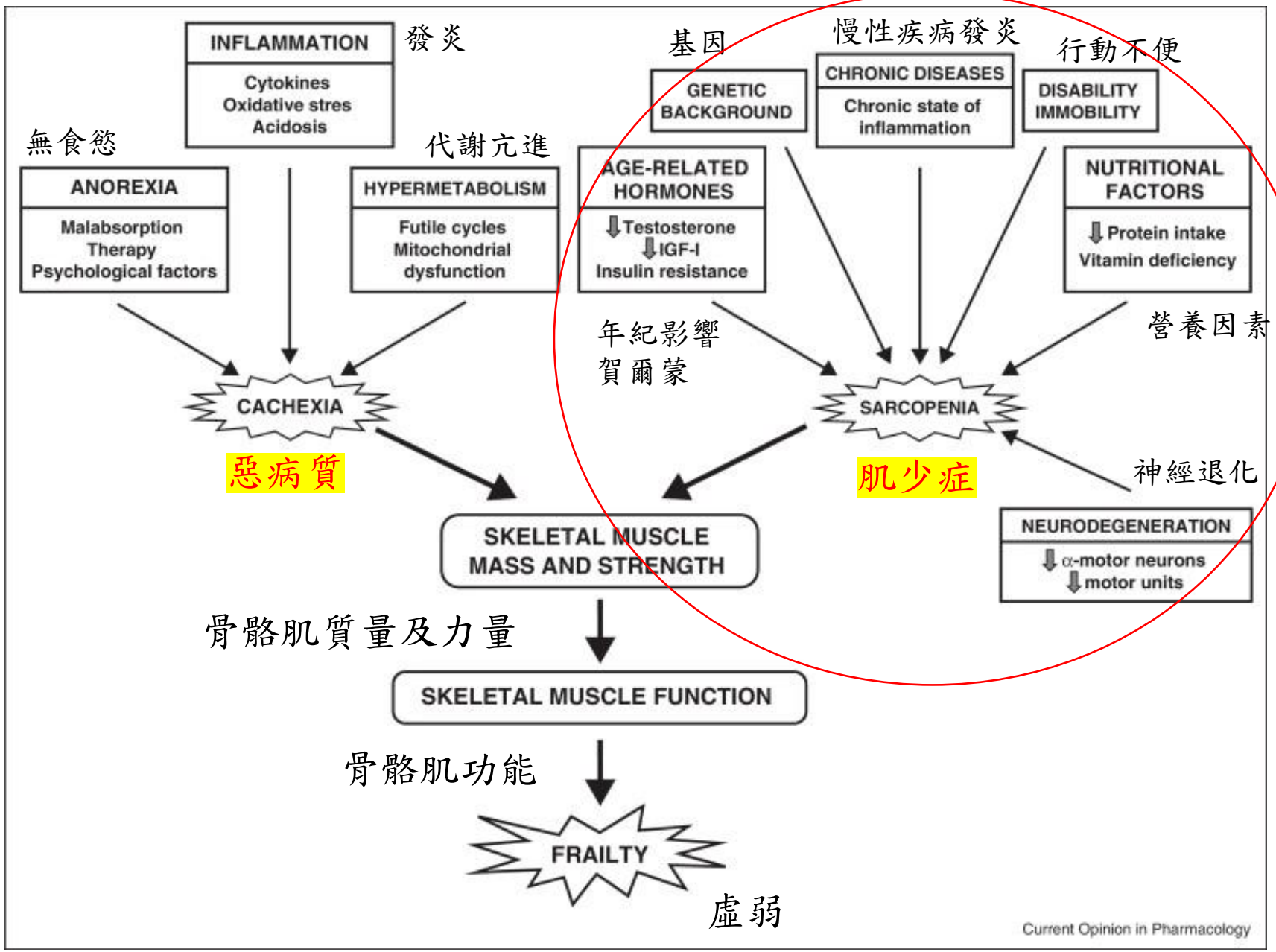
骨骼肌功能異常：
- 肌肉量、肌力、收縮力下降
- 粒線體異常
- 同化作用阻抗、胰島素合成阻抗
- 蛋白質轉換改變、凋亡
- 再生能力減少



- 疲倦
- 免疫缺陷
- 骨質疏鬆
- 換氣不良
- 心功能改變
- 身體殘疾

Josep Margilés et al. Cachexia and sarcopenia mechanisms and potential targets for intervention, Current Opinion in Pharmacology , volume 22, June 2015, p 100-106.

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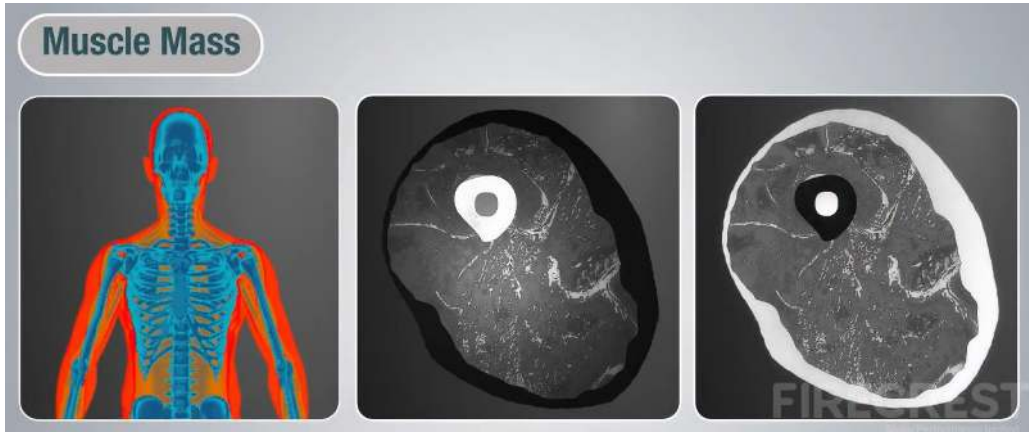
Josep M Argilés et al. Cachexia and sarcopenia mechanisms and potential targets for intervention, Current Opinion in Pharmacology , volume 22, June 2015, p 100-106.

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肌少症(sarcopenia)-human

- Diagnosis:
  - Physical performance: Gaits speed , Chair rise time (SARC-F Screening Tests)
  - Muscle strength: Hand-Grip Dynamometry
  - Muscle mass: Dual Energy X-ray Absorptiometry (DEXA), Computed Tomography (CT), Magnetic Resonance Imaging (MRI)
- Treatments:
  - Pharmacological approach
    - anabolic and anti-catabolic process of neuromuscular function
  - Nutrition
  - Omega 3 Fatty Acid
  - Modify metabolic alterations to prevent and even reverse muscle loss as the mechanisms of these syndromes are becoming better understood (e.g., ghrelin receptor agonists, myostatin antagonists)
  - Physical activity (both resistance and aerobic)

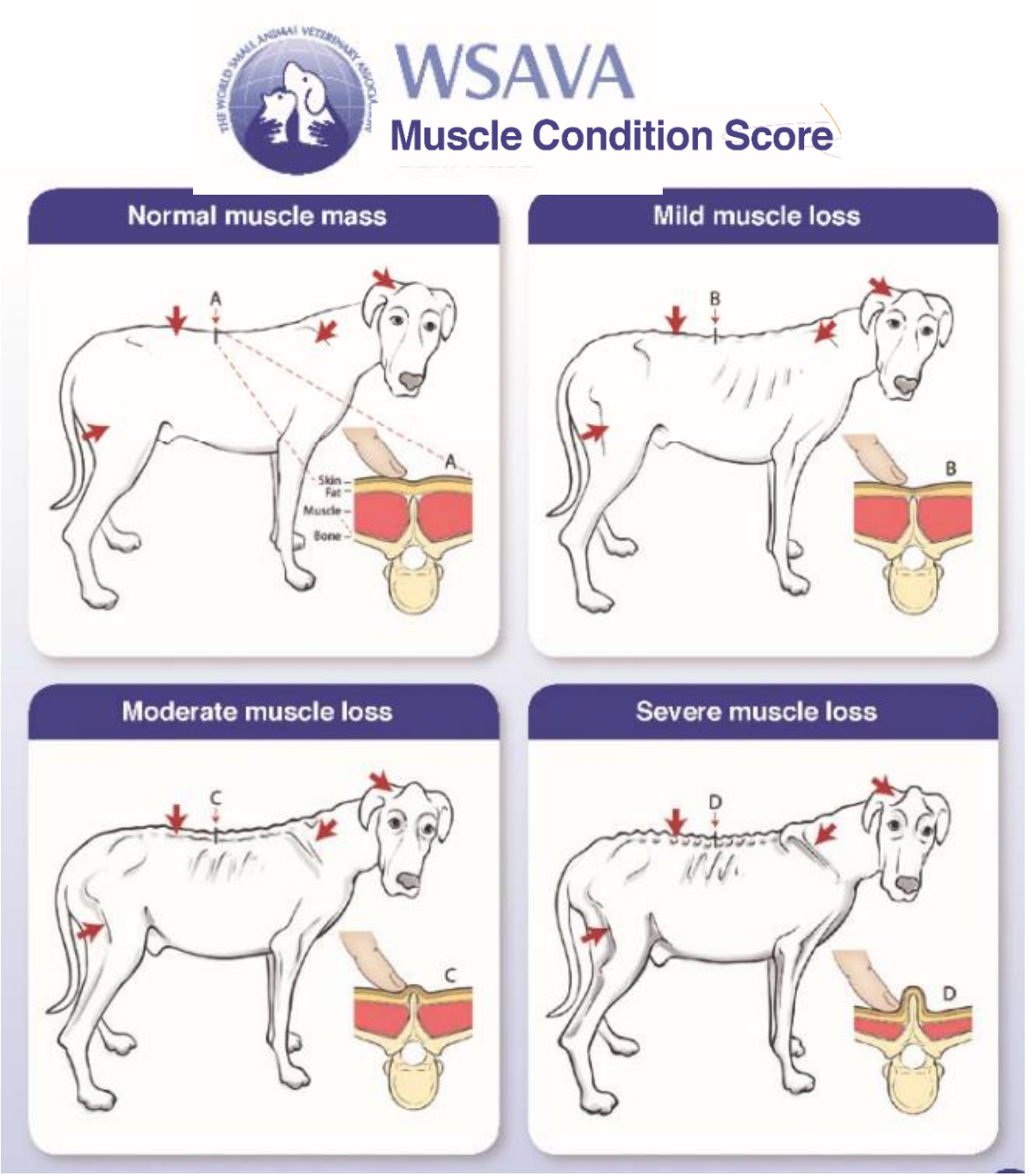




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肌少症(sarcopenia)-dog

- Diagnosis:
  - MCS-(Muscle Condition Score)
  - Muscle mass: Dual Energy X-ray Absorptiometry (DEXA), Computed Tomography (CT), Magnetic Resonance Imaging (MRI)
  - Muscle strength: C-MMT
  - Physical performance??
- Treatments:
  - Nutrition
  - Omega 3 Fatty Acid
  - Physical activity (both resistance and aerobic)



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- Strength 肌肉力測試
  - C-MMT（canine manual muscle test）徒手肌肉測試
  - Closed-chain motion 3/5 閉鎖鍊式活動分級



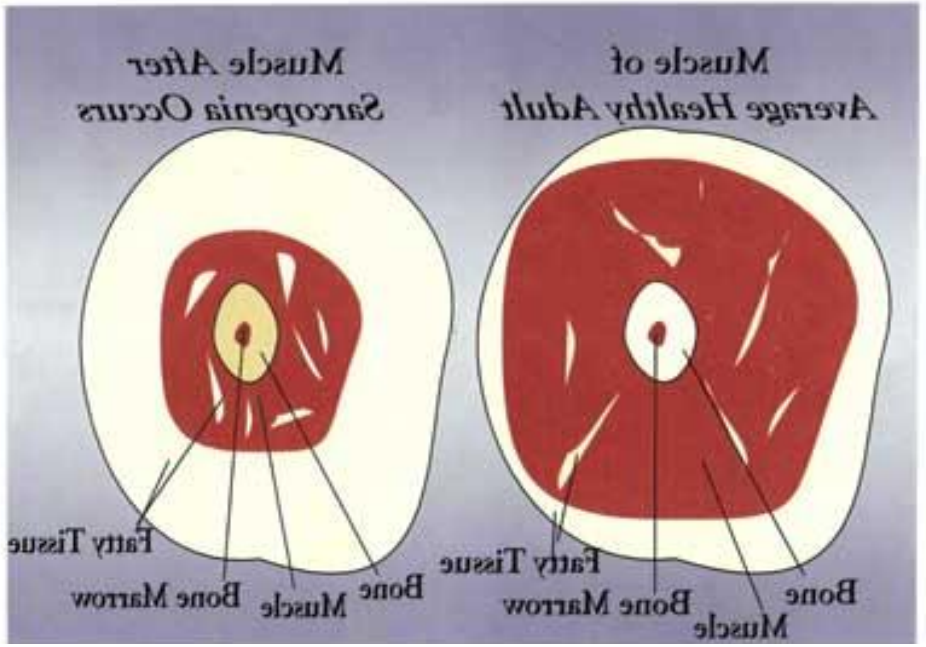


Canine Sports Medicine and Rehabilitation, 2nd Edition

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- 肌肉解剖及生理
- Type 1, 2 and 轉換

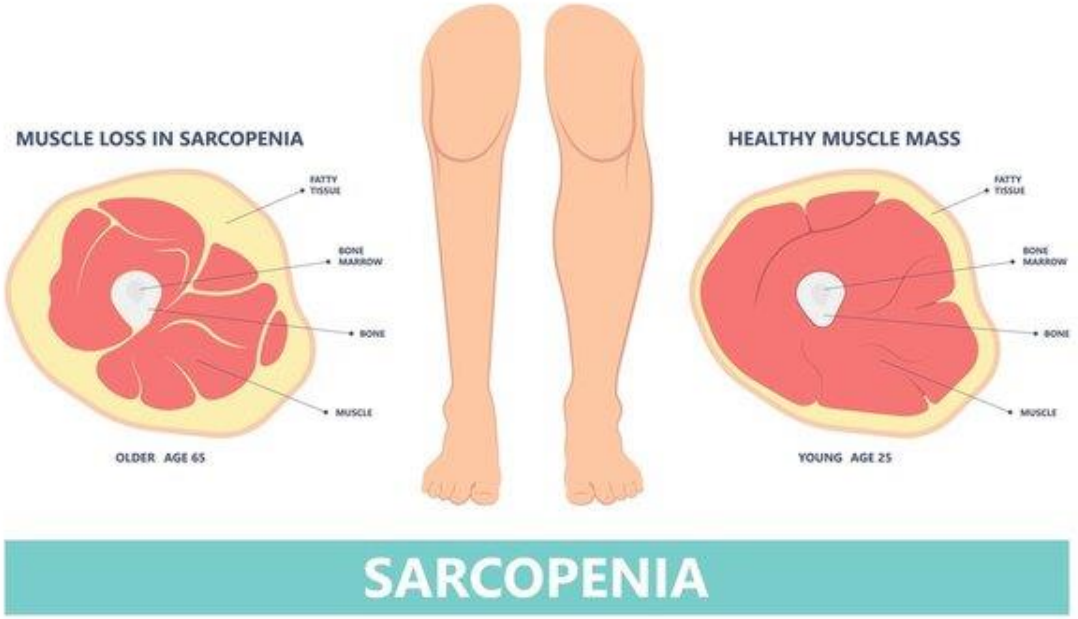




- 營養
  - Whey protein (乳清蛋白)
  - Leucine (亮胺酸)（BCAA-支鏈胺基酸）
  - Fortetropin ®
  - Ursolyx™ (ursolic acid熊果酸)
- 運動
  - Isometric 等長收縮
  - Concentric 向心收縮
  - Eccentric 離心收縮

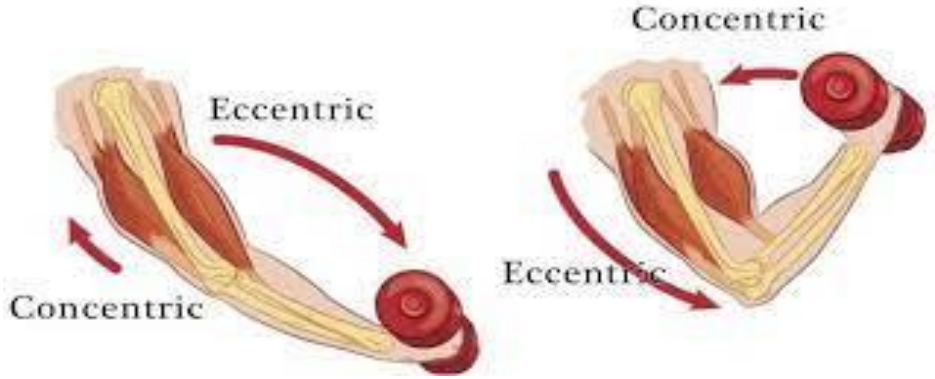
  - Anaerobic V.S. aerobic





Google.com

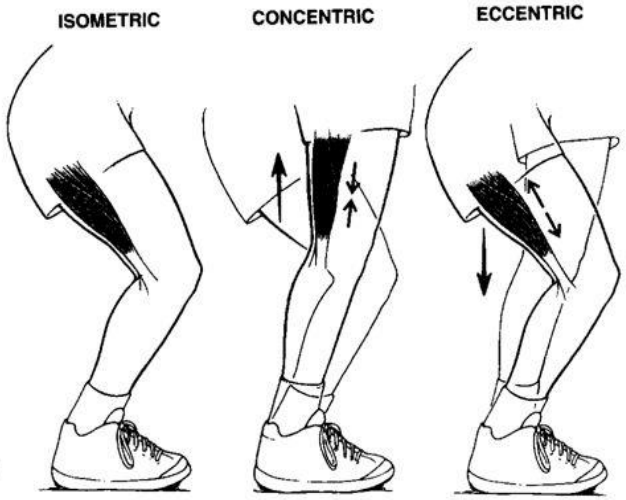
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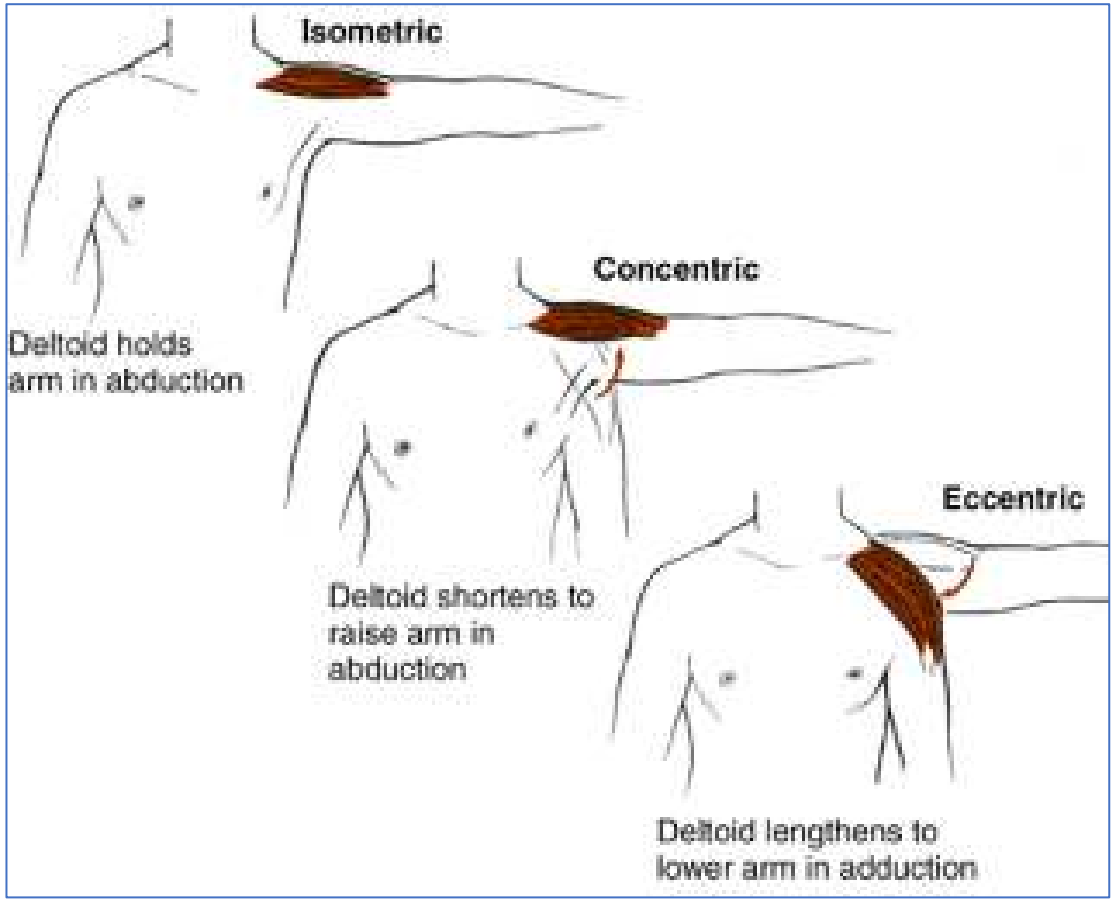


Muscular Action

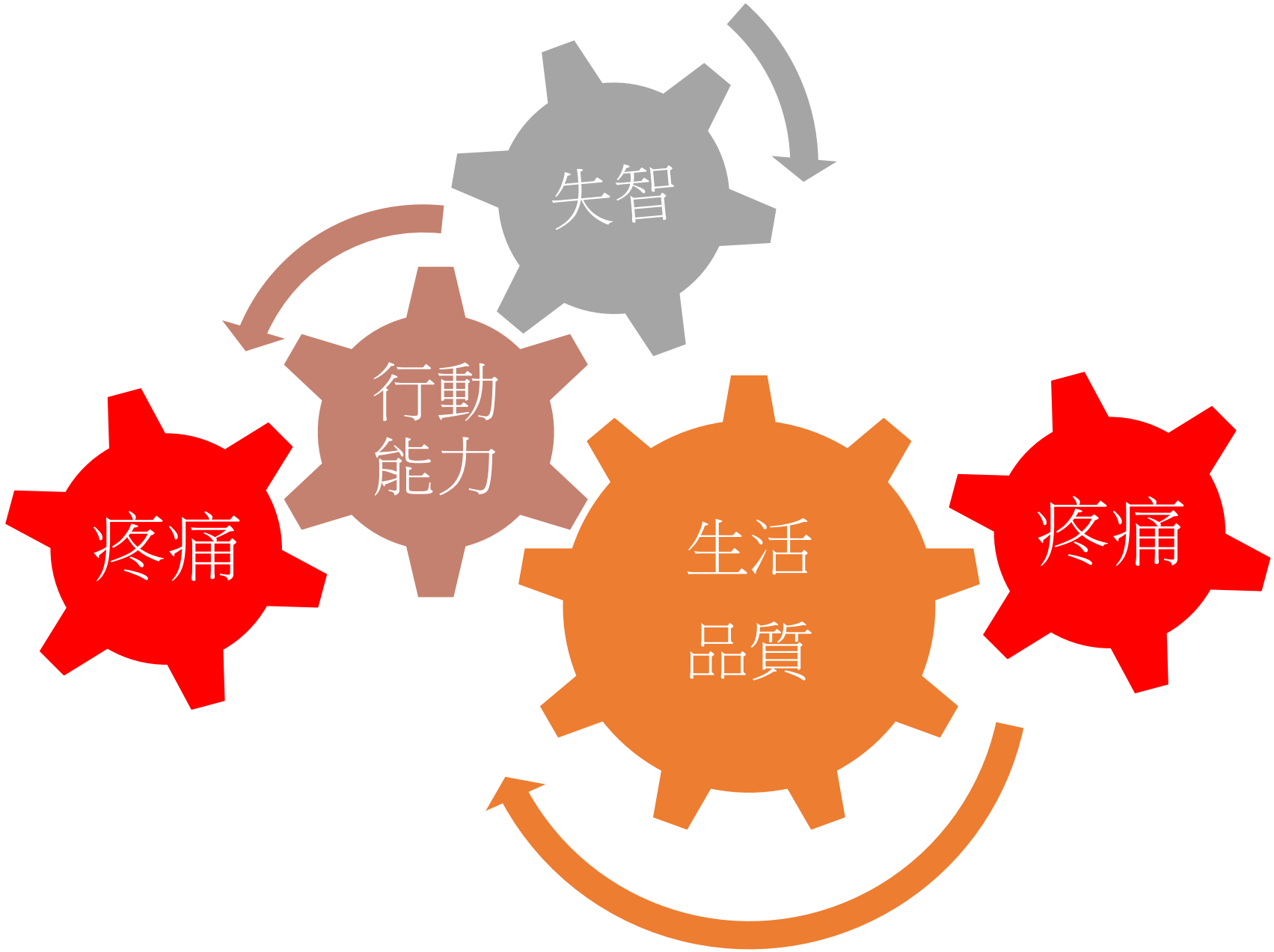
Whole Muscle

- **isometric action**
  - no change in fiber length
- **concentric action**
  - shortening of fibers to cause movement at a jt
- **eccentric action**
  - lengthening of fibers to control or resist a movement

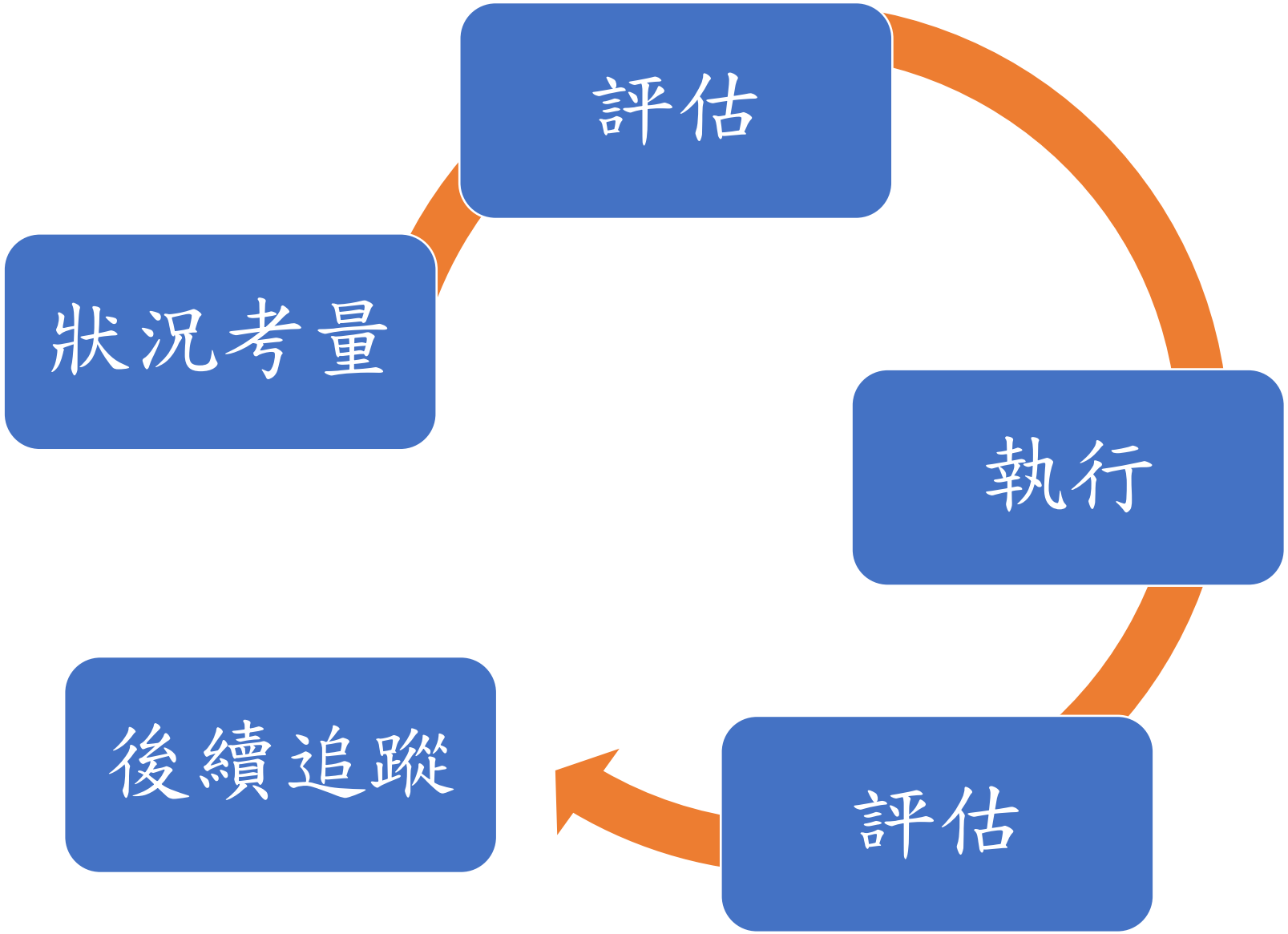




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老年狗復健治療項目

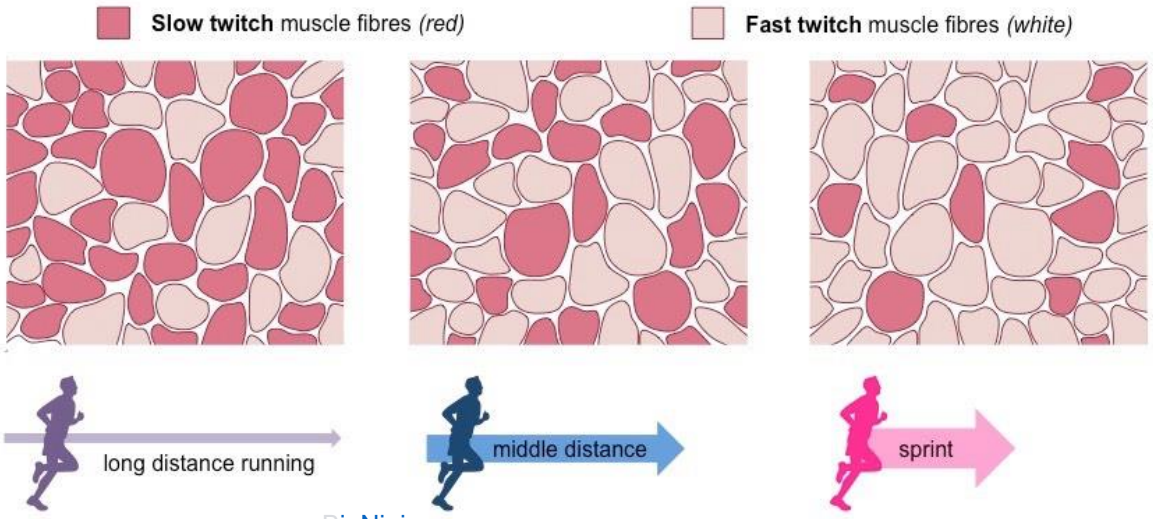
- 營養
- 理療復健
- 針灸推拿按摩
- 運動復健
- 輔具應用及居家環境調整

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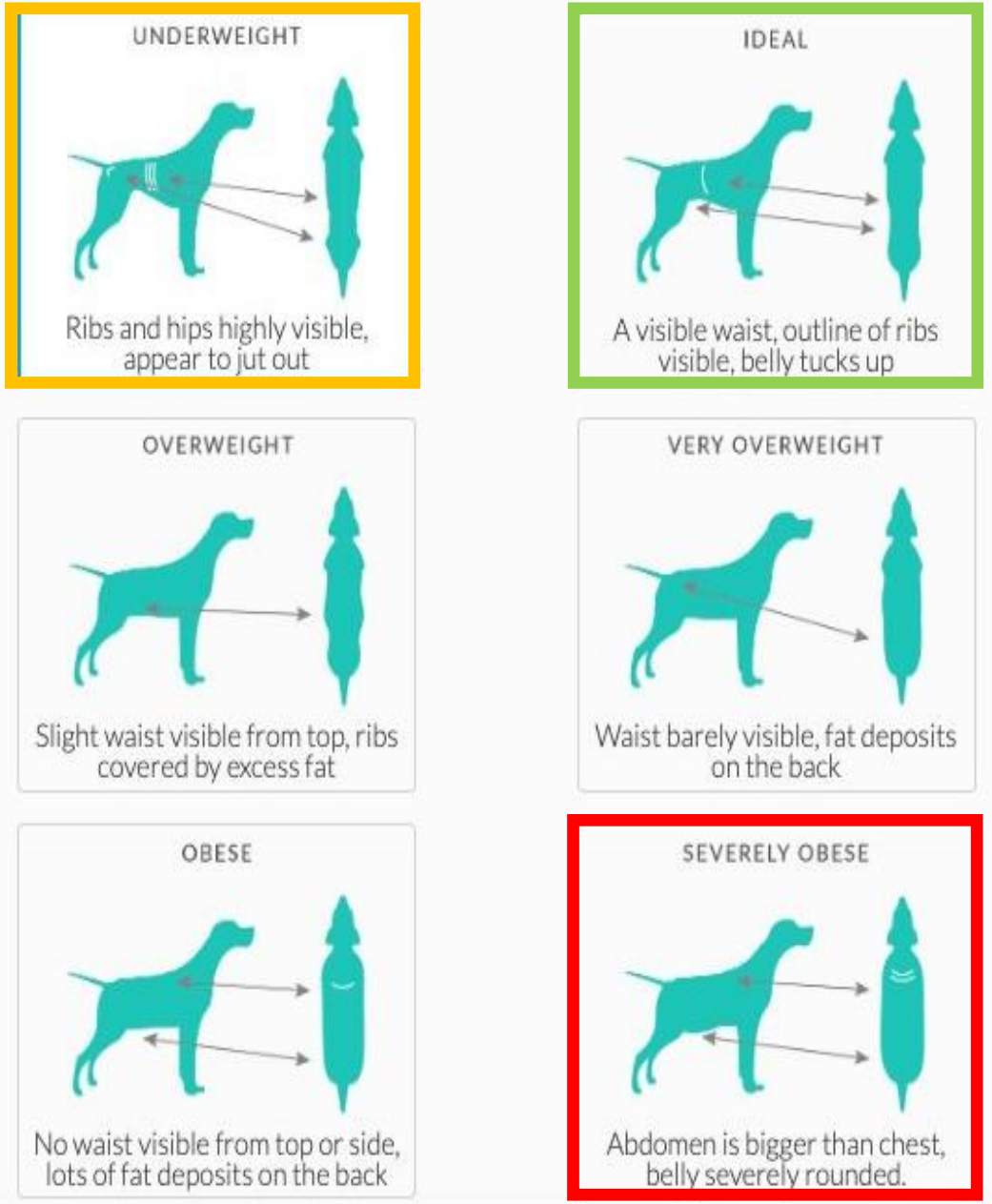
- 老年狗營養需求



- 體重控制
  - 80%靠營養控制，20%靠運動控制
  - 快肌(type 2 muscle fiber)與慢肌(type 1 muscle fiber)運動



BioNinja
Slow vs Fast Twitch | BioNinja



https://www.researchgate.net/publication/340190495/figure/fig4/AS:873380865662977@1585241633508/Body-Condition-Score-Legend-Survey-question-for-body-condition-shown-to-respondents.jpg

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- 理療復健
  - 雷射
  - 治療超音波
  - 脈衝磁圈
  - 水療





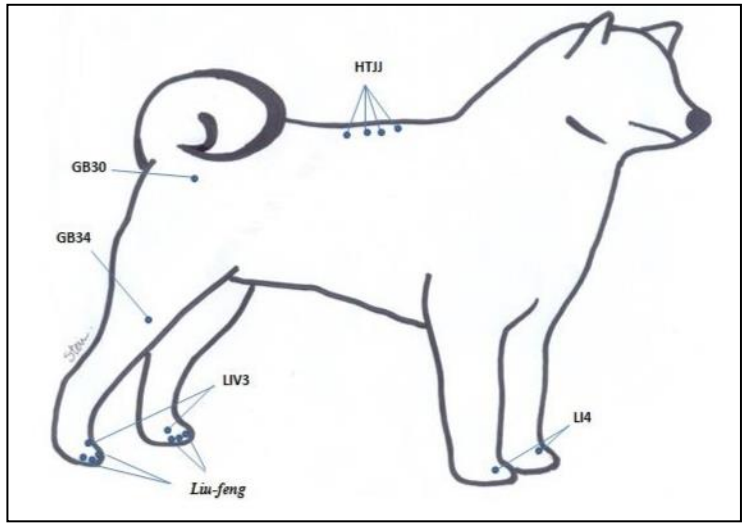


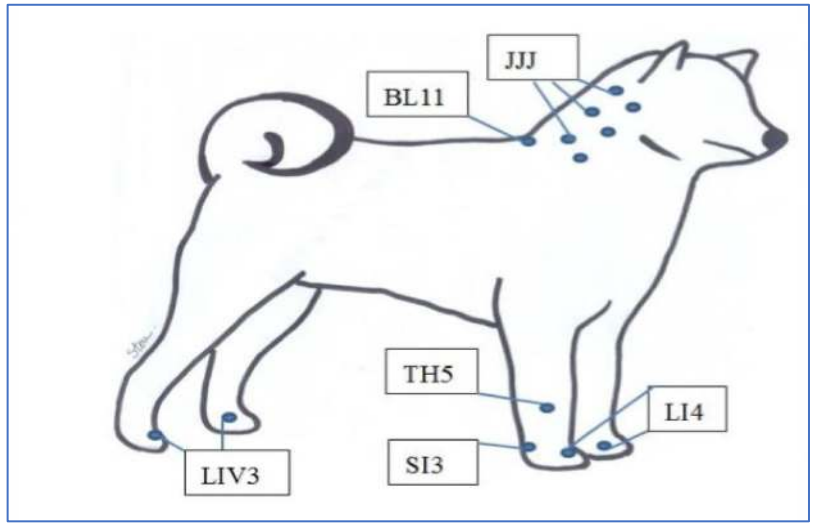




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- 針灸：
  - 止痛
  - 幫助修復
  - 恢復功能
  - 失智

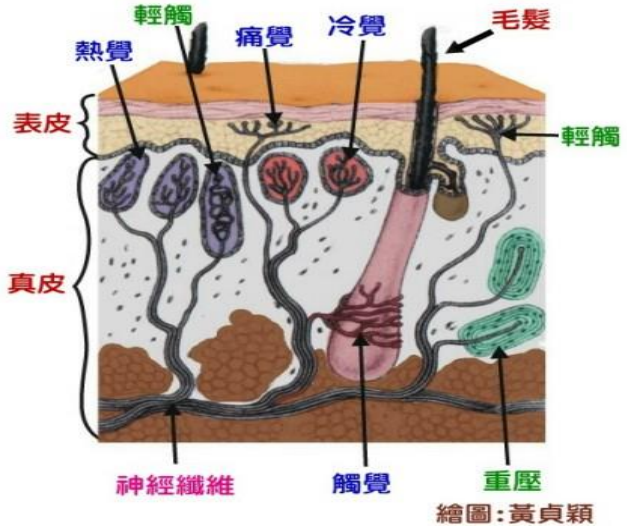






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- 推拿按摩
  - 體表刺激、經絡刺激、穴位刺激、激痛點、肌肉刺激



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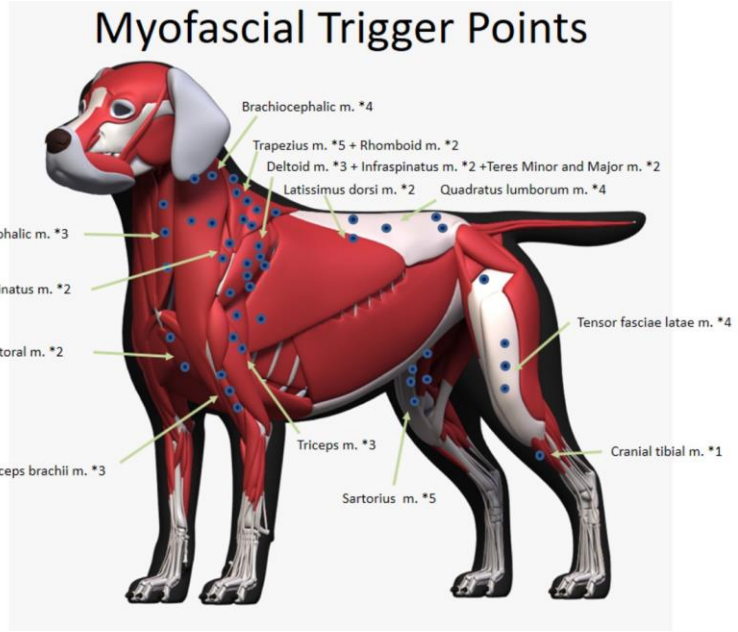


推拿按摩





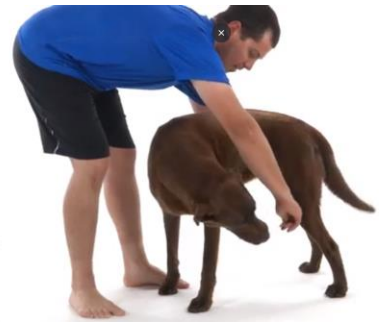




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- 老年狗運動復健目標
  - 預防或減少肌肉萎縮
  - 維持關節活動
  - 活化大腦
  - 幸福感 (psychological well-being)維持
- 老年狗運動復健原則
  - 低強度、高頻率、持續穩定增進
  - 至少選3種運動，一週進行3次
  - 運動量依每隻老狗能力做調整，一般而言每天至少一組6-8回
- 建議的居家運動項目：
  - ☐ side bends(側腰伸展),
  - ☐ diagonal or three leg standing(對角站立或三足站立),
  - ☐ wobble boards(搖擺板),
  - ☐ sit-stand on an uneven surfaces(不平穩地面坐與站),
  - ■ side stepping(側身走路),
  - ■ walking on narrow platforms(走窄平台),
  - ■ walking on short flights of stairs(走短階梯),
  - ■ crawling(匍匐前進),
  - ■ walking backwards(倒退走),
  - ■ stepping over objects(跨障礙),
  - ■ weaving around objects(繞障礙),
  - ■ Walking downhill/down a ramp(走下坡/下緩坡)
  - ■ swimming or walking in shallow water(游泳或淺水走路).
- 所有運動皆須有幾天休息要讓組織適當休息及修復，切勿過度
- 增加樂趣時段以促進認知鍛鍊(cognitive fitness) 盡量避免無聊呆板



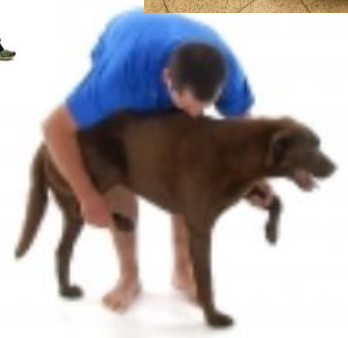










- 輔具應用及居家環境調整





















https://canineexercise.com/ ; google.com

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老人活力曲線

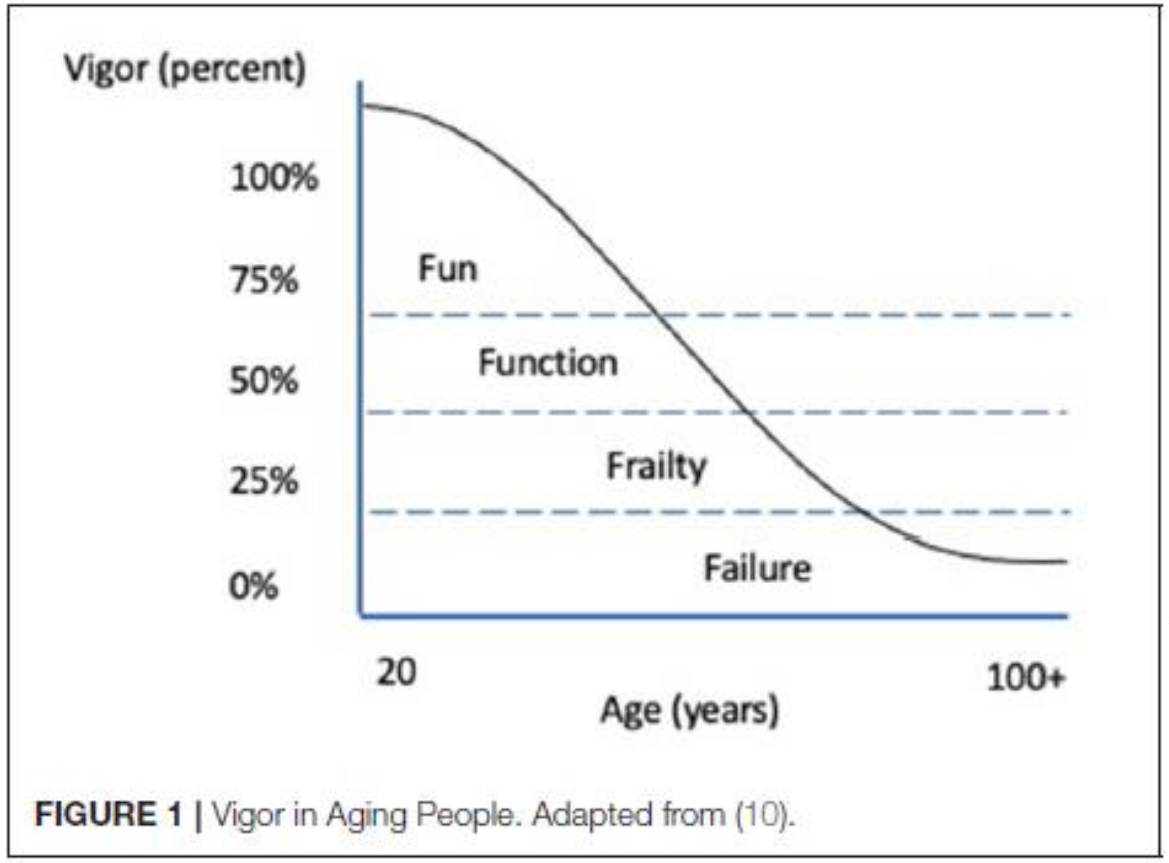


FIGURE 1 | Vigor in Aging People. Adapted from (10).

樂趣

功能

虛弱

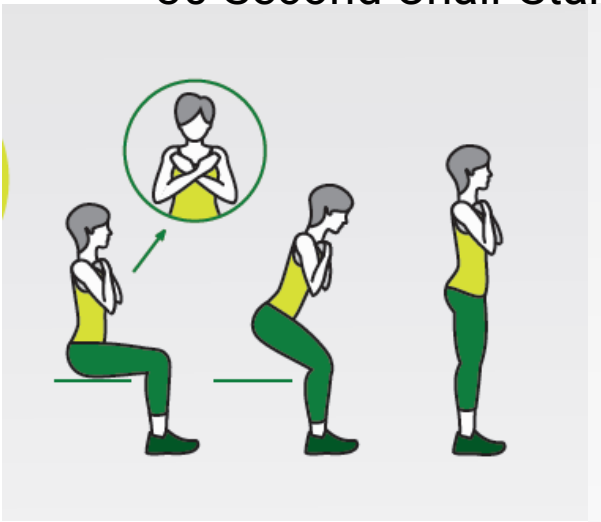
衰竭

Frye et al., 2022. Canine Geriatric Rehabilitation Considerations and Strategies for Assessment, Functional Scoring, and Follow Up. doi: 10.3389/fvets.2022.842458

行動功能評估：

- 在老年人
  - 從坐椅子站起30秒做幾次
  - 5xSST（5X Sit-to-Stand Test）從坐椅子站起做5次所需時間幾秒
  - 從坐椅子起身走3公尺，轉身，走回椅子，坐下，不超過12秒
- 在老年狗
  - 老年狗身體正趴起立，走10個身長距離的時間

30-Second Chair Stand Test



SCORING

Chair Stand
Below Average Scores

AGE	MEN	WOMEN
60-64	< 14	< 12
65-69	< 12	< 11
70-74	< 12	< 10
75-79	< 11	< 10
80-84	< 10	< 9
85-89	< 8	< 8
90-94	< 7	< 4

A below average score indicates a risk for falls.

www.cdc.gov/steadi

5xSST

Age-Matched Norms[4]:

Age Bracket	Time (sec)
60-69 yo	11.4
70-79 yo	12.6
80-89 yo	14.8

Fall Risk:

- Geriatrics
  -need for further assessment of fall risk: ≥ 12 sec[5]
  -recurrent falls: > 15 sec[6]
- Vestibular Disorders
  -fall risk: > 15 sec[7]
- Parkinson's Disease
  -fall risk: > 16 sec[8]

Canine geriatric functional score tests 老年狗行動功能測試量表

測試	描述	給分
A. TUG–timed up and go 坐站起走測試	正趴起身前進10個身長距離穩定踏步前進	0 無法執行 1 超過15秒 2 10-15秒 3 5-10秒 4 小於5秒
B. Cavaletti 跨越橫桿	牽繩引導前進跨越橫桿，橫桿高度及腳踝，去回各一次，共跨越4次橫桿	0 無法執行 1 撞擊橫桿，前進緩慢且困難 2 常碰觸橫桿，會部分調整腳步 3 偶而碰觸橫桿，會自行調整腳步，完成 4 幾乎不會碰觸橫桿，方向感正確
C. Figure 8's 8字型繞圈	牽繩引導，8字型直徑為身長，做4次	0 無法執行 1 腳背拖地，嚴重穿越困難，腳趾拖地，延遲轉身 2 有時腳背拖地，輕度至中度穿越困難，腳趾拖地，延遲轉身 3 延遲轉身，腳趾有時拖曳 4 順利完成
D. Down 趴到站	正趴起身60秒內做幾次	0 無法執行 1 小於5次秒 2 5-10次 3 10-15次 4 大於15次

總分	
0-4	差
5-8	一般般
9-12	好
13-16	優良

Frye et al., 2022. Canine Geriatric Rehabilitation Considerations and Strategies for Assessment, Functional Scoring, and Follow Up. doi: 10.3389/fvets.2022.842458

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甚麼是犬失智症(癡呆)?

- 犬失智症(Canine Dementia)正式名稱為犬認知功能障礙(Canine Cognitive Dysfunction Syndrome; CCDS)
- 大腦退化引起的學習、記憶、認知、社交互動、生活習慣、睡眠以及活動力等行為異常之漸進性疾病
- 通常11歲以上主人發現寵物記憶及學習能力下降，甚至提早到6歲
- 犬失智最常見症狀為漫無目的走路，角落發呆，白天睡多晚上不睡(佔57%)，與人或動物互動改變(佔51%)，焦慮不安(佔46%)
- 沒有特別的好發品種，門診常見為年紀大的柴犬、迷你雪納瑞、混種犬

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犬失智症診斷

- **臨床症狀**
- **失智量表**

- 腦部核磁共振攝影(MRI)
- 生物標記(Biomarkers)





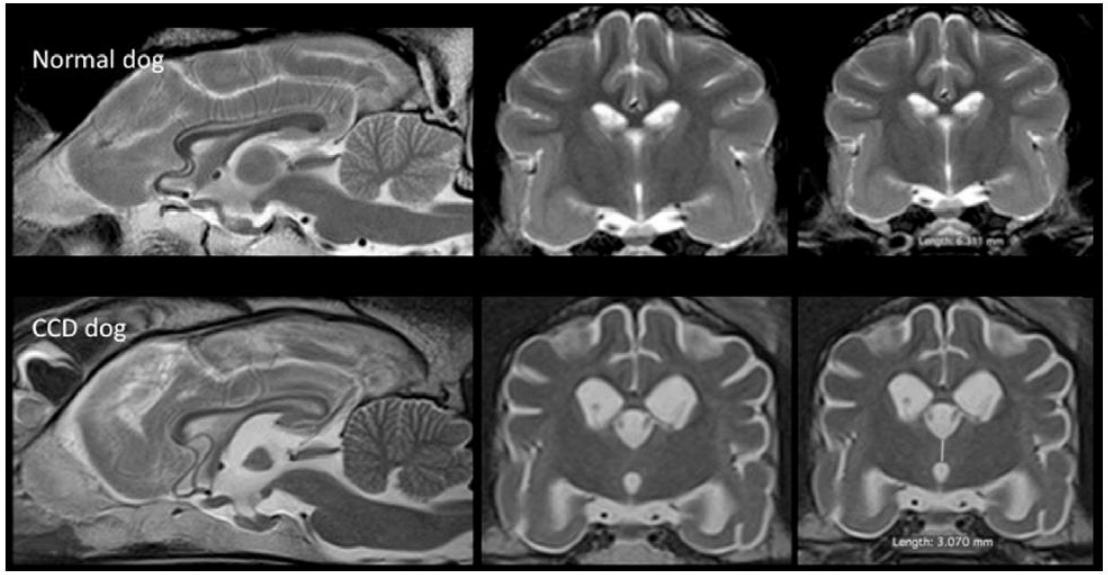


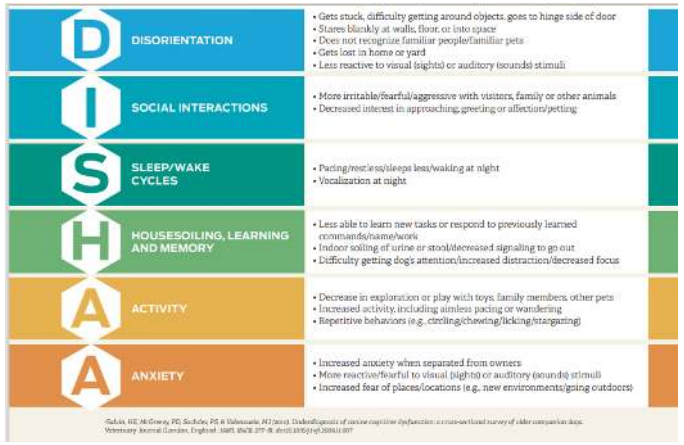
Fig. 2. Sagittal and transverse T2w MRI of normal and affected brain showing the reduced size and abnormal shape of the interthalamic adhesion.

Canine CognitiveDysfunctionPathophysiology, Diagnosis, and Treatment 2019

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犬失智症診斷

- 失智量表
  - DISHAA; 2003;2012 辨識失智臨床症狀敏感度高
  - CCDR (canine cognitive dysfunction rating scale); 2011;準確度達98.9% (n=957)
  - CADES (canine dementia scale); 2015; 預測失智嚴重度進程工具
  - CCAS (canine assessment scale)
  - PCT (practical cognitive test)





Canine Cognitive Dysfunction Rating (CCDR) Scale

	Never	Once a month	Once a week	Once a day	> Once a day	Score
How often does your dog pace up and down, walk in circles and/or wander with no direction or purpose?	☐=1	☐=2	☐=3	☐=4	☐=5	=
How often does your dog stare blankly at the walls or floor?	☐=1	☐=2	☐=3	☐=4	☐=5	=
How often does your dog get stuck behind objects and is unable to get around?	☐=1	☐=2	☐=3	☐=4	☐=5	=
How often does your dog fail to recognise familiar people or pets?	☐=1	☐=2	☐=3	☐=4	☐=5	=
How often does your dog walk into walls or doors?	☐=1	☐=2	☐=3	☐=4	☐=5	=
How often does your dog walk away while, or avoid, being patted?	☐=1	☐=2	☐=3	☐=4	☐=5	=
	Never	**1–30% times**	**31–60% times**	**61–99% times**	**Always**	
How often does your dog have difficulty finding food dropped on the floor?	☐=1	☐=2	☐=3	☐=4	☐=5	=
	Much less	**Slightly less**	**The same**	**Slightly more**	**Much more**	
Compared with 6 months ago, does your dog now pace up and down, walk in circles and/or wander with no direction or purpose?	☐=1	☐=2	☐=3	☐=4	☐=5	=
Compared with 6 months ago, does your dog now stare blankly at the walls or floor?	☐=1	☐=2	☐=3	☐=4	☐=5	=
Compared with 6 months ago, does your dog urinate or defecate in an area it has previously kept clean (if your dog has never house-soiled, tick 'the same')?	☐=1	☐=2	☐=3	☐=4	☐=5	=
Compared with 6 months ago, does your dog have difficulty finding food dropped on the floor?	☐=1	☐=2	☐=3	☐=4	☐=5	**Multiply by 2 =**
Compare with 6 months ago, does your dog fail to recognise familiar people or pets?	☐=1	☐=2	☐=3	☐=4	☐=5	**Multiply by 3 =**
	Much more	**Slightly more**	**The same**	**Slightly less**	**Much less**	
Compared with 6 months ago, is the amount of time your dog spends active?	☐=1	☐=2	☐=3	☐=4	☐=5	=
0-39 = Normal 40-49 = At risk 50+ = CCD					**Total**	=

Please return this form to:
Regenerative Neuroscience Group
Brain and Mind Research Institute
University of Sydney
94 Mallett Street,
Camperdown, NSW 2050

Contacts
Dr Sarah Toole BVSc(Hons)
sarah.toole@sydney.edu.au
0418 838 911

Dr Rosalynn Wen BVSc(Hons)
rosalynn.wen@sydney.edu.au
0409 510 657

Your name:
Email:
Pet name:
Age:
Sex:
Breed:

REGENERATIVE NEUROSCIENCE GROUP
THE UNIVERSITY OF SYDNEY

Domain/items	
A. Spatial orientation	B. Social interaction
1, disorientation in a familiar environment (inside/outside)	6, changes in interaction a man/dog, dog/other dog (playing, petting, welcoming)
2, to recognise familiar people and animals inside or outside the house/apartment	7, changes in individual behaviour of dog (exploration behaviour, play, performance)
3, abnormally respond to familiar objects (a chair, a wastebasket)	8, response to commands and ability to learn new task
4, aimlessly wandering (motorically restless during day)	9, irritable
5, a reduced ability to do previously learned task	10, expression of aggression
SCORE (0–25)	SCORE (0–25)
C. Sleep–wake cycles	D. House soiling
11, abnormally responds in night (wandering, vocalization, motorically restless)	13, eliminate at home at random locations
12, switch over from insomnia to hypersomnia	14, eliminate in its kennel or sleeping area
Score × 2 (0–20)	15, changes in signalisation for elimination activity
	16, eliminate indoors after a recent walk outside
	17, eliminate at uncommon locations (grass, concrete)
	SCORE (0–25)
Total score (A + B + C + D) (0–95)	
Clinical stage: Normal ageing (Score 0–7), Mild cognitive impairment (8–23), Moderate cognitive impairment (24–44), Severe cognitive impairment (45–95)	

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犬失智症治療與照顧

- 治療目標
  - 延緩腦神經細胞受損
  - 改善臨床症狀
- Drugs
  - Selegiline® (已停產), Eldepryl®-MAO-B inhibitor 个 dopamine
  - Propentofylline (Vivitonin®)- 可通過BBB限制自由基並抑制發炎
- Diet
  - Hills Canine b/d
  - Purina One Vibrant Maturity 7+ Formula
  - MCT
- Supplements
  - Phosphatidylserine (AKTIVAIT®腦活素, Senilife®腦力活)一種磷脂化物可以增加細胞膜流動性
  - SAMe 沙美
  - CogniCap® 康格憶
  - Zylkene® 解憂寧
  - Melatonin 褪黑激素











Senilife	Phosphatidylserine, *Gingko biloba*, vitamin B6 (pyridoxine), vitamin E, resveratrol
Activait	Phosphatidylserine, omega-3 fatty acids, vitamins E and C, L-carnitine, alpha-lipoic acid, coenzyme Q, selenium

Cognitive dysfunction syndrome: a disease of canine and feline brain aging Cognitive dysfunction syndrome: a disease of canine and feline brain aging. 2012 Jul;42(4):749-68, vii.

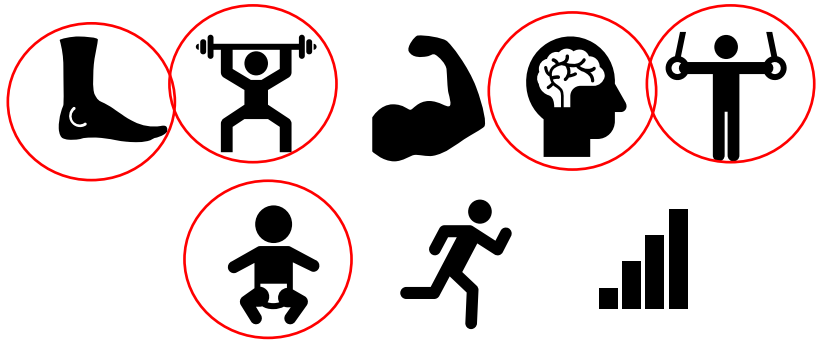
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犬失智症治療與照顧

- 中獸醫 Canine CognitiveDysfunctionPathophysiology, Diagnosis, and Treatment. Vet Clin Small Anim 49 (2019) 477–499.
  - 針灸-
    - GB-34, GV-20, LIV-3, SP-10, and ST-36 reduced the expression of microglia;
    - LI-4 and LIV-3 activated regions of the cerebrum responsible for memory and cognition
    - GV-20, GV-14, and ST-36 increased expression of synaptophysin and neurotrophin in the hippocampal region
  - 中藥方- Sai Luo Tong (賽絡通; 主成份為人參、銀杏、紅花)，複方丹參(丹參、三七、冰片)，六味地黃，八味地黃，抑肝散
- LASER-transcranial PBMT
  - decreases inflammation and increases microcirculation
  - 10Hz, 0.3-3.0 J/cm^2 on head
  - Twice/wk for 2 wks, then once/wk for 2 wks, then every 2-3 wks as needed
- 居家運動復健- 擊掌，坐到站，趴下到站立，跨越橫桿，穿梭

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主動運動



- 握手或High 5 擊掌
  - 坐著握手
  - 站著握手
  - 平衡板上握手

  - 每天1-3次，每次3-5回，視動物狀況調整



From Canine Sports Medicine and Rehabilitation 2nd Edition



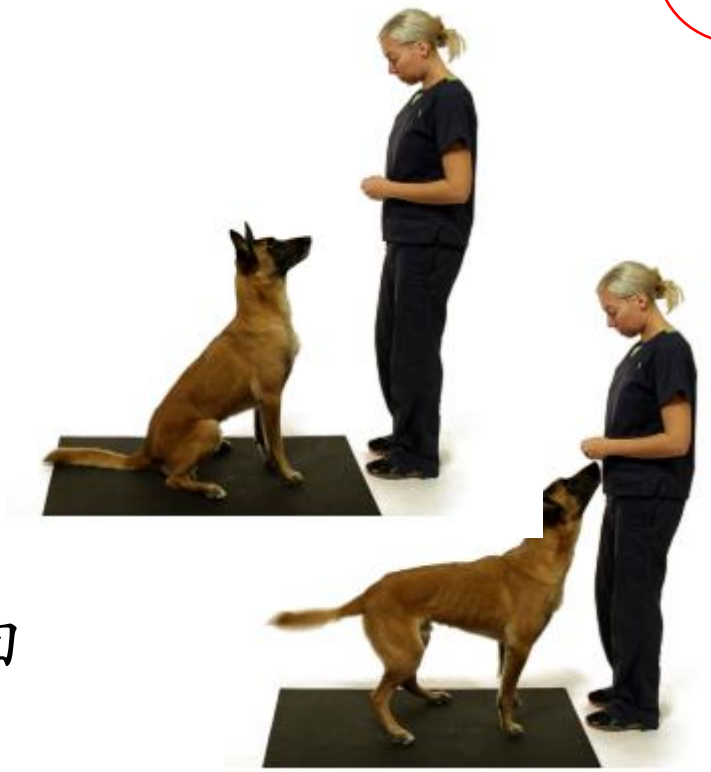
From Canine Exercise Solutions (https://www.canineexercise.com/)

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主動運動



- 由坐到站
  - 坐，5秒後，站
  - 著重於髖及膝之伸肌訓練



From Canine Exercise Solutions (https://www.canineexercise.com/)

- 每天3-5次，每次3-5回
- 可以視動物狀況增加至每次15回



From Canine Rehabilitation and Physical Therapy

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主動運動



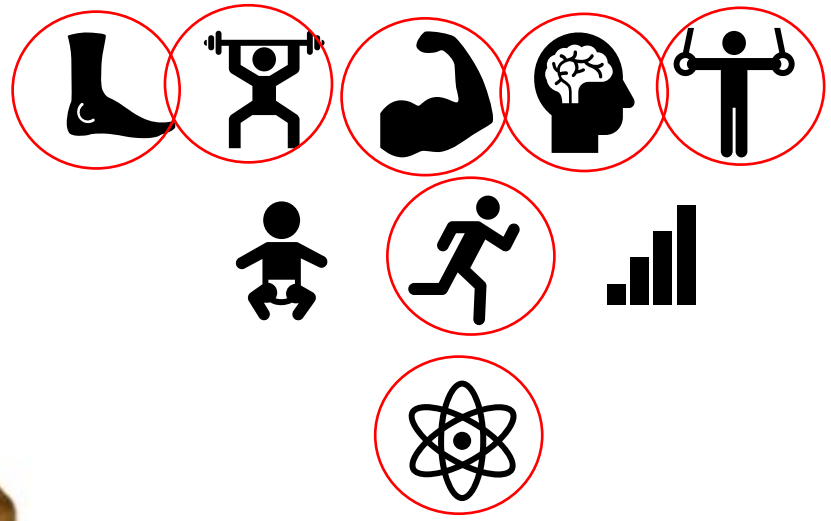
- 由趴下到站立
  - 類似由坐到站
  - 趴下到站立著重於髖及膝之伸肌訓練，以及肩與肘之伸肌訓練
  - 每天3-5次，每次3-5回
  - 可以視動物狀況增加至每次10-15回



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主動運動

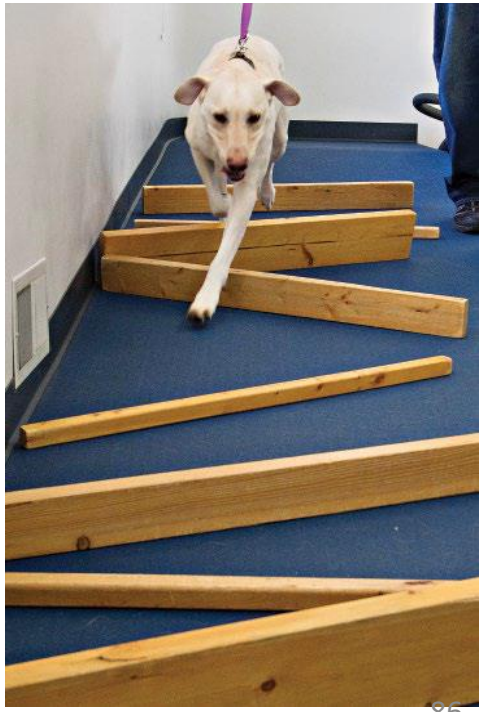


- 跨越橫桿
  - 等高，愈高難度越高
  - 等距，愈窄難度越高
  - 不規律



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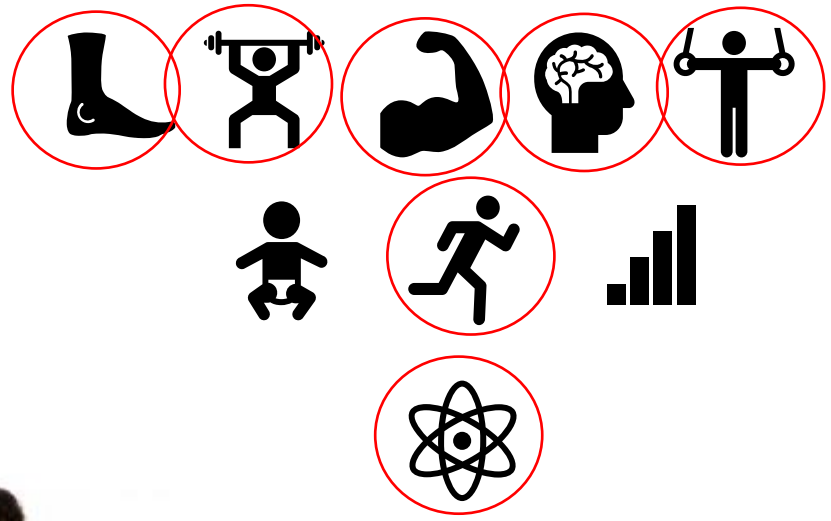
  - 每天2-3次，每次3-5趟
  - 視動物狀況可以逐漸增加至每天次10-20趟



From Canine Sports Medicine and Rehabilitation 2nd Edition

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主動運動



- 穿越穿梭錐，繞圈，8字形
  - 大圈與小圈，愈窄難度越高
  - 每天2-5次，總共2-3分鐘
  - 視動物狀況可以逐漸增加至每天不超過5分鐘







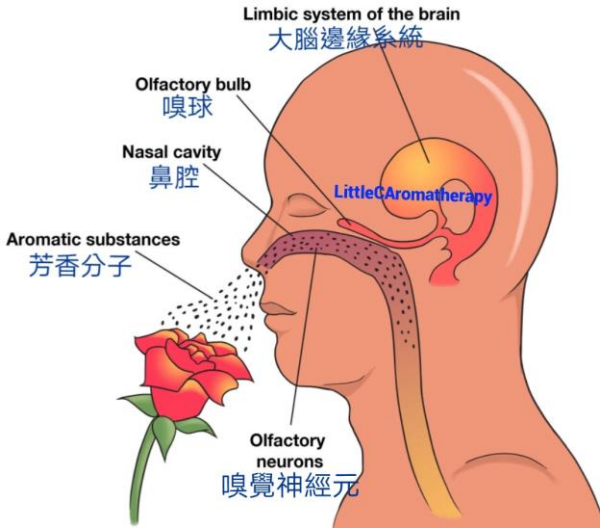
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犬失智症治療與照顧



- 精油芳香療法(aroma therapy)
  - 日:柑橘類-尤加利、檸檬-振奮、醒腦作用
  - 夜:薰衣草、洋甘菊-安撫作用
  - 豐富環境及安全防護- 規律戶外散步、新玩具、嗅聞盤



littlec-aromatherapy.blogspot.com



芳療





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犬失智症最新資訊

- GedaCure® (crisdesalazine)
- 可在一定程度上顯著減少阿茲海默症的病理特徵，包括神經元凋亡、tau蛋白病變和澱粉樣蛋白斑塊。甚至在crisdesalazine給藥4週後，犬隻就能重新識認主人並向主人擺尾，夜眠安好，活動力改善



http://www.businesswire.com/news/home/20210210005036/en

- LASER
  - 660 and 810 nm penetrate to brain
  - Class 3b or class 4- time↓
  - Class 1 time~ 5-10 minutes (home use)
- Biomarkers 生物標記
  - Aβ42(Amyloid β42) ↓, Aβ40↓ during 11-15.6 year CDS
  - neurofilament light chain (NFL)神經絲輕鏈↑



Fig. 4. Application of tPBMT to a dog using a class 1 laser (photo courtesy of Iker Asteinza DVM).

- Transcranial photobiomodulation (laser) therapy for cognitive impairment: A review of molecular mechanisms and potential application to canine cognitive dysfunction (CCD). *Open Veterinary Journal*, (2022), Vol. 12(2): 256–263.
- Novel Diagnostic Tools for Identifying Cognitive Impairment in Dogs: Behavior, Biomarkers, and Pathology. Front. Vet. Sci., 15 January 2021 Jan 15;7:551895.
- Plasma Amyloid Beta Concentrations in Aged and Cognitively Impaired Pet Dogs. Molecular Neurobiology 2021 Feb;58(2):483-489.

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監控進展及評估要點

1. 定期血檢、尿檢以監控使用NSAIDs以及動物身體代謝狀況
2. 定期專業復健治療每週1-2次，包括治療性注射、針灸、按摩以及理療復健如雷射、治療超音波、水療
3. 每4-6週根據動物狀況調整目標及適度的復健
4. 每4-6週體重監控
5. 使用評估量表以及步態分析等客觀評估設備

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5H2M生活品質量表

Quality of Life Scale (HHHHHMM Scale)

Using a scale of 0 to 10 (0 = Unacceptable, 10 = Excellent), patients can be evaluated for their quality of life.

Score	Criterion
0-10	**Hurt**—Is the patient in pain, including distress from difficulty in breathing? Can the pet's pain be successfully managed? Is oxygen necessary?
0-10	**Hunger**—Is the pet eating enough? Does hand-feeding help? Does the pet require a feeding tube?
0-10	**Hydration**—Is the pet dehydrated? Are subcutaneous fluids once or twice daily enough to resolve the problem? Are they well tolerated?
0-10	**Hygiene**—The pet should be kept brushed and clean, particularly after elimination. Does the pet have pressure sores?
0-10	**Happiness**—Does the pet express joy and interest? Is he responsive to things around him (family, toys, etc)? Is the pet depressed, lonely, anxious, bored, or afraid? Can the pet's bed be near the kitchen and moved near family activities to minimize isolation?
0-10	**Mobility**—Can the pet get up without assistance? Does the pet need human or mechanical help (eg, a cart)? Does she feel like going for a walk? Is she having seizures or stumbling? *Note: Some caregivers feel euthanasia is preferable to amputation, yet an animal with limited mobility may still be alert and responsive, and can have a good quality of life as long as the family is committed to quality care.*
0-10	**More Good Days than Bad**—When bad days outnumber good days, the pet's suffering is appreciable and quality of life might be too compromised. When a healthy human-animal bond in no longer possible, the caregiver must be made aware that the end is near.
Total: A total of > 35 points is acceptable quality of life for pets.	

Adapted from Canine and Feline Geriatric Oncology: Honoring the Human–Animal Bond, *Villalobos A, Kaplan L—Hoboken, NJ: Wiley-Blackwell, 2007, with permission.*

Published in the May 2008 issue of *Clinician's Brief* (Decision-Making Issues with Euthanasia, p 23).

生活品質評估量表(5H2M)

給分 0 到 10（0 表示無法接受，10 表示很好）

分數　　意義

☐ Hurt 疼痛(0-10)

寵物感到疼痛嗎？因呼吸困難而不舒服嗎？疼痛有獲得控制嗎？需要提供氧氣嗎？

☐ Hunger 飢餓(0-10)

寵物吃得夠嗎？需要以手餵食嗎？需要裝置胃管餵食嗎？

☐ Hydration 水合狀態(0-10)

寵物有脫水嗎?每日皮下輸液 1-2 次足夠嗎？皮下輸液寵物受得了嗎？

☐ Hygiene 整潔及衛生狀態(0-10)

寵物身體必須維持每日梳洗及清潔，特別是排便排尿後，寵物有褥瘡嗎？

☐ Happiness 快樂狀態(0-10)

寵物表現愉快及對事物興致高昂嗎？對周遭事物如家人或玩具有反應嗎？是否呈現沮喪、孤單、焦慮、無聊或害怕？可否將寵物床移至靠近廚房或家人活動處以減少疏離感？

☐ Mobility 活動能力(0-10)

寵物能否自行站立不需協助？需要家人或輔具輪椅幫忙助行嗎？寵物會想散步嗎？有癱瘓或走路跌跌撞撞嗎？有些照顧者會認為安樂死比截肢好，但是寵物行動力雖然受限仍舊有活力及有精神有注意力，只要好好照顧仍舊有好的生活品質。

☐ More good days than bad 好日子比壞日子多(0-10)

當壞日子比好日子多而且寵物處於受罪狀態，與人的互動幾乎喪失，表示寵物生命終點將近。

☐ 加總分數若高於 35 分，代表還有生活品質

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- Canine health related quality of life survey-21 (CHQLS-21)犬生活品質量表

CANCER TREATMENT FORM

SURVEY DATE _____ PET OWNER_____
NAME OF PERSON COMPLETING SURVEY_____
PET NAME _____ WEIGHT _____ SPECIES_____

INSTRUCTIONS: Please indicate your assessment by circling the number on the scale next to each question, providing your opinion on your pet's CURRENT health status.
EXAMPLE: 1 2 (3) 4 5

	Disagree		Neutral		Agree
HAPPINESS					
My pet wants to play	1	2	3	4	5
My pet responds to my presence	1	2	3	4	5
My pet enjoys life	1	2	3	4	5
MENTAL STATUS					
My pet has more good days than bad days	1	2	3	4	5
My pet sleeps more, is awake less	1	2	3	4	5
My pet seems dull or depressed, not alert	1	2	3	4	5
PAIN					
My pet is in pain	1	2	3	4	5
My pet pants frequently, even at rest	1	2	3	4	5
My pet shakes or trembles occasionally	1	2	3	4	5
APPETITE					
My pet eats the usual amount of food	1	2	3	4	5
My pet acts nauseous or vomits	1	2	3	4	5
My pet eats treats / snacks	1	2	3	4	5
HYGIENE					
My pet keeps him/herself clean	1	2	3	4	5
My pet smells like urine or has skin irritation	1	2	3	4	5
My pet's hair is greasy, matted, rough looking	1	2	3	4	5
WATER INTAKE (HYDRATION)					
My pet drinks adequately	1	2	3	4	5
My pet has diarrhea	1	2	3	4	5
My pet is urinating a normal amount	1	2	3	4	5
MOBILITY					
My pet moves normally	1	2	3	4	5
My pet lays in one place all day long	1	2	3	4	5
My pet is as active as he/she has been	1	2	3	4	5
GENERAL HEALTH					
General health compared to last evaluation	1	2	3	4	5
	Worse		Same		Better
General health compared to initial diagnosis of cancer	1	2	3	4	5
	Worse		Same		Better
Current quality of life	Very Poor				Excellent

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